GlobalGeoTree Dataset Visualisation Project

Exploring source bias, temporal trends, and regional patterns in a global tree observation dataset

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1 Introduction

This report analyses the GlobalGeoTree dataset, a global collection of more than 6 million georeferenced tree observations covering species, taxonomic groups, and geographical locations of trees combined from several sources from across the world.

This dataset was originally published in the paper 'GlobalGeoTree: A Multi-Granular Vision-Language Dataset for Global Tree Species Classification' (Mu et al., 2025).

GlobalGeoTree was assembled from existing datasets rather than from a new field survey. The authors combined tree-species lists from TreeGOER, GlobalTreeSearch, and other open sources, then standardised the taxonomy with the GBIF Species API. This produced an initial catalogue of 87,845 possible tree species.

For these species, the authors downloaded georeferenced occurrence records from GBIF — Global Biodiversity Information Facility. They kept openly licensed human observations from 2015–2024, removed duplicates and records with location problems, and excluded observations outside areas classified as forest (Mu et al., 2025).

The high-level goals of our project were to identify the sources of this dataset, examine how those sources are distributed, assess geographic and source-related biases, evaluate how well different regions are represented, and explore whether the data can support practical tree-related applications.

The report is organised in three main parts.

Part I examines global patterns: which sources dominate the dataset and how they shape what we see, how observations have changed over time, and how taxonomic composition and species richness vary across the world.

Part II zooms into Germany as a whole and then Berlin in detail.

Part III examines the United States, the most-represented country in the dataset, analysing species composition and geographic patterns at the national level.

2 Methodology

This section explains the analytical choices that we implemented throughout the project.

Data scope. Rows with missing country codes or species names are removed during loading. Coordinates are converted to numeric values and validated before the continent spatial join; records with missing or out-of-range coordinates, unmatched continents, or open-ocean assignments are excluded from the spatially analysed dataset.

Source classification. Each observation carries a free-text source field with thousands of distinct values. A keyword-matching function (`classify_source()`) assigns each source name to one of six groups: iNaturalist, Pl@ntNet, Other citizen-science apps, Institutional & survey data, Reference & thematic databases, and Other / unknown. The rules are applied in order, and the first match wins. This approach works well for English-language source names and major platforms but can misclassify source names in other languages or those using project-specific abbreviations. The "Other / unknown" group is a catch-all for everything that did not match any rule. This approach is demonstrated in the corresponding report section.

Geographic boundaries. For the Germany analysis, administrative boundaries at the NUTS1 level — the federal states (Bundesländer) — and the NUTS3 level (districts) are loaded from GISCO, the geographic information system of the European Commission. Berlin local-area boundaries come from a separate open-data project, listed under sources. If a local copy of these files exists, it is used directly; otherwise the file is downloaded from the public URL on first render.

Visualisation and subsampling. The dataset contains more than 6 million observations, so plotting every record would be slow, produce very large files, and add little visual value. Point-based maps therefore use reproducible random subsamples with a fixed seed of 42. Global maps display up to 40,000 observations, while source-specific and density maps use up to 50,000 records per group or map. US point and ridge figures use a 50,000-row figure-wide budget allocated proportionally across their displayed groups. To keep the rarest category legible, a small per-group minimum is applied, so on the point maps the scarcest categories are shown slightly above their true proportional share; the US period map applies no such minimum and uses the same inclusion rate in every panel, so its visible density remains proportional and comparable over time. Germany maps retain their region-specific limits, and the dataset is often small enough that all German observations can be shown. The sampling affects only the visualisations. All counts, rankings, species-richness tables, rarefaction analyses, and source summaries use the full relevant dataset.

Part I: Global Patterns

3 Data Preparation

3.1 Data Loading and Cleaning

The CSV file was loaded using `fread()`, which is designed to handle large datasets efficiently.

The text fields were cleaned by removing extra spaces, and columns such as coordinates and years were converted into the correct numeric format.

Records without a country code or species name were excluded.

The coordinates were matched to a world map to identify the continent of each observation.

Two manual corrections were added: Russia was divided at 60°E between Europe and Asia, and Réunion was assigned to Africa.

Measure	Value
Observation file	data/GlobalGeoTree.csv
Dataset period	2015–2024
Rows analysed	5,843,499
Distinct sources	2,397
Distinct source groups	6
Distinct countries	165
Distinct species	20,446
Germany rows	203,728

The table above confirms that the dataset loaded correctly and gives an overview of its scope. The analysis uses 5,843,499 observations in total.

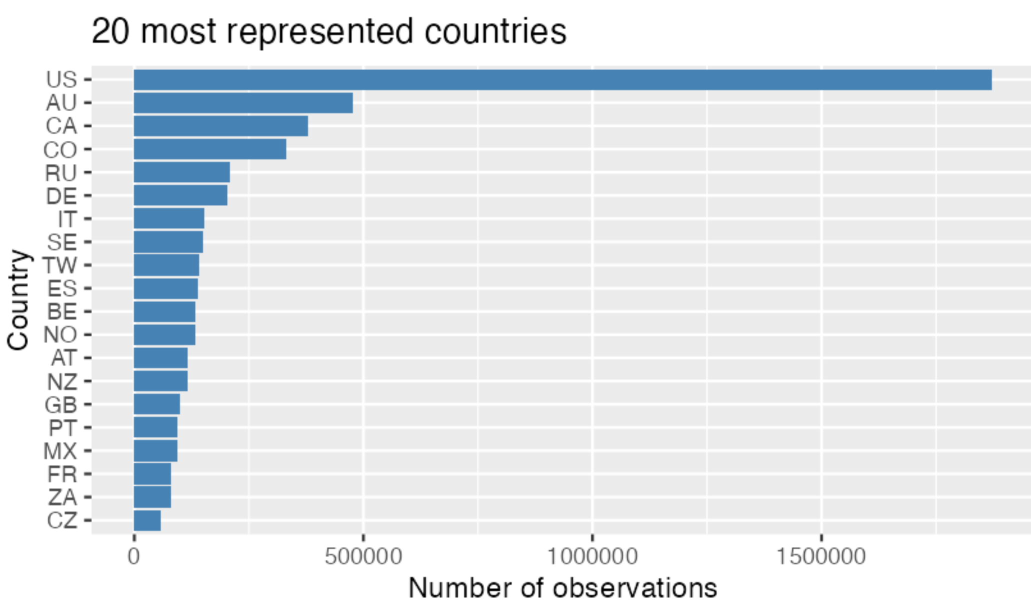
3.2 Dataset Overview

Column	Description
`sample_id`	Entry ID
`country_code`	Code of the country (e.g. US for the USA)
`level0`	Broad vegetation category grouping plants by shared ecological and physiological traits (e.g. Evergreen Broadleaf)
`level1_family`	Taxonomic Family — a group of related genera sharing common evolutionary characteristics (e.g. Asphodelaceae)
`level2_genus`	Taxonomic Genus — a group of closely related species (e.g. Aloe)

Column	Description
`level3_species`	Taxonomic Species — the most specific classification level (e.g. Aloe africana)
`location`	Country name
`source`	Source from which the occurrence record was obtained (e.g. iNaturalist Research-grade Observations)
`species_key`	Species identifier
`year`	Year of the observation
`longitude`	Geographic longitude
`latitude`	Geographic latitude

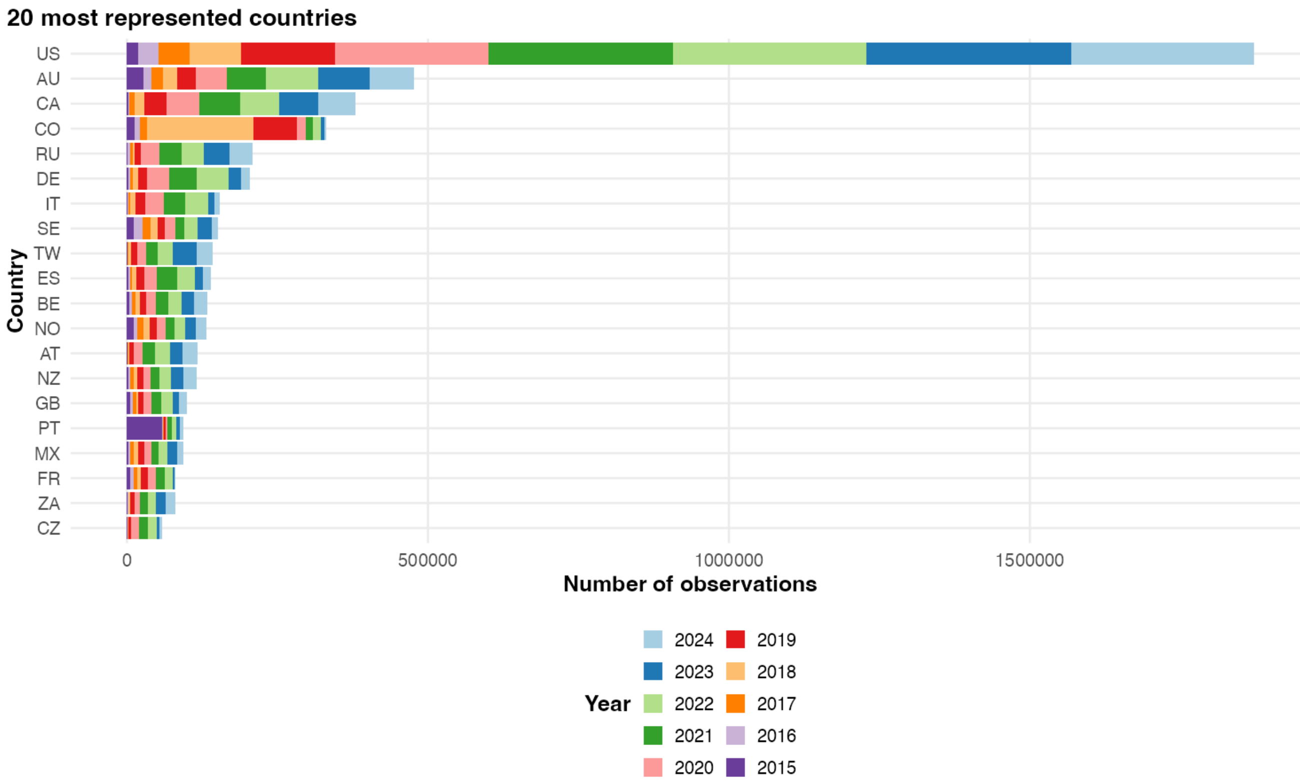
3.3 Distribution of the data across the world

Data distribution across the different countries:



We see that most of the data comes from developed western countries, and especially from the US, which has a very large number of observations compared to the other countries.

Therefore the following research questions arise: is there any evolution of the data collected throughout the years? Do some countries present large imbalance in the data distribution? How could this affect our analysis?



The graph shows that observations across different countries were collected over a ten-year period. To understand these records in more depth, we analyse their data sources and examine the worldwide distribution of observations in the dataset.

4 Source and Platform Bias

In this section of the report we were trying to answer the following research questions:

- *How many distinct sources are in the dataset, and how can they be grouped?*
- *Is the dataset dominated by a few platforms, or do many sources contribute evenly?*
- *What does source concentration mean for the rest of the analysis?*

This section starts from the raw `source` column in the dataset and works outwards: first what the column actually contains, then how its thousands of names can be grouped, which sources contribute the most observations, how source composition varies by country and plant family, and finally whether different sources leave distinct geographic footprints.

4.1 Exploring the Raw Source Field

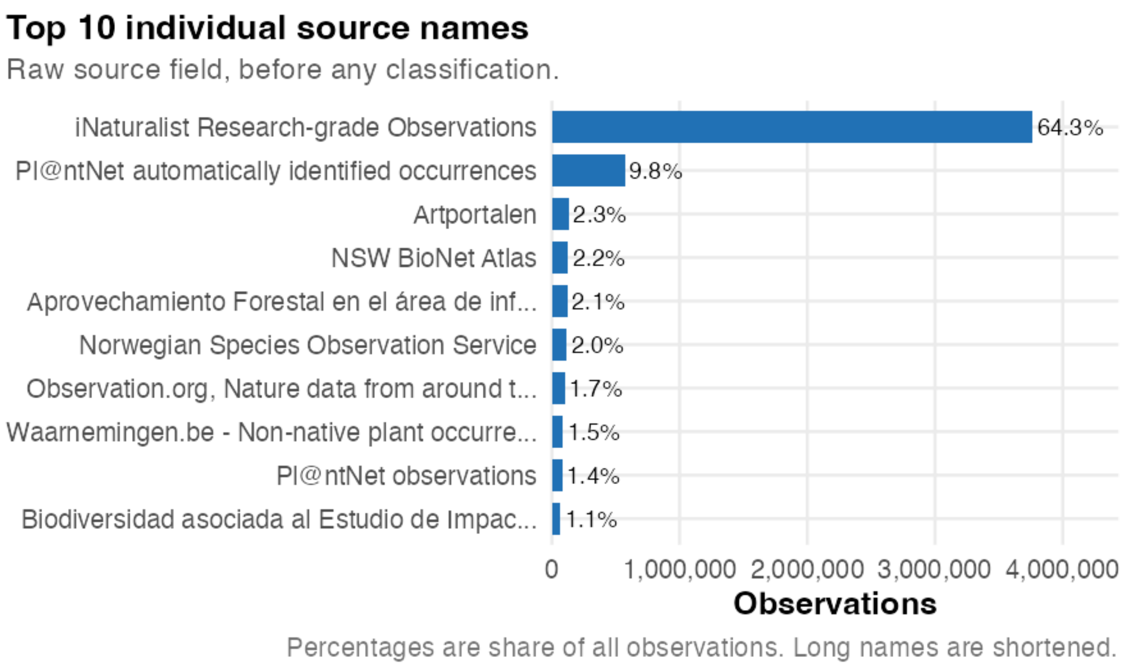
The `source` column contains free-text entries. The dataset does not specify how many providers contributed records, what types of organisations they are, or whether the same platform is always written in the same way.

Because all later analysis in this section depends on these details, the source field is examined first, before any comparisons are made.

The field contains **2,397 distinct source names** across 5,843,499 observations. They are not evenly sized: the ten largest names alone account for **88.4%** of all observations, while 2,052 names contribute fewer than 100 observations each and together make up just 0.5% of the data.

Observations per source	Distinct sources	Observations	Share of observations
100,000	7	4,930,405	84.4%
10,000 – 99,999	16	559,024	9.6%
1,000 – 9,999	83	257,080	4.4%
100 – 999	239	70,571	1.2%
< 100	2,052	26,419	0.5%

This is the main problem: a small number of very large sources provide most observations of the dataset, while hundreds of smaller sources provide only a very small part of it. Therefore, an analysis should not assume that all sources contribute equally. Doing so would focus too much on the many small sources, even though the few large sources have the greatest influence on the results.



Reading the actual names is what makes the grouping possible. The most common sources belong to several recurring provider types: two global identification apps, several national observation portals, an aggregator category, forest inventory databases, and floristic archives.

The same provider may appear under different spellings, and some source names are written in languages other than English. Both issues affect the classification method explained in the next section.

4.2 Source Classification

4.2.1 From Raw Names to Source Groups

The 2,397 distinct source names are too numerous to compare individually. Grouping them makes broader questions answerable, such as whether a pattern is linked to citizen-science apps or formal surveys, although some detail is lost within each group.

The groups were derived from the source names rather than defined in advance. The ten most frequent names already account for 88.4% and revealed several recurring provider types. Each type became a group, identified through keywords. The `classify_source()` function applies these rules in order, assigns the first match, and uses a final catch-all category.

Keyword matching is more practical than exact matching because provider names are often inconsistent, for example "iNaturalist", "iNat", and "iNaturalist Research-grade Observations". It also avoids manually listing the 2,052 rare names.

GBIF is not one of the groups, even though it is the aggregator every record passed through. The authors downloaded the entire dataset from GBIF, so the `source` field names the underlying dataset that GBIF redistributes — "iNaturalist Research-grade Observations", "EOD – eBird Observation Dataset" — and never GBIF itself.

"Other / unknown" should not be treated as fully unclassifiable. Multilingual names and project codes may fail to match rules based mainly on English-language terms. The category is therefore examined further rather than discarded.

The resulting six groups (ordered from most to least specific):

- **iNaturalist** — the largest citizen-science biodiversity platform worldwide.
- **Pl@ntNet** — a plant-identification app with a stronger presence in French-speaking countries and the Global South.
- **Other citizen-science apps** — regional portals (e.g., Artportalen, iRecord, Naturgucker, NatureMapr).
- **Institutional & survey data** — herbarium/museum collections and systematic field campaigns.
- **Reference & thematic databases** — compiled floristic atlases, global plant databases (e.g., BIEN, RAINBIO, Tropicos), Wikidata, and invasive-species tracking.
- **Other / unknown** — everything that did not match any keyword rule.

The table below shows the rules themselves, in the order `classify_source()` evaluates them, together with the largest source names each rule actually captures. The examples are read from the data rather than written by hand, so they stay accurate if the rules or the dataset change.

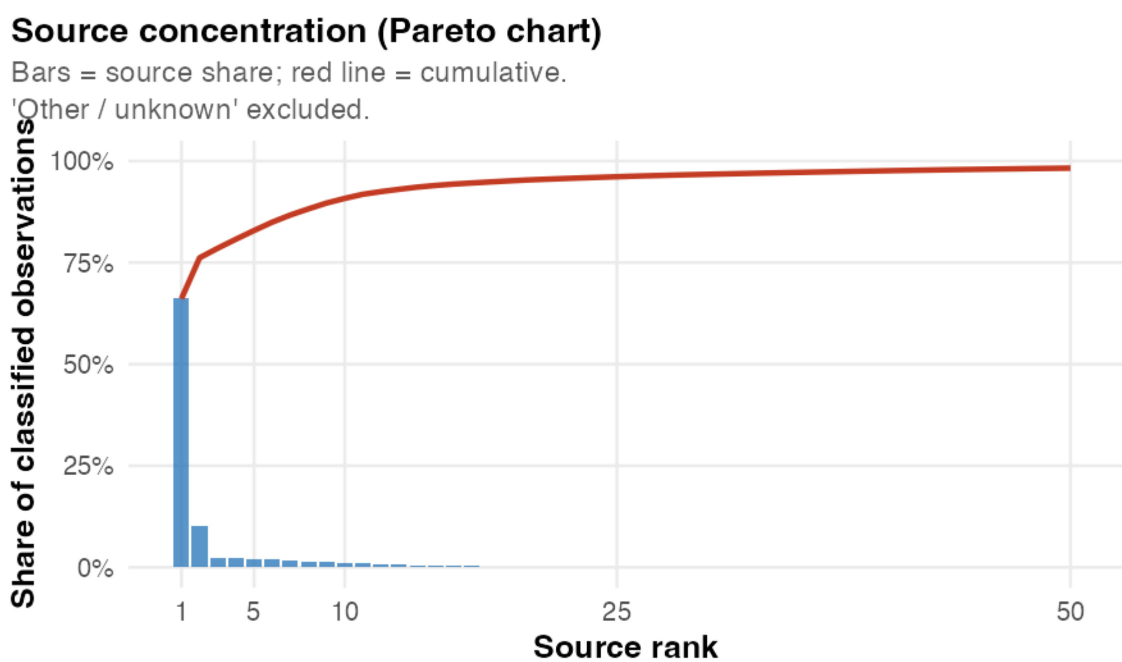
4.2.2 Classification coverage

Of all 5,843,499 observations, **97.3%** are assigned to one of the five named groups, and **2.7%** fall into "Other / unknown". The higher the coverage rate, the more reliably the source-group comparisons throughout this report represent the full dataset.

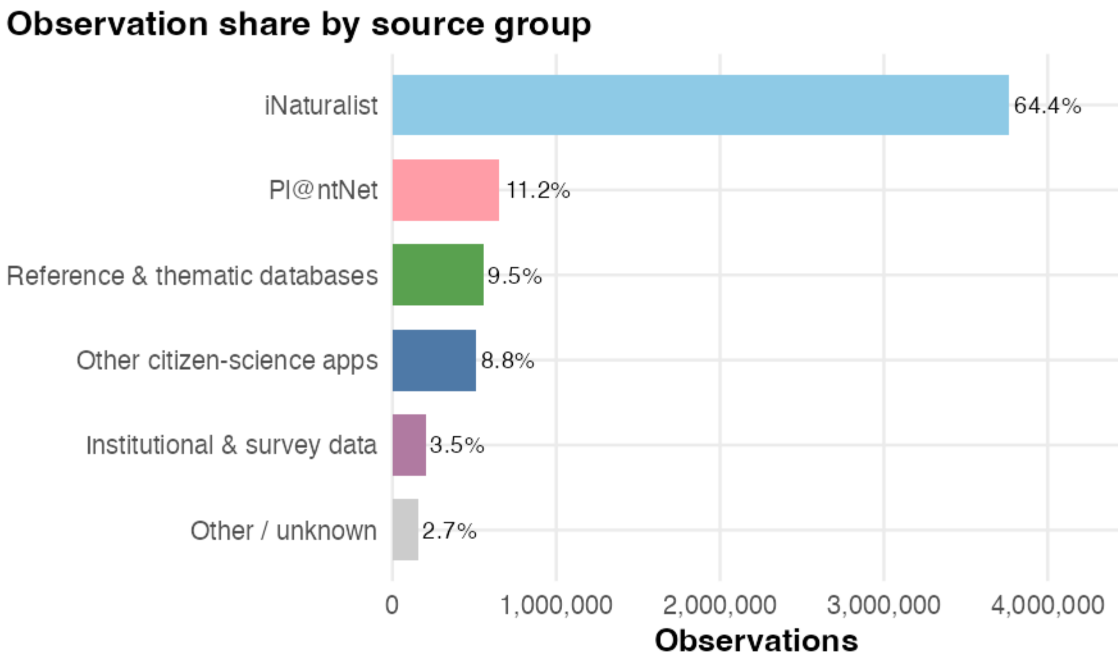
Order	Keyword rule (abridged)	Assigned group	Example matched sources
1	inaturalist, inat	iNaturalist	iNaturalist Research-grade Observat...; ""Flora of Russia"" on iNaturalist:...
2	plantnet, pl@ntnet	Pl@ntNet	Pl@ntNet automatically identified o...; Pl@ntNet observations
3	artportalen, observation.org, naturgucker, irecord, …	Other citizen-science apps	Artportalen; Norwegian Species Observation Service
4	herbar, museum, botanic, inventory, survey, monitoring, …	Institutional & survey data	Forestry Inventory 2015; Danish government nature monitoring...
5	atlas, flora, bien, rainbio, tropicos, wikidata, invasive, aprovechamiento forestal, impacto ambiental, …	Reference & thematic databases	NSW BioNet Atlas; Aprovechamiento Forestal en el área...
6	— no rule matched —	Other / unknown	EOD – eBird Observation Dataset; SA Flora

4.3 Source Concentration

iNaturalist alone contributes **64.4%** of all observations; the three citizen-science groups together account for **84.3%**.



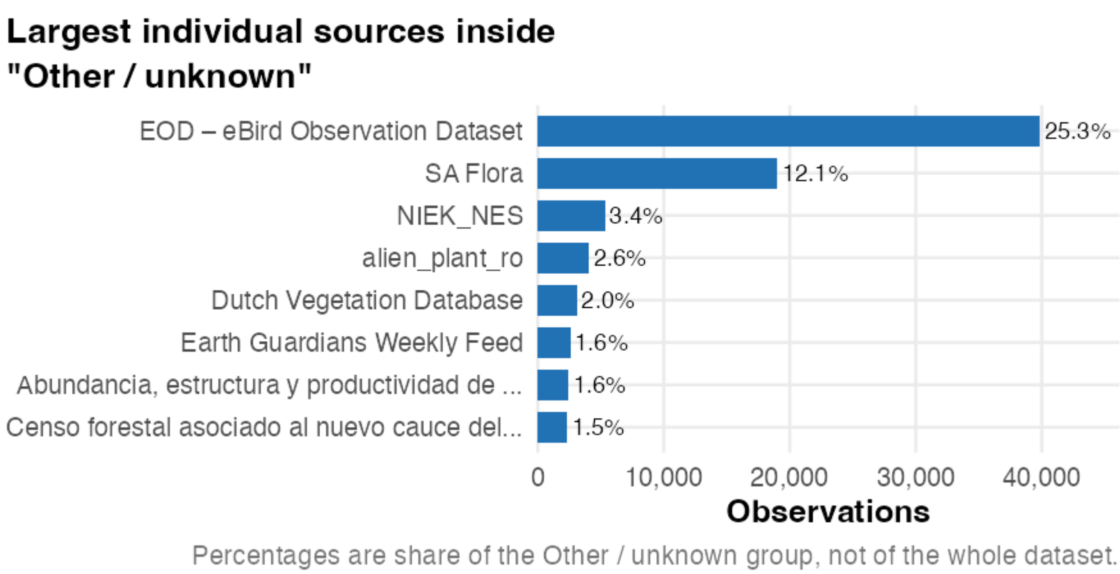
The Pareto chart shows high source concentration: a small number of sources provide most of the classified observations, and the cumulative share rises quickly. As a result, any consistent bias in the largest sources can strongly affect the full dataset.



The bar chart also shows that citizen-science platforms provide the largest share of observations, with iNaturalist as the main contributor. Institutional and reference sources are smaller, but they may cover regions and species that citizen-science platforms do not.

4.4 What Is in "Other / Unknown"?

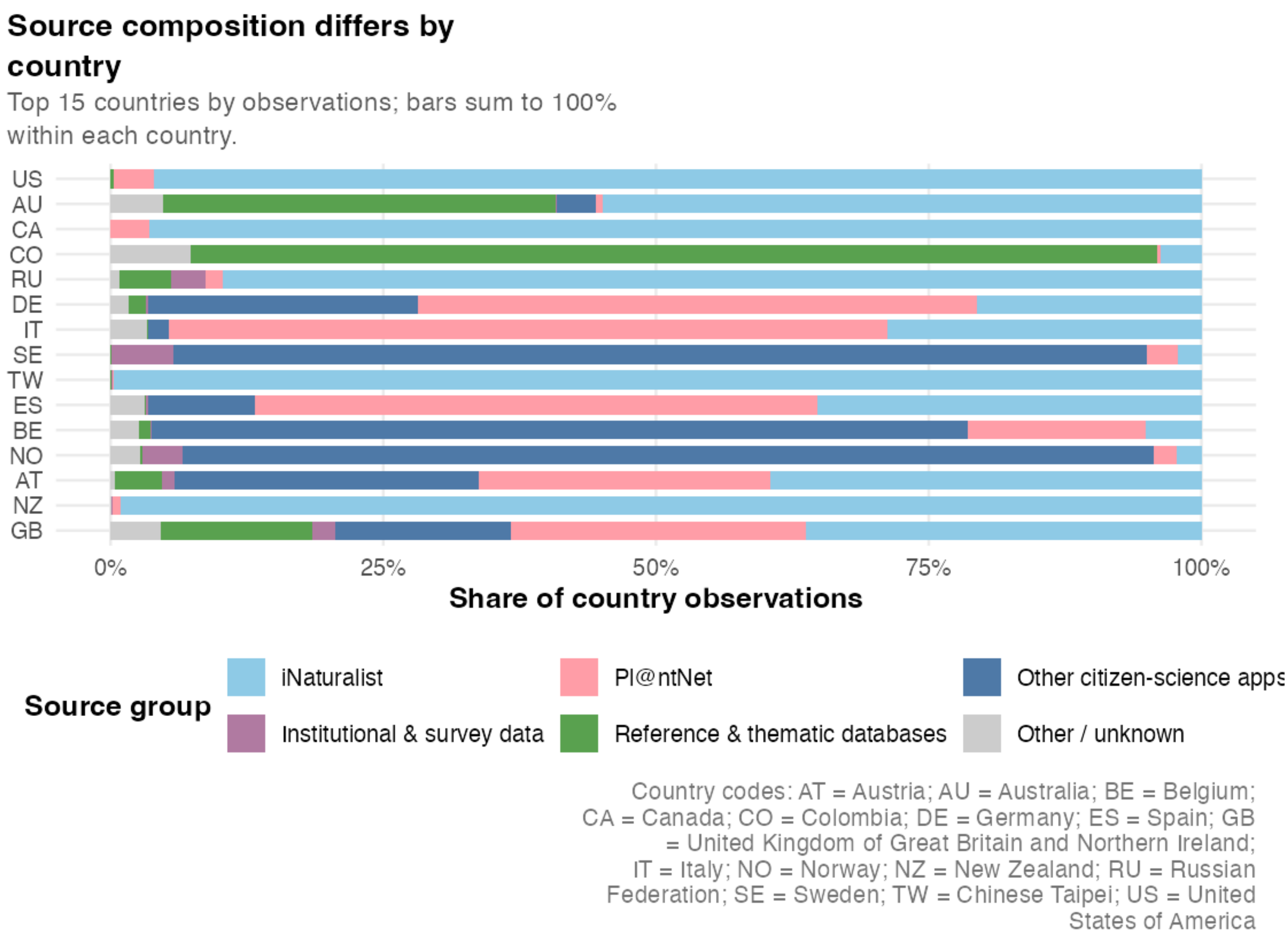
"Other / unknown" contains 1,426 individually named sources that did not match any keyword rule. The chart below shows the largest contributors within this heterogeneous catch-all group.



4.5 Countries and Regional Coverage

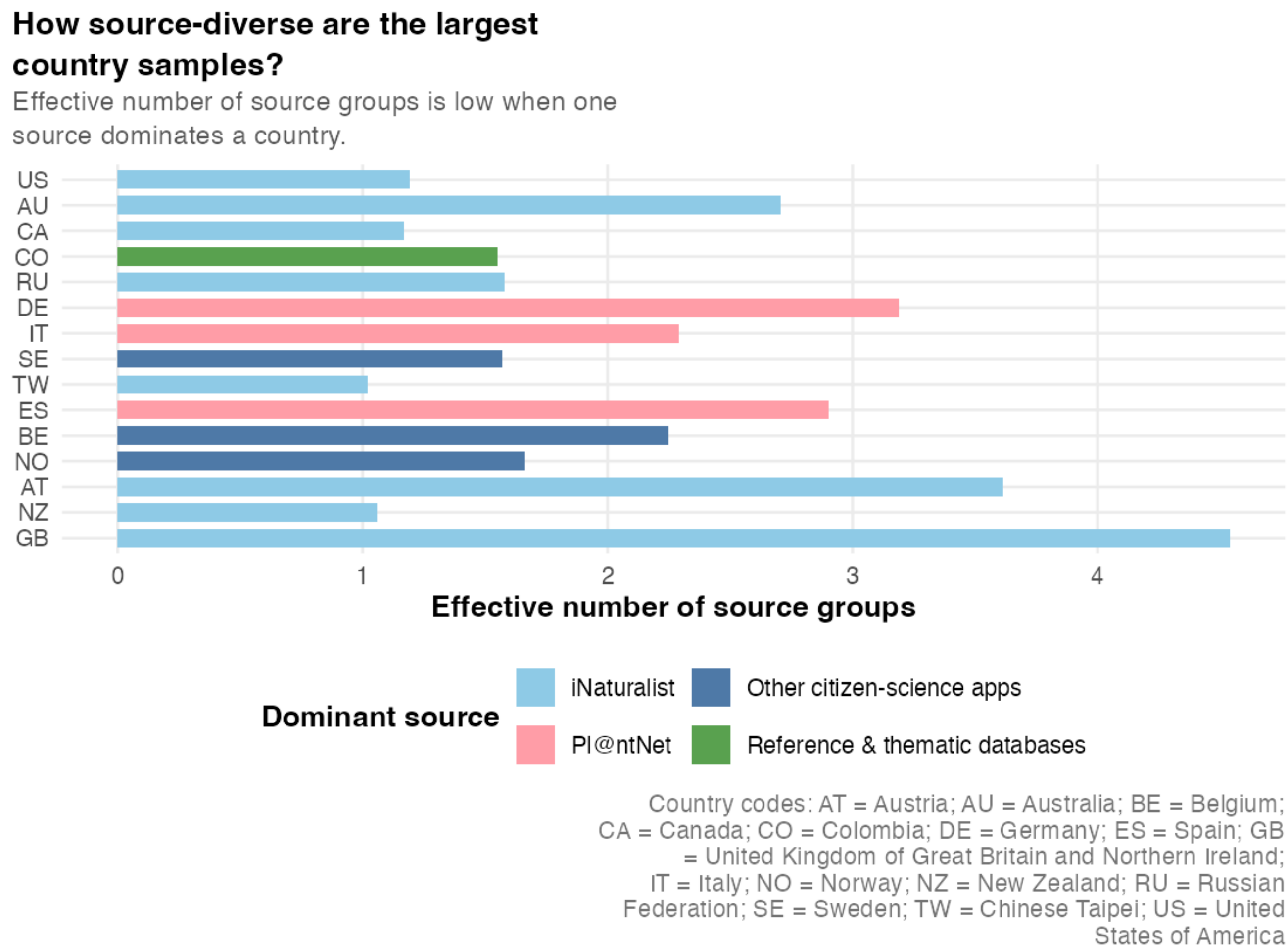
- *Does source composition vary by country, or are some countries dominated by a single platform?*

When most observations from a country come from one platform, that platform strongly influences the recorded species and locations. The charts below compare the source composition of the most-observed countries and measure how diverse their sources are.



The stacked bar chart shows large differences between countries. Some, such as the United States, rely mainly on iNaturalist, while others, including the United Kingdom and Germany, use a more balanced mix of sources. Countries dominated by one source are more exposed to its geographic and taxonomic biases.

Source diversity is measured using Shannon entropy, converted into an "effective number of source groups". A value close to 1 means that one source group dominates.



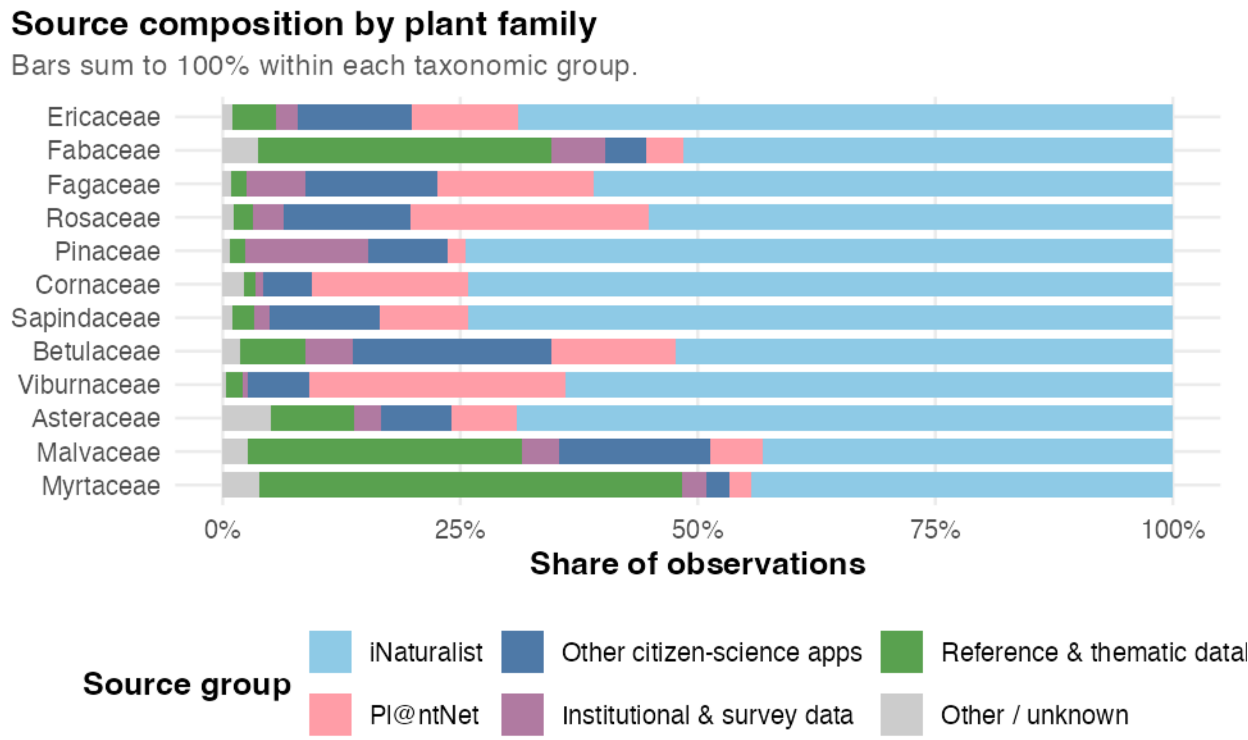
Countries with low effective source diversity, close to 1, depend almost entirely on a single source. Their recorded species and locations therefore reflect the users and sampling patterns of that platform. Countries with higher source diversity combine several independent data sources, making their results less affected by the biases of any one platform.

Austria, Germany, Spain and Portugal draw on several source groups. In contrast, the US, Canada, Taiwan and New Zealand rely mostly on one source.

4.6 Sources and Taxonomic Groups

- *Do different sources record different plant families, or is the source mix uniform across plant families?*

When one source type dominates, apparent geographic or temporal patterns may reflect gaps in that source's coverage rather than real ecological differences.



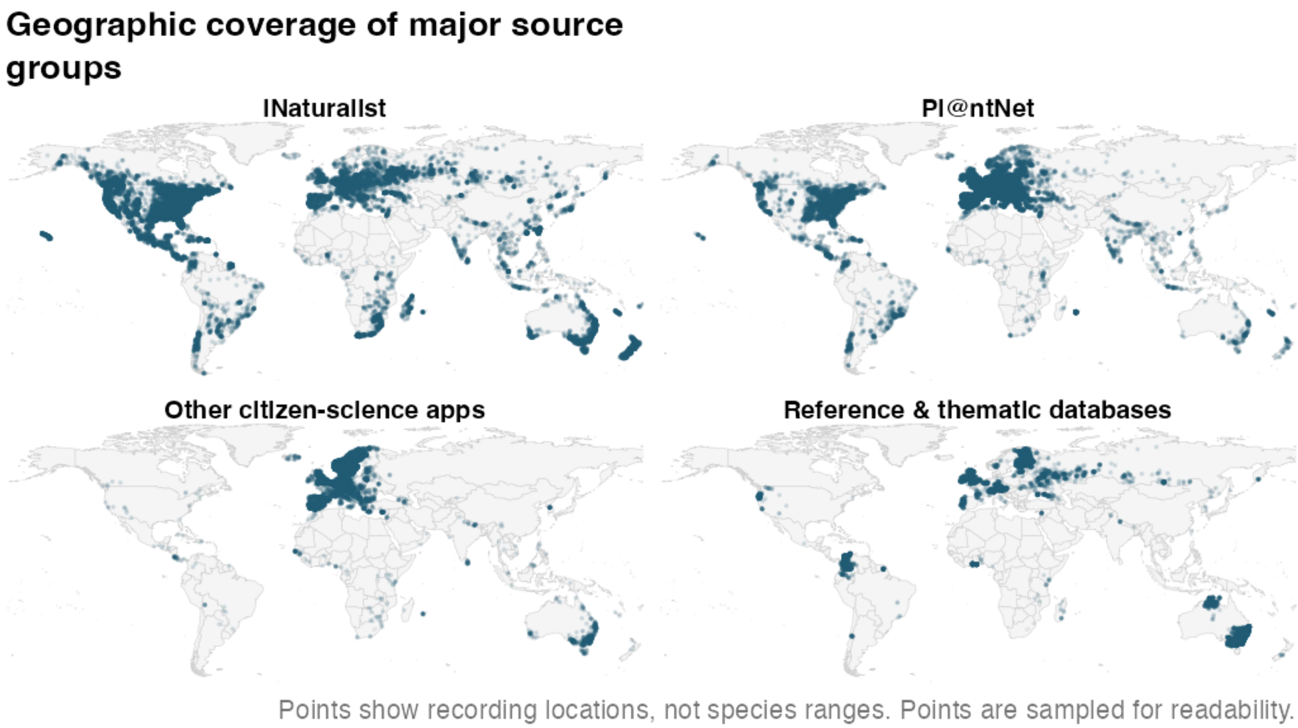
The source mix differs across plant families. Some families are recorded mainly through citizen-science apps, while others depend more on institutional surveys or reference databases.

iNaturalist is the largest source for all plant families, but its share varies. Some families also receive many records from Pl@ntNet, reference databases, surveys or other apps.

4.7 Geographic Footprints of Data Sources

- *Do the major sources cover the same regions, or do they each focus on different parts of the world?*

If sources are geographically complementary, the combined dataset may be broader than any single source. If they overlap heavily, gaps in one source may persist in the whole dataset.



The faceted map reveals different spatial footprints across source groups. iNaturalist provides the most geographically widespread coverage but is concentrated in North America, Europe, and Oceania. Pl@ntNet complements it with stronger presence in French-speaking Africa and parts of South America, while still showing high data coverage for Europe. Institutional and survey sources tend to fill in localised regions that citizen-science platforms miss. These differences mean that where the dataset is geographically strong or weak depends directly on which sources are active in a given region.

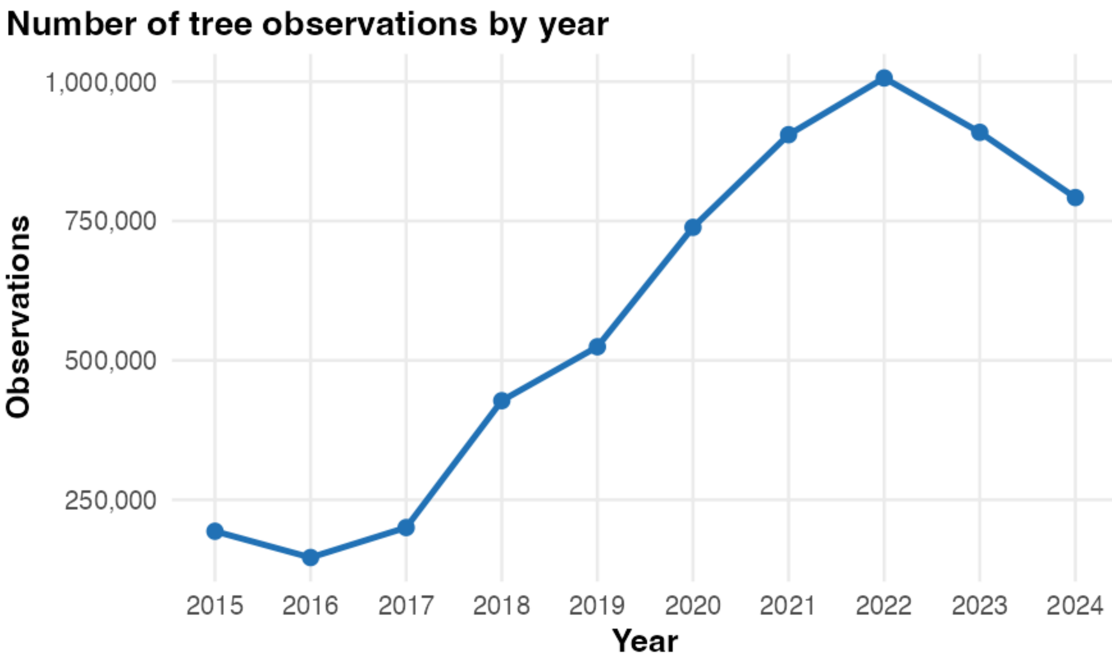
5 Temporal Development

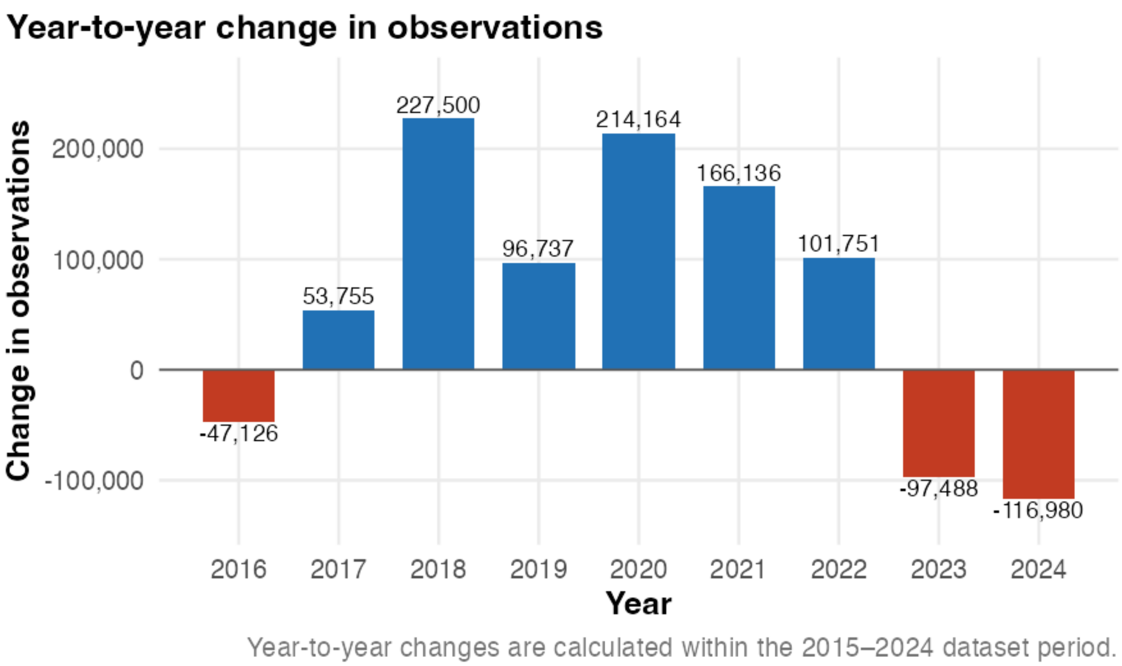
- *How do observation volume and geographic representation differ by observation year?*
- *Which sources are associated with the largest increases in dated observations?*

This section traces observations across the dataset time period, decomposes the pattern by source group, and examines whether the geographic footprint broadens or mainly becomes denser.

5.1 Observation Volume Over Time

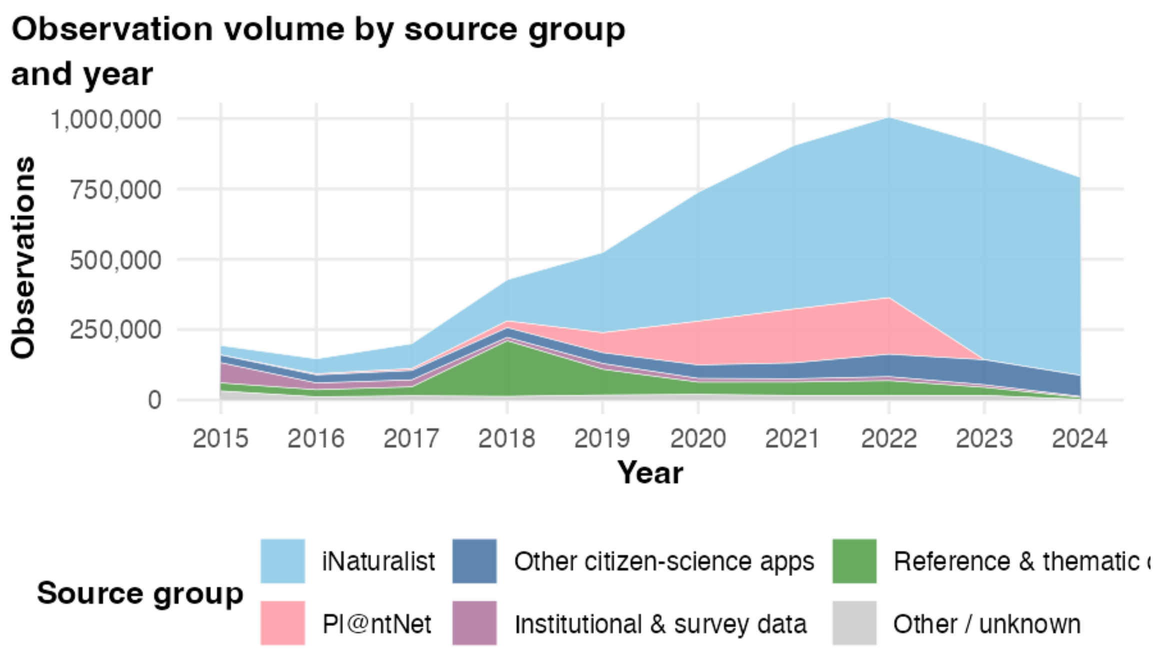
Between 2015 and 2024, observations changed from **193,601** to **792,050**, an overall change of **309.1%**.



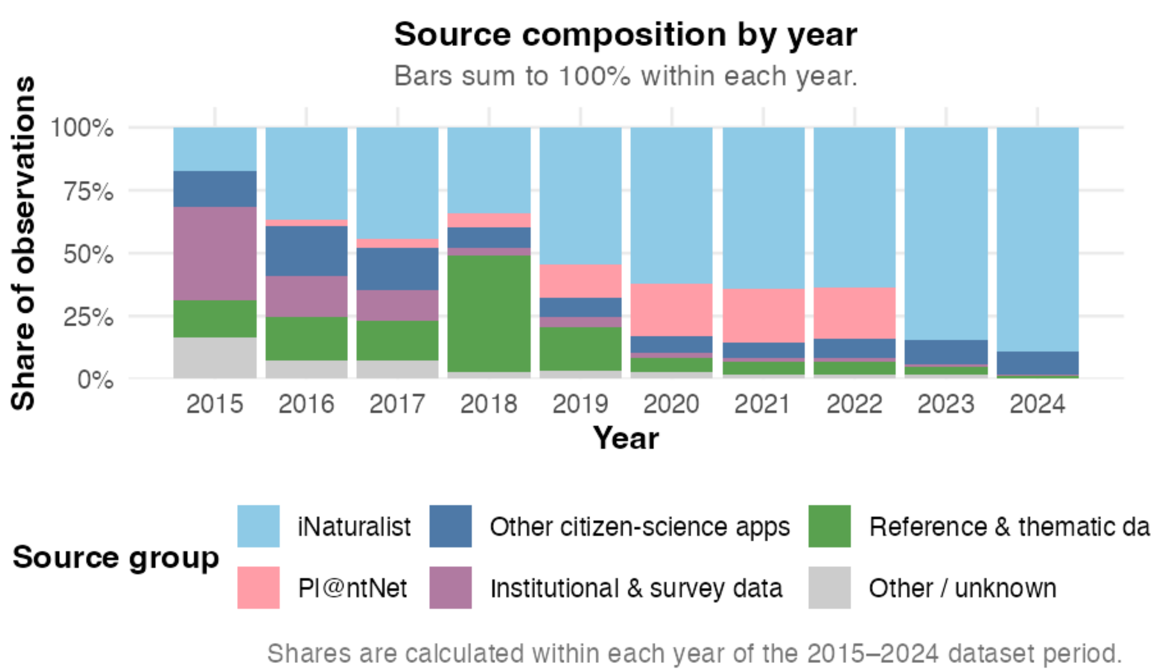


The strongest growth occurs in **2018** (+227,500 observations), and the strongest decline in **2024** (-116,980).

5.2 Which Sources Drive the Temporal Trend?

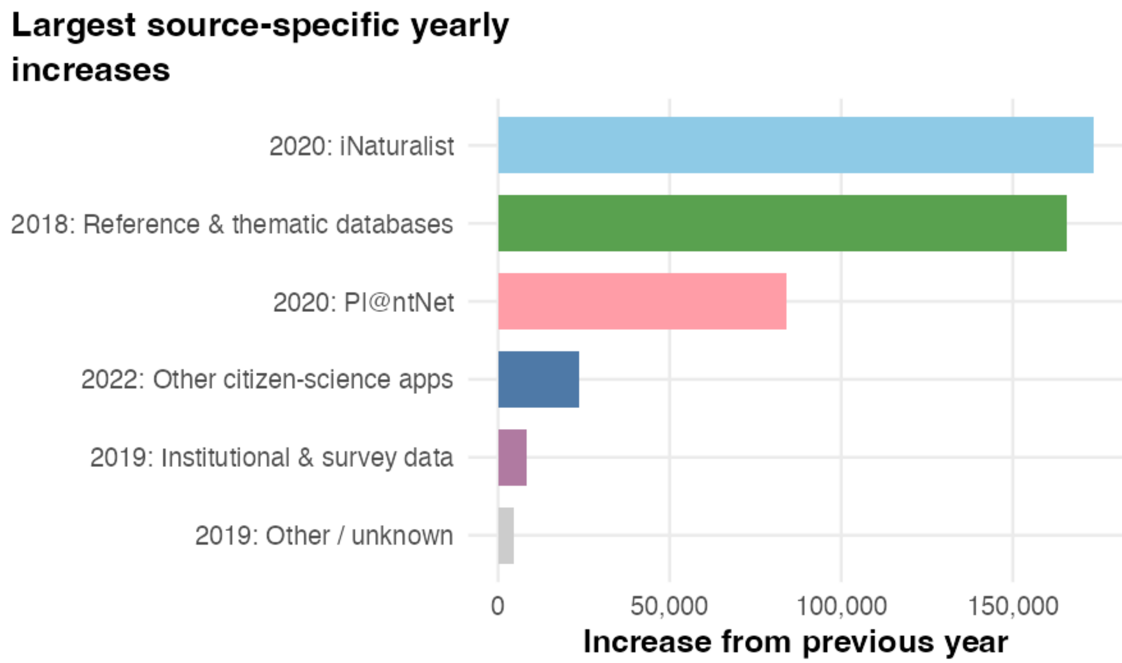


The stacked area chart shows that the growth in dated observations is largely linked to citizen-science platforms, especially iNaturalist. As iNaturalist's contribution increases, the total number of observations rises as well, reflecting changes in platform activity and geographic coverage. Before 2018, observations were distributed more evenly across source groups. The chart also shows a small increase in reference and thematic database records in 2018, followed by a noticeable rise in Pl@ntNet observations between 2020 and 2022.



The year-by-year source share view shows whether the source mix is shifting over time. If one source group's share is growing while others shrink, the dataset's overall character is changing —

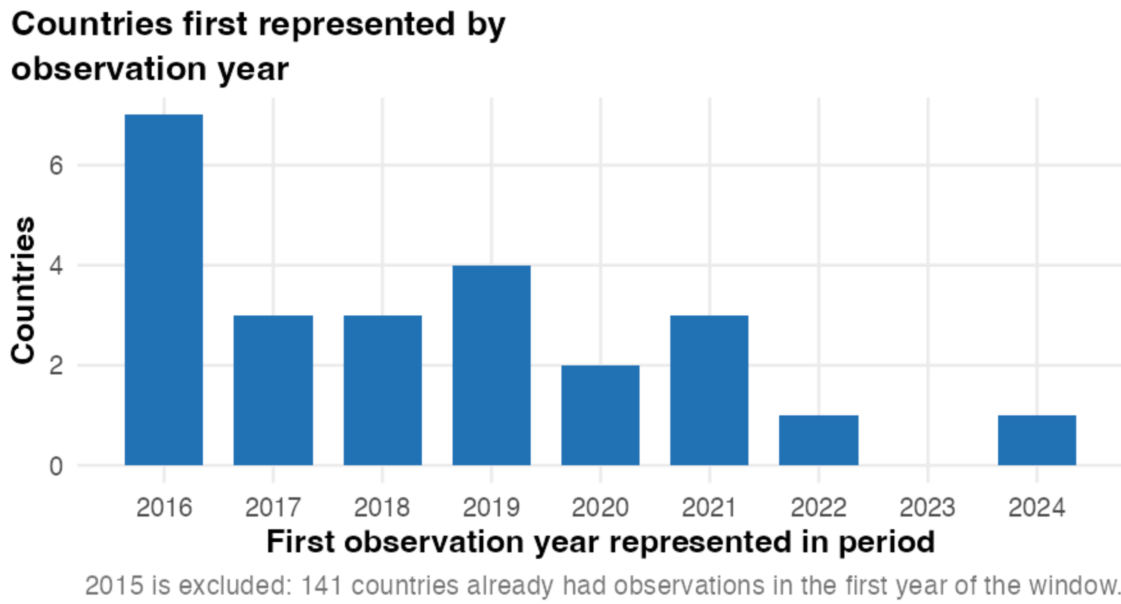
even if total volume increases. A stable composition would suggest that all sources are growing at roughly the same rate.



The chart above highlights the single largest year-over-year jumps for each source group, making it easy to identify which platforms drove the biggest surges in data volume and when those surges occurred.

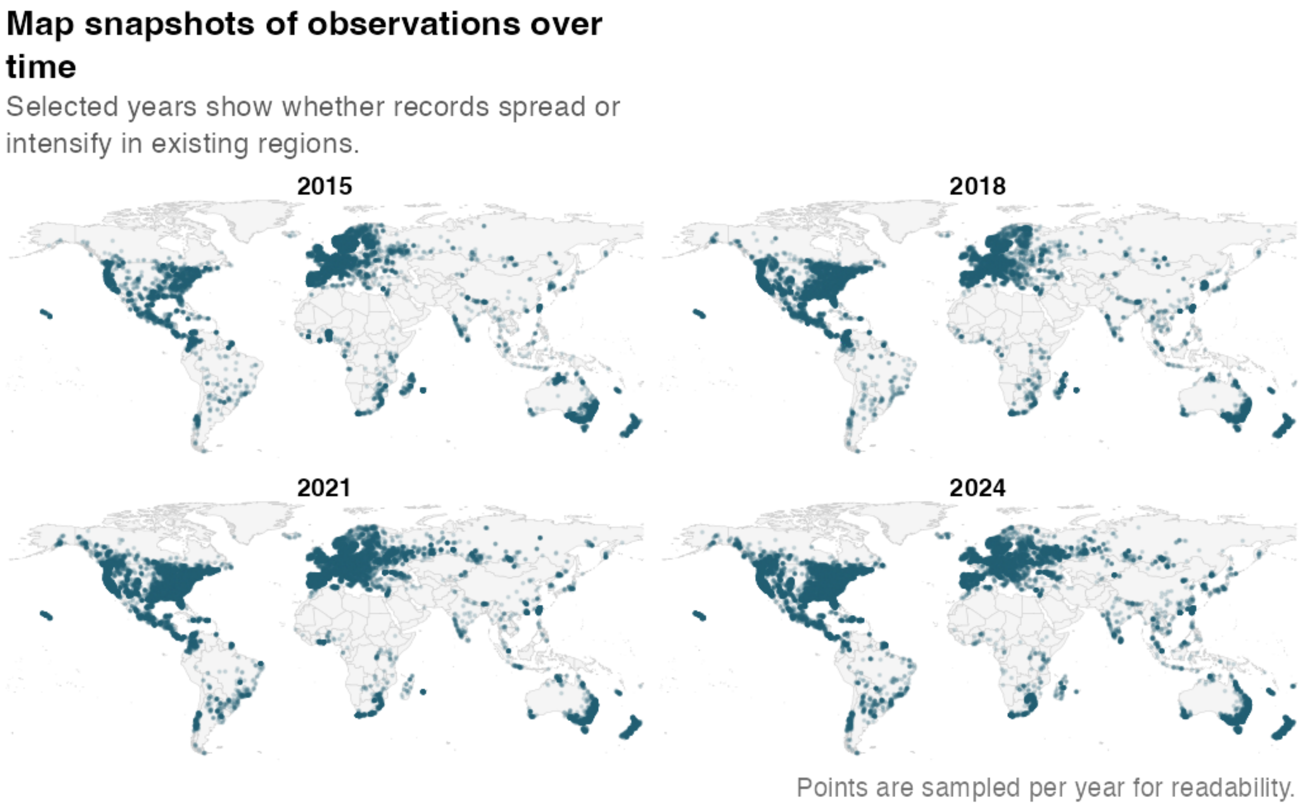
5.3 Geographic Coverage Over Time

This subsection examines how the geographic reach of observations dated from 2015 to 2024 differs over time. It compares the first observation year represented for each country within this period and maps observation locations at selected snapshot years.



This chart shows when countries first appear among the observations dated within the selected period. A declining count suggests that later observation years add relatively few countries not already represented by earlier-dated records.

The maps below show observation locations at four snapshot years, revealing whether records spread into new regions or simply become denser in existing ones.



Comparing the four snapshot years shows that the geographic footprint of the dated observations was already broadly established early in the period: North America, Europe, and parts of Oceania and South America are visible from the first panel. Later panels show a clear increase in point density within these same regions, but comparatively little expansion into previously unrepresented areas such as central Africa, Central Asia, or the Amazon interior. Within the observation-year window, the pattern is therefore more consistent with intensification than with geographic expansion.

6 Taxonomic Composition and Geographic Patterns

6.1 Species observed in this dataset

What are the different tree families referenced in this dataset?

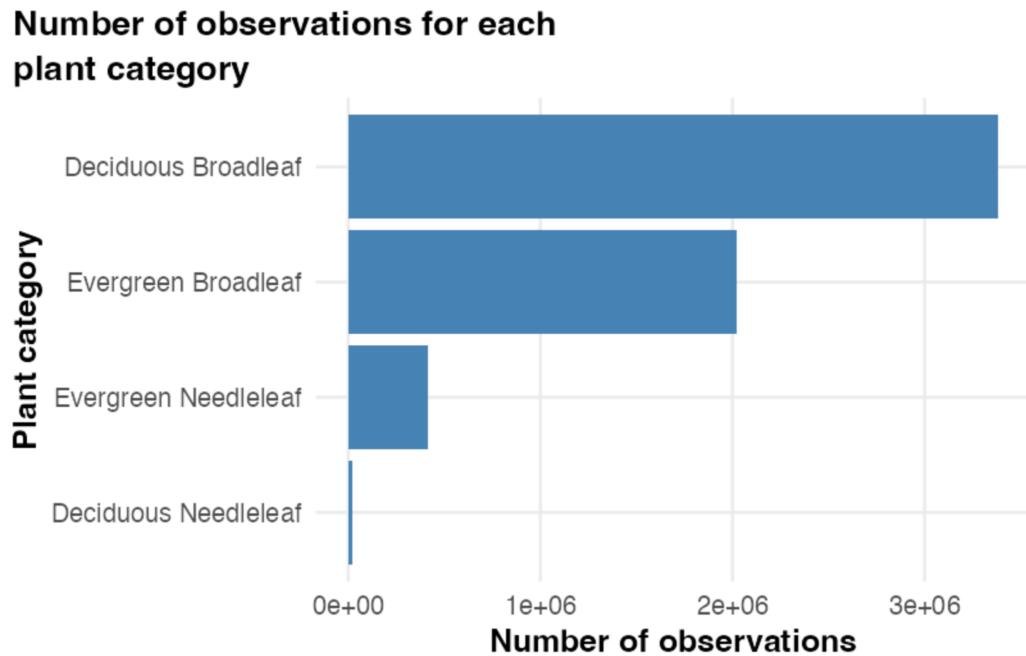


Figure 1: Observation counts for the four foliage categories in GlobalGeoTree.

Definition of each tree category:

- Evergreen Needleleaf: An evergreen needleleaf refers to woody vegetation that retains needle-like or scale-like foliage year-round
- Evergreen Broadleaf: An evergreen broadleaf is a plant or tree that has wide, flat leaves (rather than needles or scales) and retains its foliage year-round
- Deciduous Broadleaf: a diverse group of flowering plants (angiosperms) characterised by flat leaves that are shed annually before winter (Oak, Maple, Beech, Birch, and Poplar/Aspen)
- Deciduous Needleleaf: a unique group of conifers that, unlike their evergreen counterparts, shed all their needles annually in the autumn (rare trees)

Deciduous Broadleaf is the largest foliage category among observations in this dataset. Deciduous Needleleaf accounts for a much smaller share and appears in more geographically restricted areas. These values describe the composition of GlobalGeoTree observations rather than the worldwide abundance of each foliage type.

What are the different taxonomic families represented in this dataset and in which proportion are they represented?

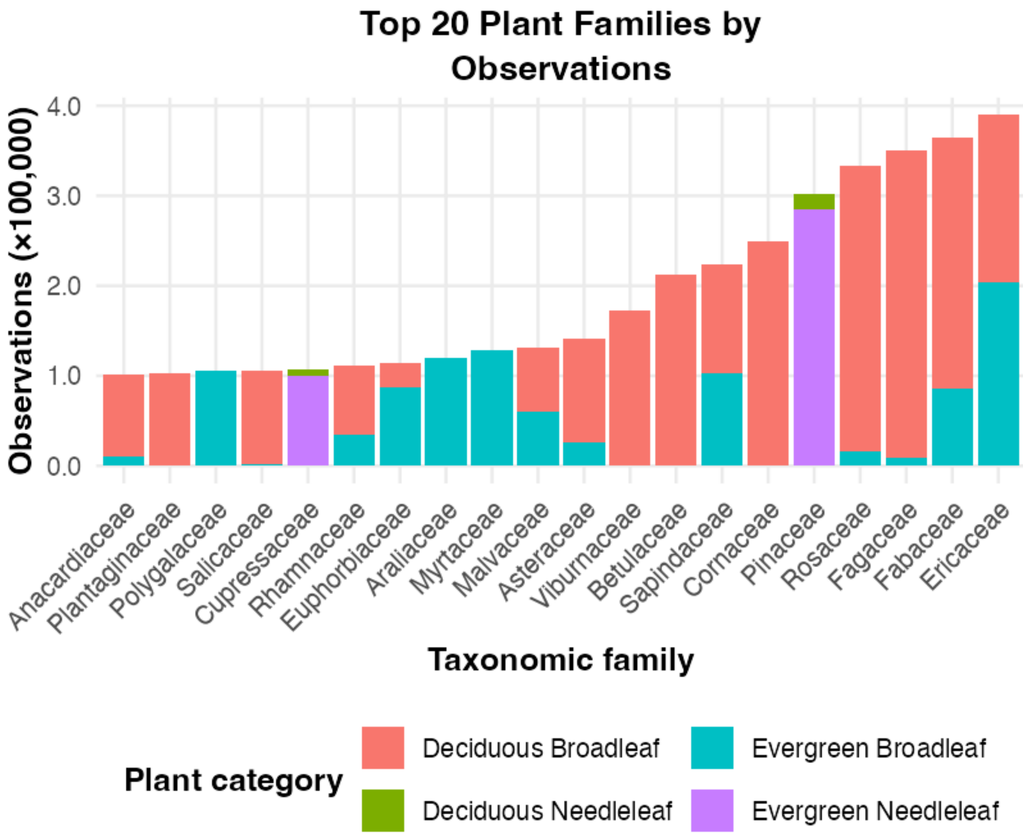
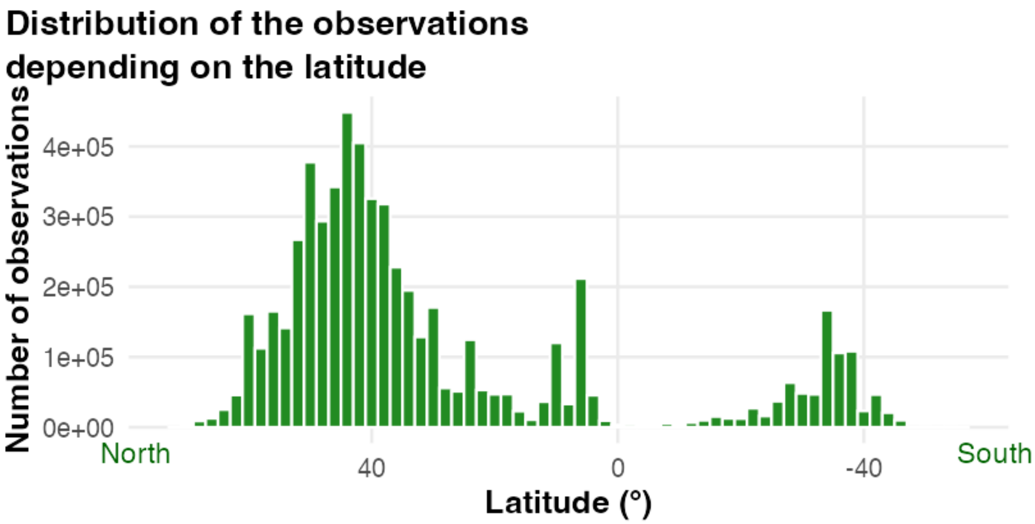


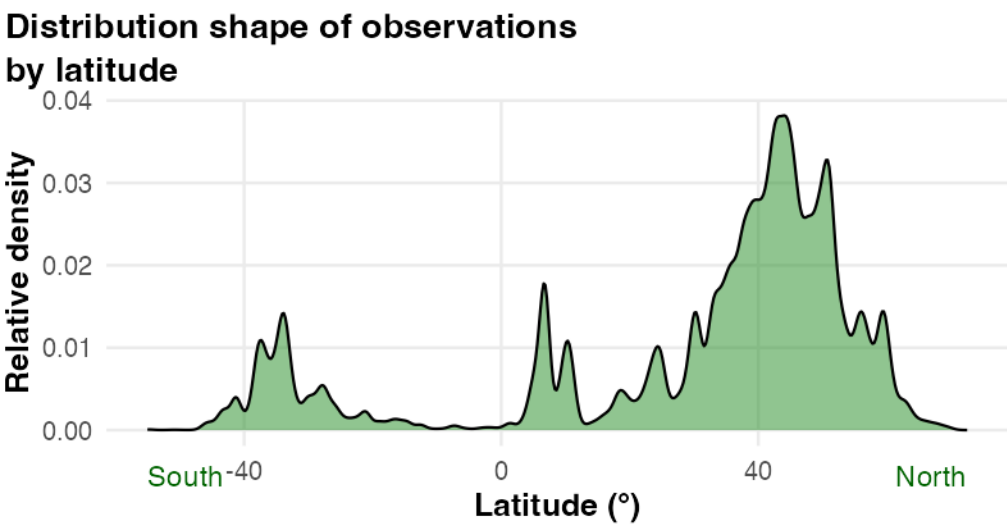
Figure 2: The 20 taxonomic families with the largest observation counts in the dataset.

Again, here we see that the most represented taxonomic families in this dataset are also the most represented ones across the world, especially in North America, Europe and the north of Asia, where most of the data come from. However, as the source bias analysis above shows, this apparent representativeness should be interpreted with caution: the dataset reflects where observers and platforms are most active, not a systematic global survey.

6.2 Geographical distribution of the observations

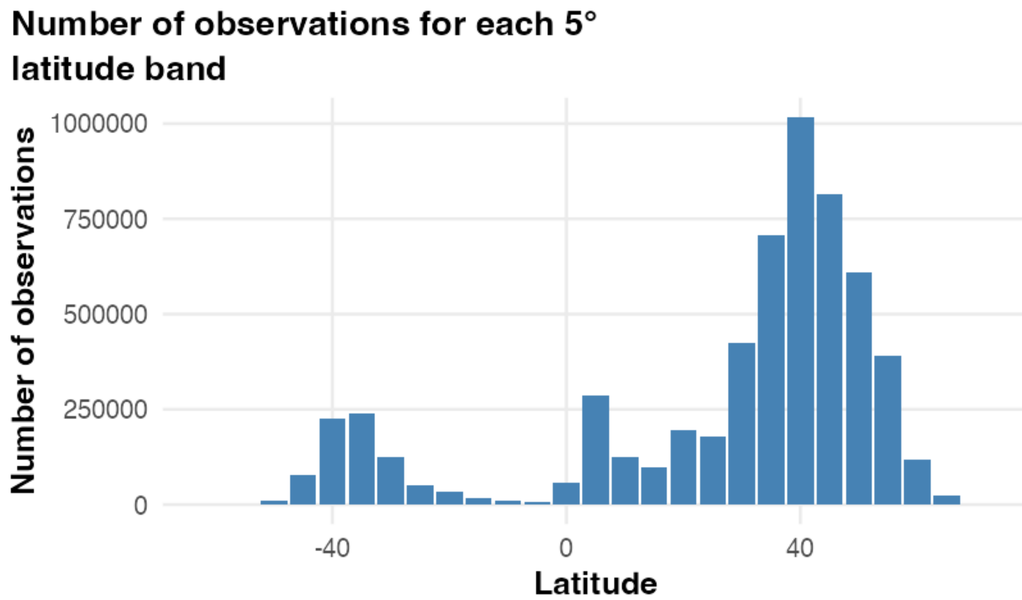
How are the trees distributed depending on the latitude?





The histogram preserves absolute observation counts in each latitude interval, while the density curve emphasises the overall distribution shape. Both show substantially more recorded observations in the Northern Hemisphere, which may reflect differences in land area and sampling effort as well as tree distribution.

Coarser aggregations are also often used in ecological studies, because 5° latitudinal bands carry a sufficient amount of information without showing too much noise.

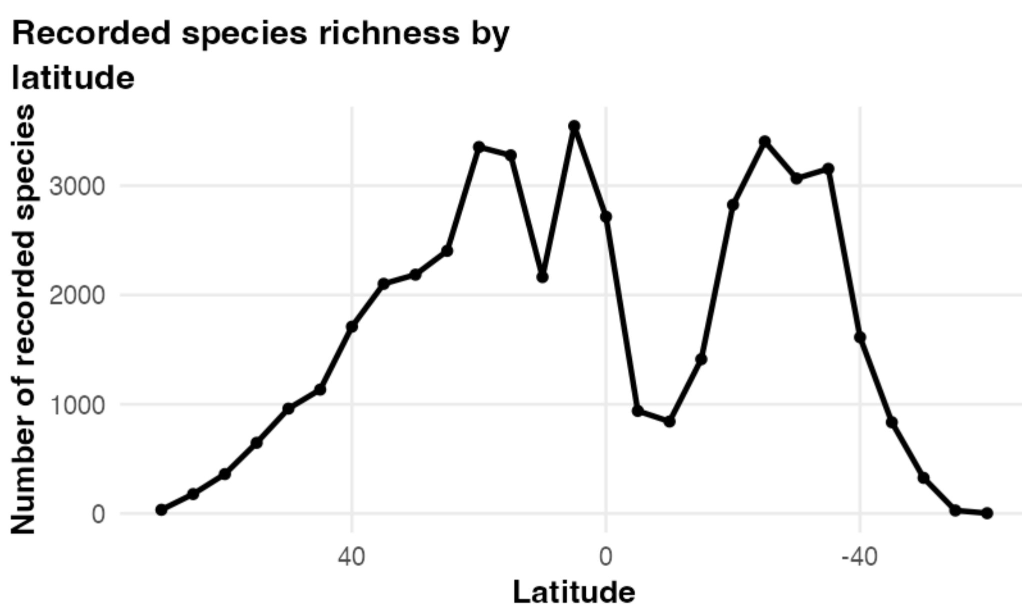


Binned into 5° bands, the distribution shows the same pattern as the histogram above: recorded observations concentrate heavily in the northern temperate latitudes.

We usually take 5° slices because they are narrow enough to reveal local tendencies and geographical patterns (climatic gradients, biodiversity, species distribution, and so on), but wide enough to avoid the noise and uninformative detail that finer bands introduce.

6.3 Geographical observations of species

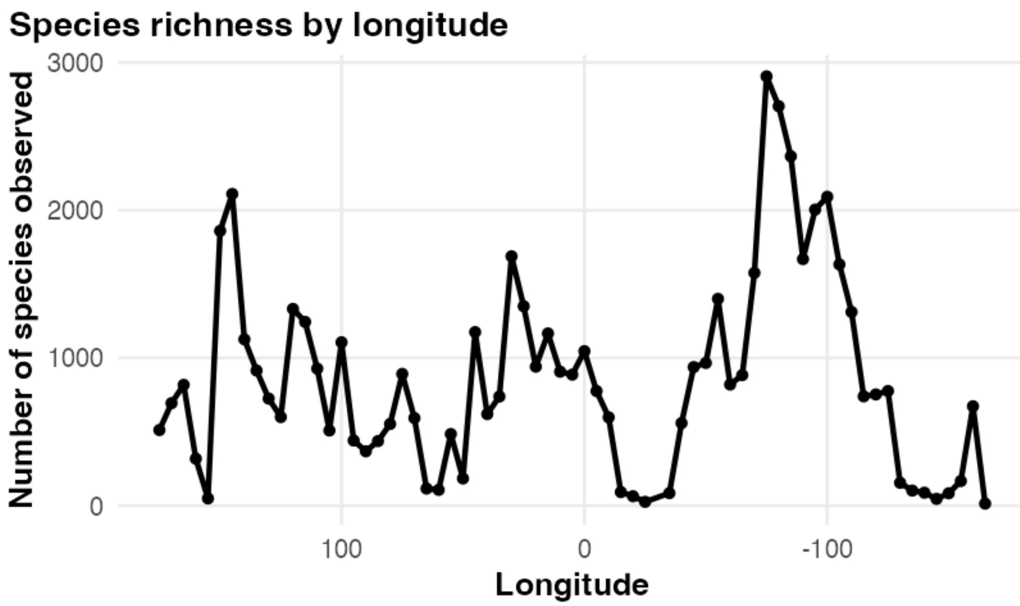
How does latitude influence the different species observed (latitude and species richness)?



Species richness is the number of distinct species recorded in a particular area. Here, recorded richness is higher across many Northern Hemisphere latitude bands and lower near the equator and poles; because observation effort is uneven, this is not a direct estimate of biological richness.

We see here that the maximum species richness is very close in the two hemispheres, but that the distribution is broader in the Northern Hemisphere than in the Southern Hemisphere (partly because trees grow at higher latitudes in the Northern Hemisphere than in the Southern Hemisphere). Similar environmental conditions can be found in the two hemispheres but they are not spread following the same pattern.

How does longitude influence the different species observed (longitude and species richness)?



Areas with high recorded species richness correspond to continental areas, but little more can be concluded from longitude alone.

What is the local recorded species richness across the world?

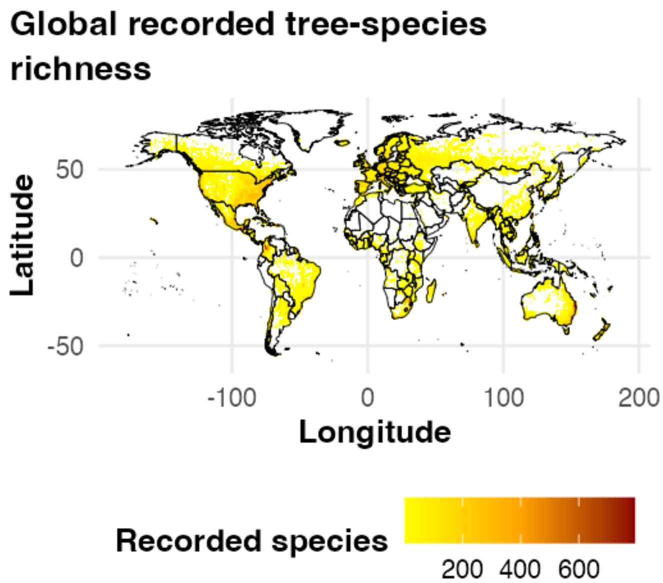
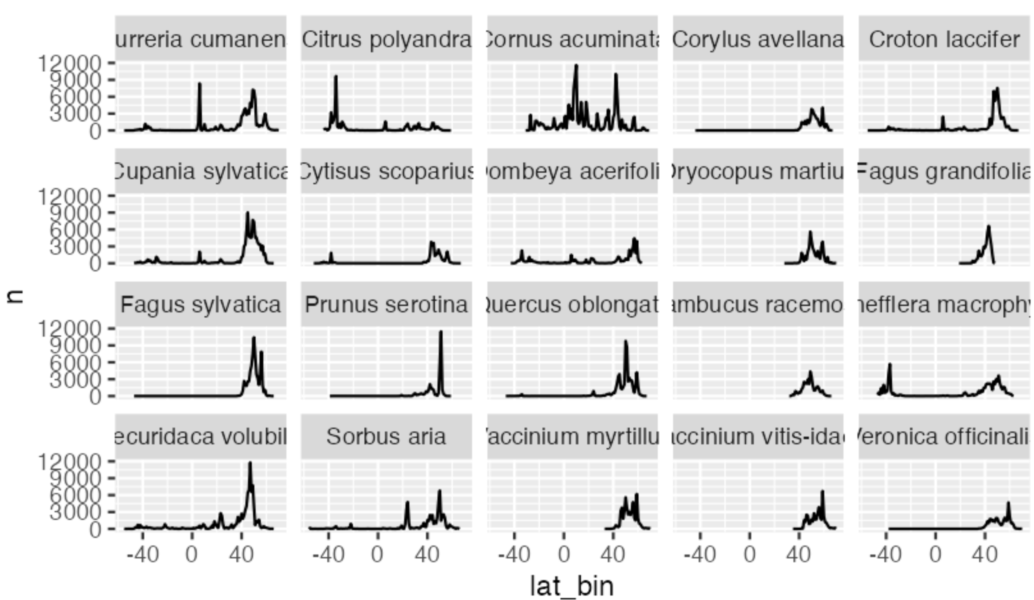


Figure 3: Recorded species richness per 1° grid cell. Counts use all observations; the map shows recorded richness rather than estimated biological richness.

Within sampled areas, many grid cells show similar recorded richness, while the species involved vary geographically. Observation effort also varies strongly between cells, so the map should not be read as showing homogeneous underlying biodiversity.

How are the different tree species distributed depending on the latitude? Here, we only check for the most represented species.



With this figure, we see that the most represented species in this dataset have different world distributions. Most are mainly present in the Northern Hemisphere, but some are also present in the Southern Hemisphere (such as *Dombeya acerifolia*), and some are almost evenly present in the two hemispheres, such as *Cornus acuminata*.

Another representation could be a heatmap representing the species depending on the latitude:

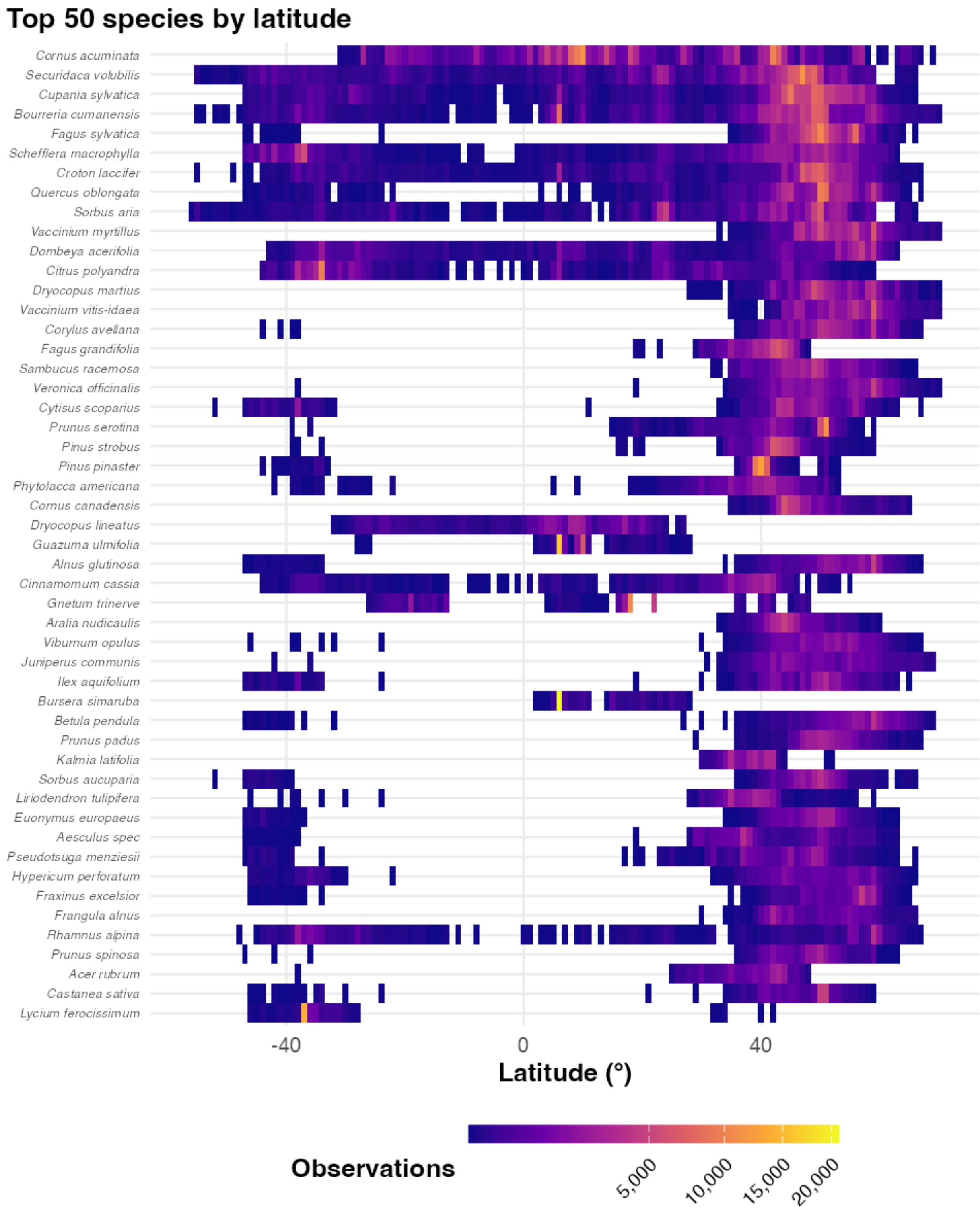
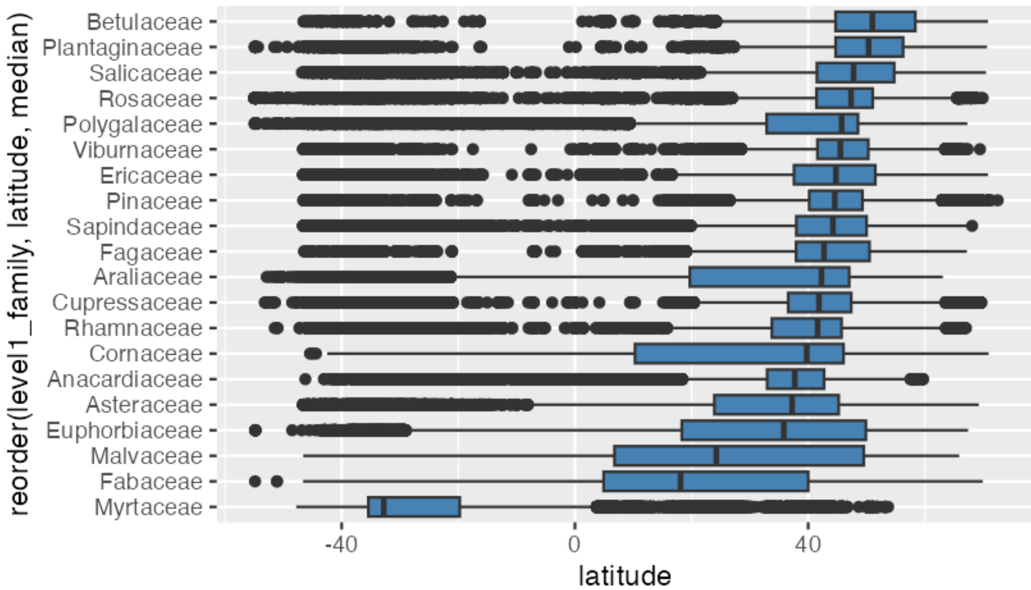


Figure 4: Observation counts by latitude for the 50 most recorded species.

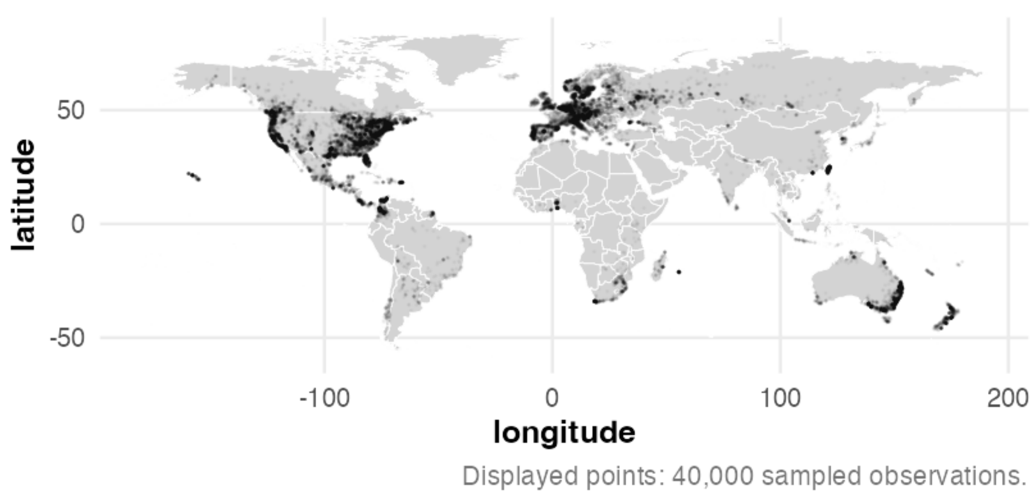
Figure 4 shows again that most species are present mostly in the Northern Hemisphere, but some can also be present in both hemispheres, or only in the Southern Hemisphere or near the equator

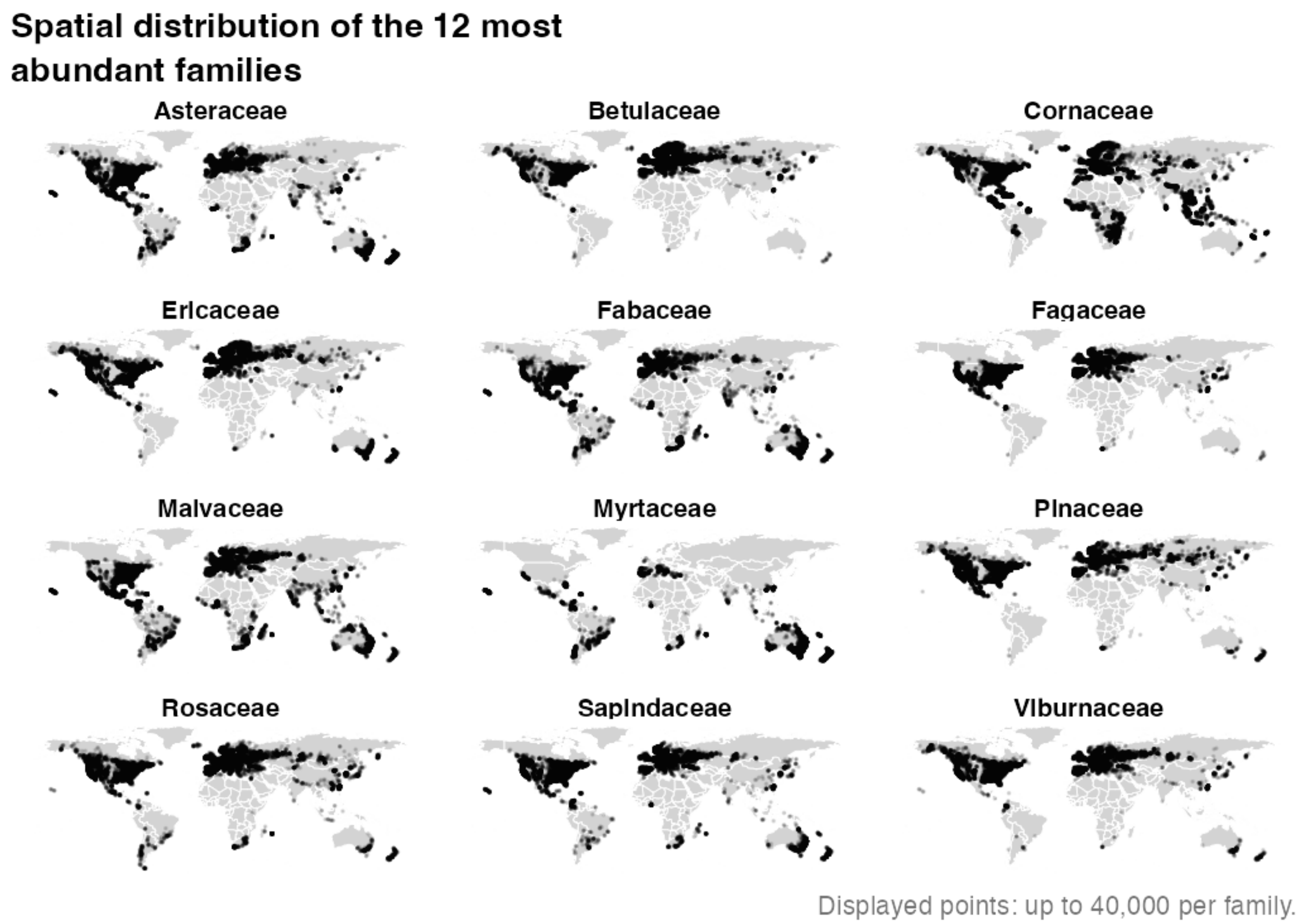
(*Guazuma ulmifolia*, a tree largely present in Latin America and very rare in other areas).

The boxplot below compares the latitudinal spread of the twenty most recorded families, ordered by median latitude, so that families centred on temperate bands can be separated from those reaching towards the equator.



The same families are shown geographically below, which reveals the longitudinal spread that a latitude-only view cannot. Both maps below use the shared worldwide point-map subsample (see Methodology for subsampling details).





On this visualisation, we can see the distribution of the different taxonomic families across the world:

- Some families are very widely spread across the world, such as the Asteraceae. This family contains a very large number of different species that live in temperate areas and can survive a range of non-extreme conditions.
- Other families are observed in more restricted areas, such as the Ericaceae and Pinaceae. These families are well adapted to cold and inhospitable places such as the tundra and boreal forests, or to places at high altitude.

7 Worldwide Distribution of the Data

How are observations distributed globally, and what does this reveal about sampling bias and taxonomic diversity? This section addresses four guiding questions:

1. Where are observations geographically concentrated, and does the coverage reflect ecological reality or observer activity?
2. How does the distribution of foliage types vary across latitudes and continents?
3. How does taxonomic richness compare across continents once differences in sampling effort are accounted for?

sample_id	country_code	level0	level1_family	level2_genus	level3_species	source	year	continent
1	MW	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	iNaturalist Research-grade Observations	2024	Africa
2	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	iNaturalist Research-grade Observations	2018	Africa
3	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	iNaturalist Research-grade Observations	2022	Africa
4	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	iNaturalist Research-grade Observations	2015	Africa
6	KE	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	naturgucker	2023	Africa
9	NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Flora of Coastal Environment in Cross-River State, Nigeria	2019	Africa

Table 4: An overview of the first few rows of the dataset

4. Has the spatial coverage of the dataset expanded or merely intensified over the years?

This distribution shapes how the rest of the dataset should be interpreted: an over-representation of certain regions may reflect sampling bias — the activity of contributors in the field — rather than the true state of global tree biodiversity.

7.1 Pre-Processing

Observations are grouped by continent rather than by country, so that comparisons are not fragmented across more than 210 individual countries. The continent for each observation was derived by spatially joining the point coordinates (longitude, latitude) with a world countries shapefile (rnaturalearth). This relies on administrative boundaries, which do not always align with the geography relevant to this study, so two adjustments were necessary:

- Russia is administratively classified as a single entity, but spans two geographical continents. Observations were reassigned to Europe or Asia based on the conventional Ural Mountains boundary (60°E longitude).
- Réunion, despite being a French overseas territory, is geographically located in the Indian Ocean off the African coast, and was therefore reassigned to Africa rather than being grouped with other European territories.

Observations for which no continent could be determined, as well as those located over open ocean (e.g., isolated island territories not covered by the country shapefile), were excluded from the dataset.

The resulting dataset, with the added continent column, is previewed in Table 4.

To ensure visual consistency throughout this section, each continent is assigned a fixed colour, shown in Figure 5. This colour scheme is derived from the Okabe-Ito palette, chosen for its colour-blind-friendly properties, and is applied consistently across all subsequent maps and charts to facilitate comparison between figures.

Figure 5: Colour scheme used to represent the continents in the following figures

7.2 Global overview

Figure 6 projects a reproducible subsample of observations on a world map, and Figure 7 quantifies the total per continent. Figure 8 then resolves density *within* continents, separating a continent sampled evenly across its territory from one where the same volume sits in a few localised hotspots.

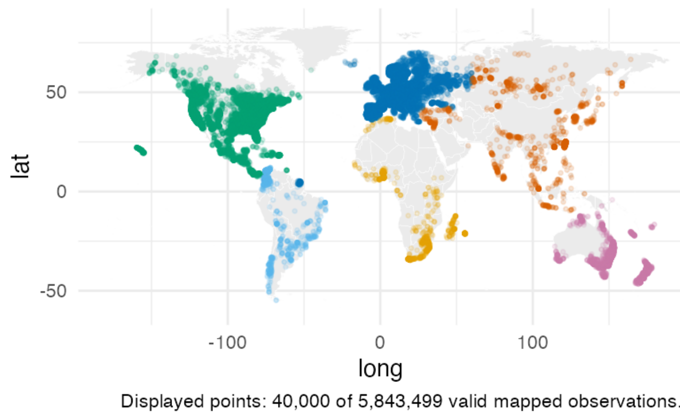


Figure 6: Geographical projection of observations by continent

Figure 6 shows a strongly uneven geographical coverage of observations. North America (green) and Europe (blue) display a dense, near-continuous coverage, closely following major population centres and road networks (a pattern typical of citizen-science platforms like iNaturalist, where sampling effort is driven by observer density rather than tree distribution alone). Asia (orange) covers a considerable geographical extent, but coverage is markedly uneven: it is dense in East Asia, India, and along the Trans-Siberian corridor, yet visibly sparse across Central Asia and Siberia's interior. Africa (yellow) shows a similar contrast, with observations concentrated in South Africa, East Africa, and coastal West Africa, while the Congo Basin and the Sahara remain largely unsampled. South America (light blue) is dominated by coastal Brazil and the Andean corridor, with the Amazon interior almost entirely absent from the dataset despite hosting one of the highest levels of tree biodiversity on Earth. Oceania (pink) is essentially limited to Australia's populated coastline and a scattering of Pacific islands. Overall, this map already suggests that observation density mirrors human accessibility and citizen-science activity more closely than it reflects actual tree biodiversity.

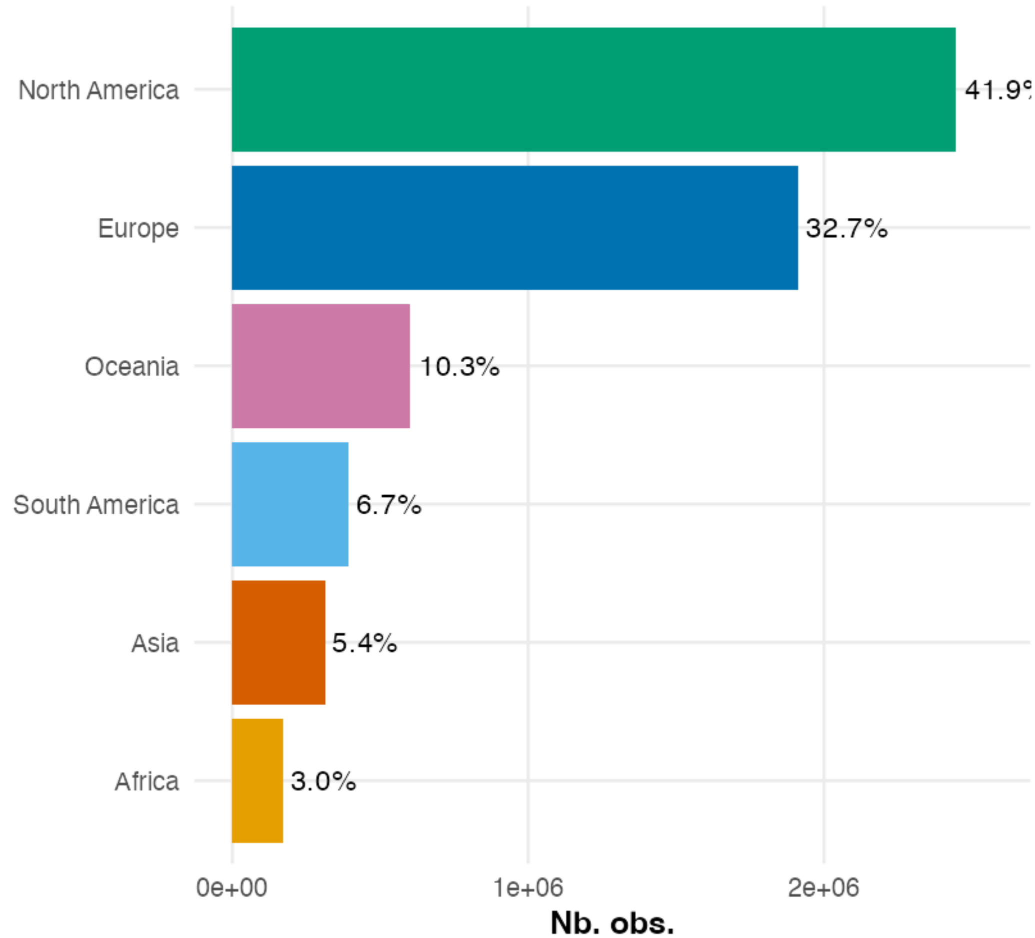


Figure 7: Total number of observations by continent

Figure 7 quantifies the imbalance already suggested by Figure 6. North America and Europe together account for 74.6% of all observations (41.9% and 32.7% respectively), despite representing only a fraction of the world's land area and tree biodiversity. Oceania follows at 10.3%, largely driven by Australia, before a sharp drop to South America (6.7%), Asia (5.4%) and Africa (3.0%). This ranking reveals a nuance compared to the previous map: although Asia appeared visually extensive in Figure 6, spanning a wide longitudinal range with a dense cluster of points, it accounts for only 5.4% of total observations, far below Europe or North America. Points can appear numerous and spread out on a map while still representing a small share of a dataset containing several million observations.

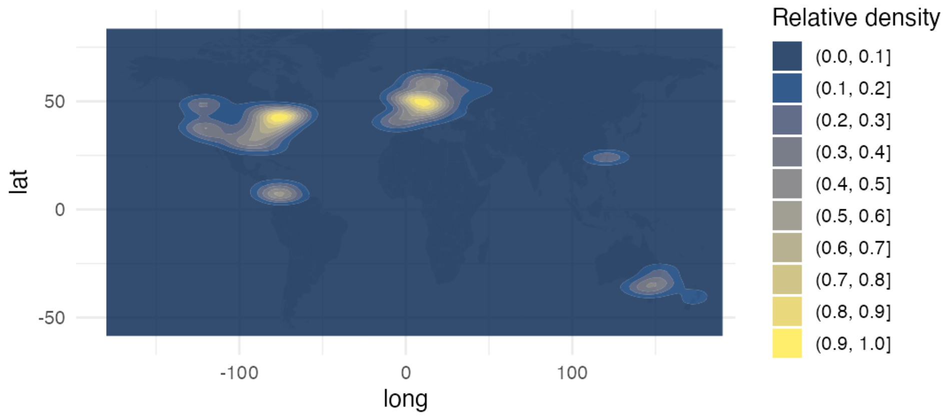


Figure 8: Geographical density of observations

Note: the density estimate is based on a reproducible random subsample of 50,000 observations (drawn with the report-level map seed from the full dataset), rather than all 5.84 million records. Passing the full dataset to `geom_density_2d_filled()` *caused a crash in ggplot2 3.5.2 due to memory limitations in the internal stat pipeline. At this sample size, kernel density estimation produces a visually identical result, as the smoothing operation is stable well below this threshold.*

Figure 8 reveals that observations are not only concentrated on certain continents, but highly clustered within a small number of localised hotspots. The most intense concentration is found in Western Europe (particularly around France, Germany and the United Kingdom), followed closely by the northeastern United States. Secondary hotspots appear along the US West Coast, in Central America, in southeastern Australia, and around Taiwan / southern Japan, while the vast majority of the remaining land surface, including most of Africa, Asia and South America, shows minimal to no density. This reinforces the hypothesis raised earlier: sampling effort appears strongly correlated with population density and citizen-science activity rather than with the underlying distribution of tree biodiversity. Even within high-observation continents such as North America (Figure 7), density is far from uniform — observations concentrate along the East and West Coasts, while large inland areas remain comparatively under-sampled.

7.3 Spatial distribution of foliage types

Each tree in the dataset is classified into one of four foliage types (Evergreen Broadleaf, Evergreen Needleleaf, Deciduous Broadleaf, or Deciduous Needleleaf). This classification is closely tied to climate, since foliage strategy reflects how trees adapt to seasonal variation in temperature and water availability. The question is whether these types are distributed uniformly across the sampled regions or follow the climatic gradients expected from ecological theory, and whether that pattern is still visible despite the sampling bias identified above.

Figure 9 maps a subsample of observations for each foliage type and Figure 10 compares foliage composition across continents. Figure 11 then drops continental boundaries altogether for a continuous latitudinal gradient, since climate — and therefore foliage strategy — is structured by latitude rather than by administrative divisions.

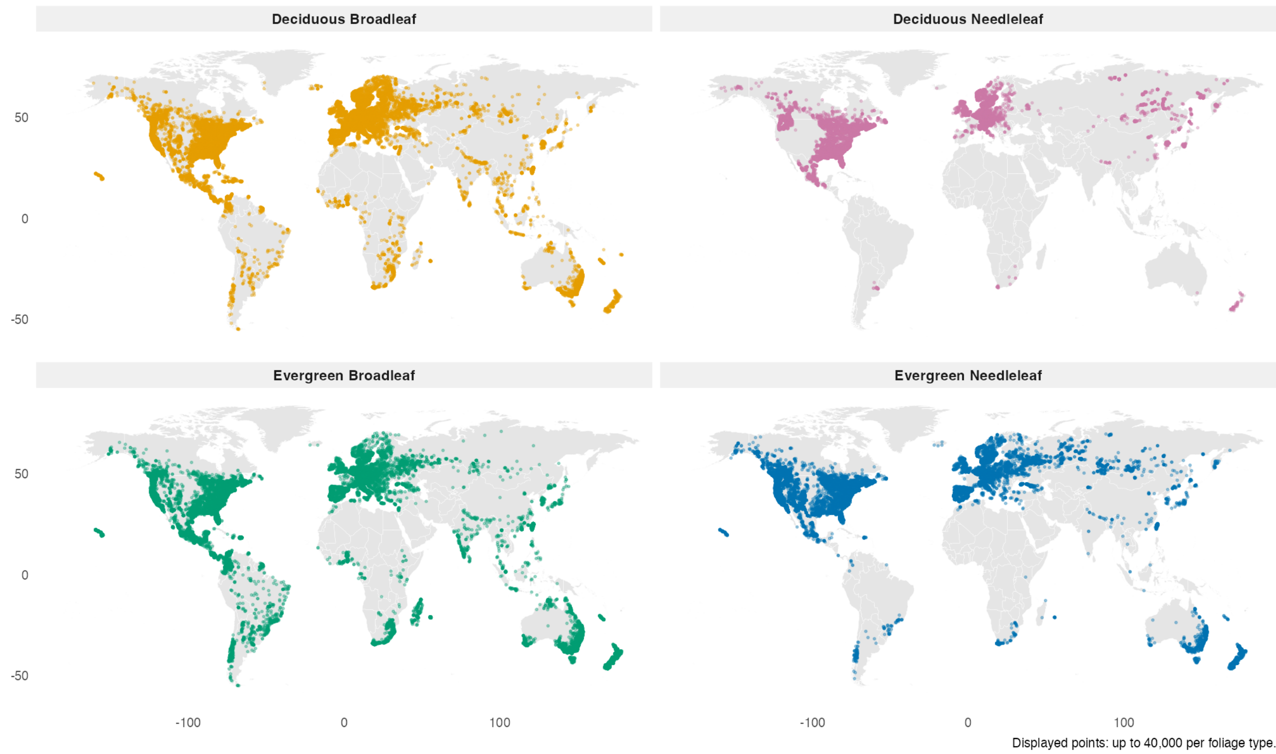


Figure 9: Geographical distribution by type of foliage

Figure 9 reveals that Deciduous Broadleaf (orange) and Evergreen Broadleaf (green) share a broadly similar geographical footprint, both present across nearly all sampled continents and spanning a wide range of latitudes, from temperate to tropical regions. This similarity suggests that, at the continental scale visualised here, geographical presence alone does not clearly separate these two categories; a finer, latitude-based view is needed to detect any underlying climatic gradient between them. Evergreen Needleleaf (blue) is most heavily concentrated across the boreal and temperate zones of the Northern Hemisphere (North America, Europe, and northern Asia) but is also present, at lower density, in the southern parts of South America, southern Africa, and Oceania, suggesting an association with temperate conditions more broadly rather than strictly cold northern climates. Deciduous Needleleaf (pink) remains overwhelmingly concentrated at high latitudes, primarily across Siberia and parts of Canada, with only a lower-density secondary presence in the United States and Europe, confirming that, unlike Evergreen Needleleaf, this foliage type stays largely confined to high-latitude regions overall.

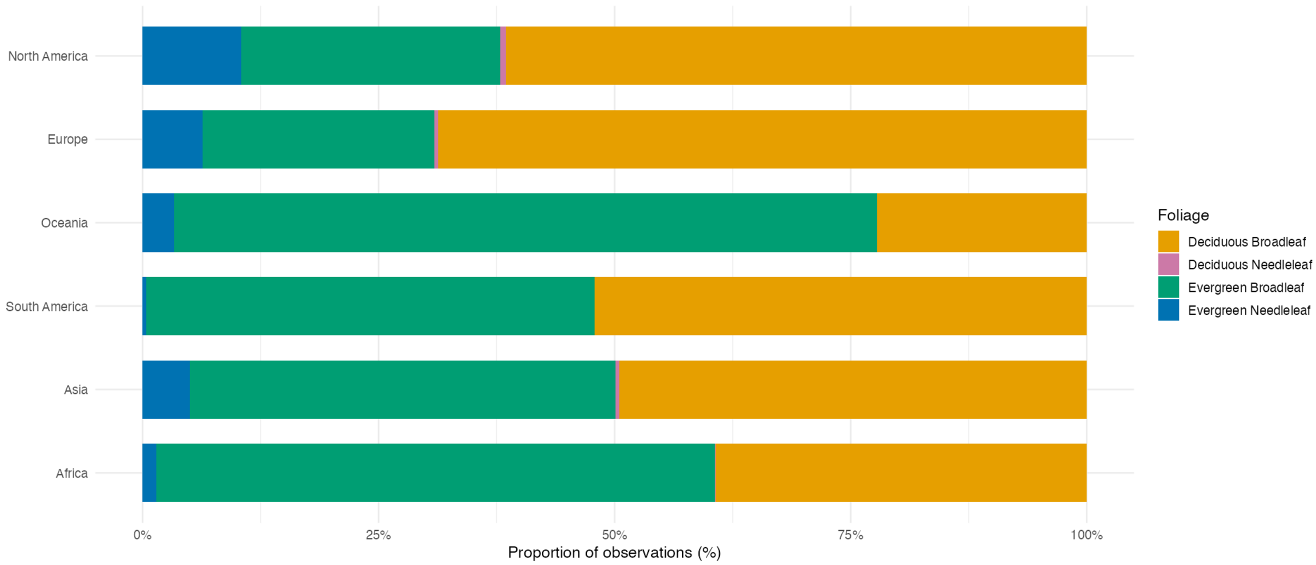


Figure 10: Geographical distribution by type of foliage

Figure 10 quantifies the relative composition of foliage types for each continent, independent of its overall sampling volume, and a clear contrast emerges between temperate and tropical/subtropical continents. North America and Europe share a similar profile, dominated by Deciduous Broadleaf (around 60–70% of observations), followed by Evergreen Broadleaf (roughly 25–30%), with Evergreen Needleleaf making up a smaller but still notable share (around 10%) not found at comparable levels on any other continent, consistent with the boreal/temperate conifer forests characteristic of these regions. Oceania and Africa, by contrast, are dominated by Evergreen Broadleaf, which accounts for the majority of observations in both (over 75% in Oceania, around 60% in Africa), with Deciduous Broadleaf as a smaller secondary category. South America and Asia present an intermediate profile, with Evergreen Broadleaf and Deciduous Broadleaf nearly evenly split at close to 50% each, suggesting these continents span a broader mix of climatic zones rather than being dominated by a single climate regime. Across all four, Evergreen Needleleaf drops to a marginal share compared to North America and Europe, and Deciduous Needleleaf is nearly absent except for a very thin presence in North America, Europe, and Asia, consistent with the restricted geographical range already observed in Figure 9. This distinction by continent, however, remains an approximation, since large countries such as the United States or China span multiple climatic zones.

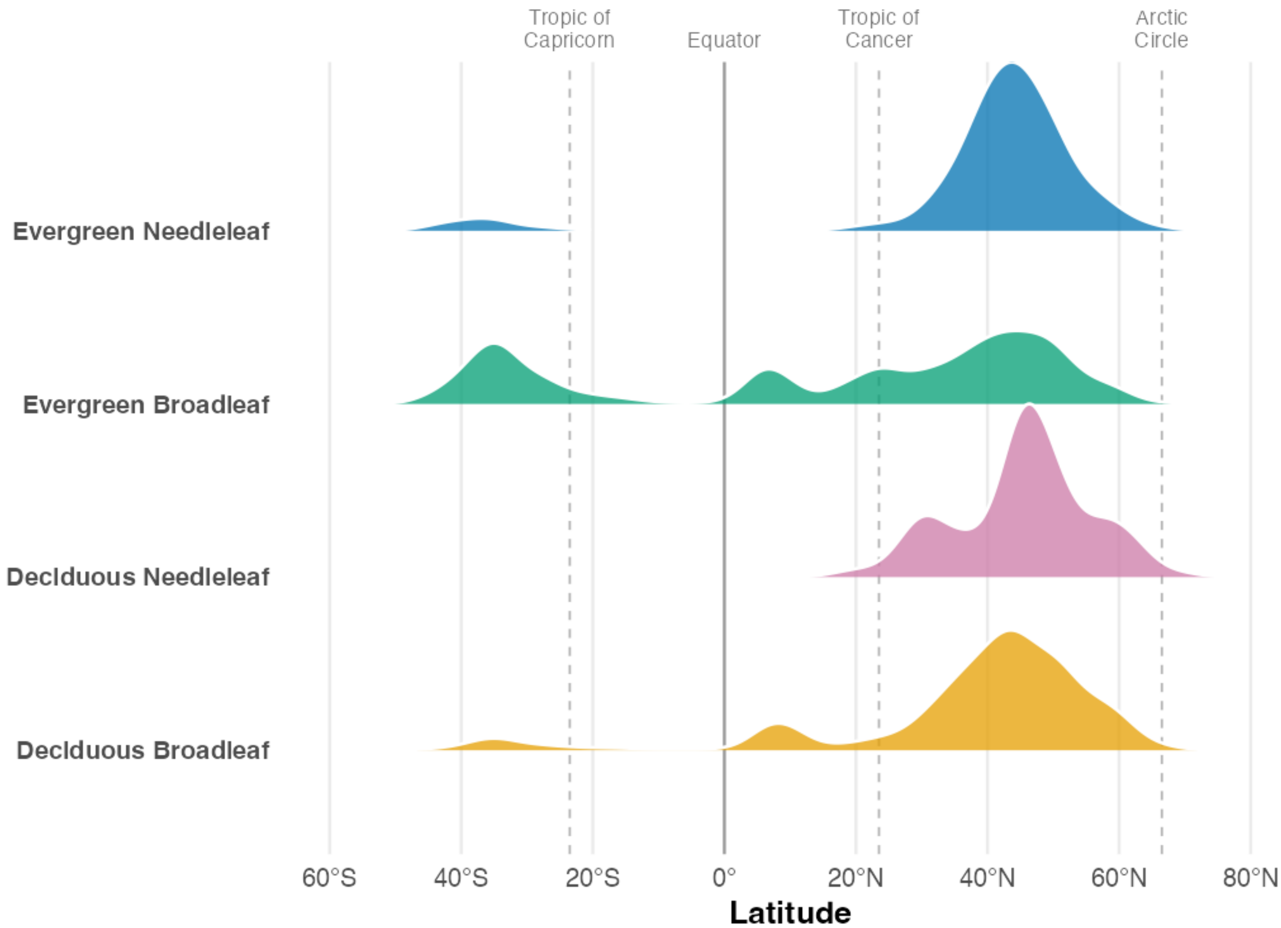


Figure 11: Distribution of foliage types by latitude

Figure 11 refines the patterns suggested by the previous map and bar chart. Evergreen Needleleaf (blue) shows a sharp peak around 45–50°N, with only a small secondary bump around 30–40°S. Deciduous Needleleaf (pink) is similarly concentrated at high northern latitudes, with a pronounced peak around 45–55°N and little recorded presence in the Southern Hemisphere. Deciduous Broadleaf (orange) has a large peak around 45–50°N, while Evergreen Broadleaf (green) is more widely distributed and has a more substantial tropical presence. Unlike the continent-level chart, the ridge plot exposes the continuous latitudinal shape of each category. These patterns are consistent with an association between foliage strategy and climate, but the plot alone cannot establish latitude as the primary driver because observation effort and other environmental variables are not controlled.

7.4 Taxonomic richness by continent

How much taxonomic diversity does each continent actually contain? Counting distinct species or genera per continent cannot answer this directly, because of the sampling imbalance identified above: a continent with more observations is mechanically more likely to have recorded more distinct taxa, even if its underlying biodiversity is not higher. Comparing raw taxonomic counts between continents therefore risks conflating genuine biodiversity differences with differences in sampling effort alone.

Table 5 and Figure 12 report the raw number of distinct taxa per continent, which at this stage remains directly influenced by each continent's sampling volume (Figure 7). Figure 13 addresses that limitation through rarefaction curves, which estimate the expected number of unique species as a function of sample size, allowing continents to be compared on an equal footing regardless of how many observations were collected for each.

continent	Foliage	Families	Genus	Species	Complete_taxa
Africa	4	168	972	3723	3723
Asia	4	179	951	3749	3749
Europe	4	165	661	2128	2128
North America	4	202	1218	6484	6484
Oceania	4	174	876	4833	4833
South America	4	185	1104	4977	4977

Table 5: Table of the taxonomic richness by continent

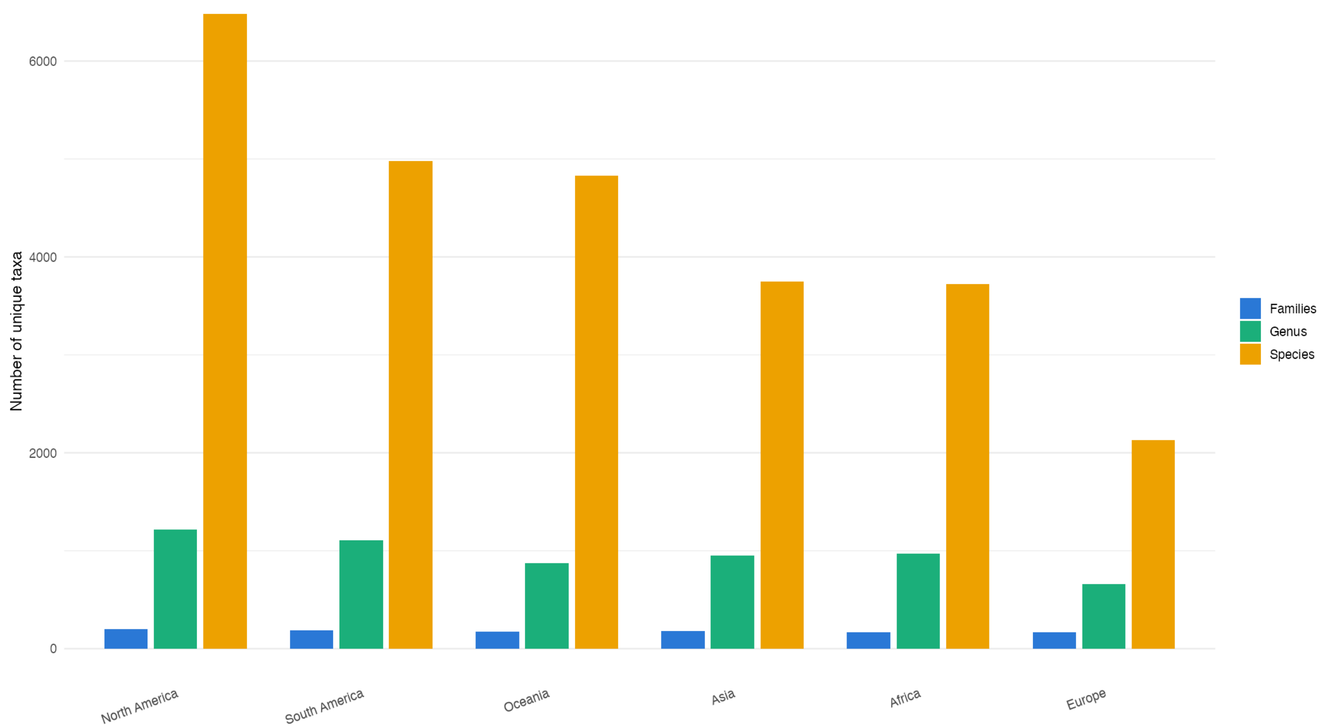


Figure 12: Taxonomic richness by continent (raw values)

Table 5 and Figure 12 report the raw number of distinct taxa recorded per continent, at three taxonomic levels: family, genus, and species. All six continents show a comparable number of families (165–202), suggesting that the dataset captures a broadly similar range of major taxonomic groups everywhere. This similarity breaks down markedly at finer taxonomic resolution: genus counts range from 661 (Europe) to 1,218 (North America), and species counts range even more widely, from 2,128 (Europe) to 6,484 (North America). North America stands out as the richest continent at every taxonomic level, while Europe consistently ranks lowest.

Comparing this ranking to the observation volume documented in Figure 7 reveals an interesting discrepancy. Europe, despite being the second most heavily sampled continent overall (32.7% of observations), records the lowest species richness of all six continents (even lower than Africa and Asia, which together account for 8.4% of total observations). Conversely, South America and Oceania, which together represent 17.0% of observations, show species counts (4,977 and 4,833 respectively) roughly 2.3 times that of Europe, despite Europe having been sampled roughly 3.8 times more intensively.

This pattern suggests that raw taxonomic richness, as shown here, is not simply proportional to sampling volume, but it does not yet establish whether these differences reflect genuine biodiversity patterns or are still partly shaped by uneven sampling effort within each continent.

Methodological note: The `vegan::rarefy()` function calculates the expected number of recorded species in a random sample of n observations, drawn without replacement from each continent's species-abundance counts. It evaluates this expectation analytically rather than performing repeated random simulations. Calculating the expectation across increasing sample sizes produces one rarefaction curve per continent. This standardises the number of observations being compared, but it does not correct for geographic coverage, source composition, detectability, or other differences in sampling design.

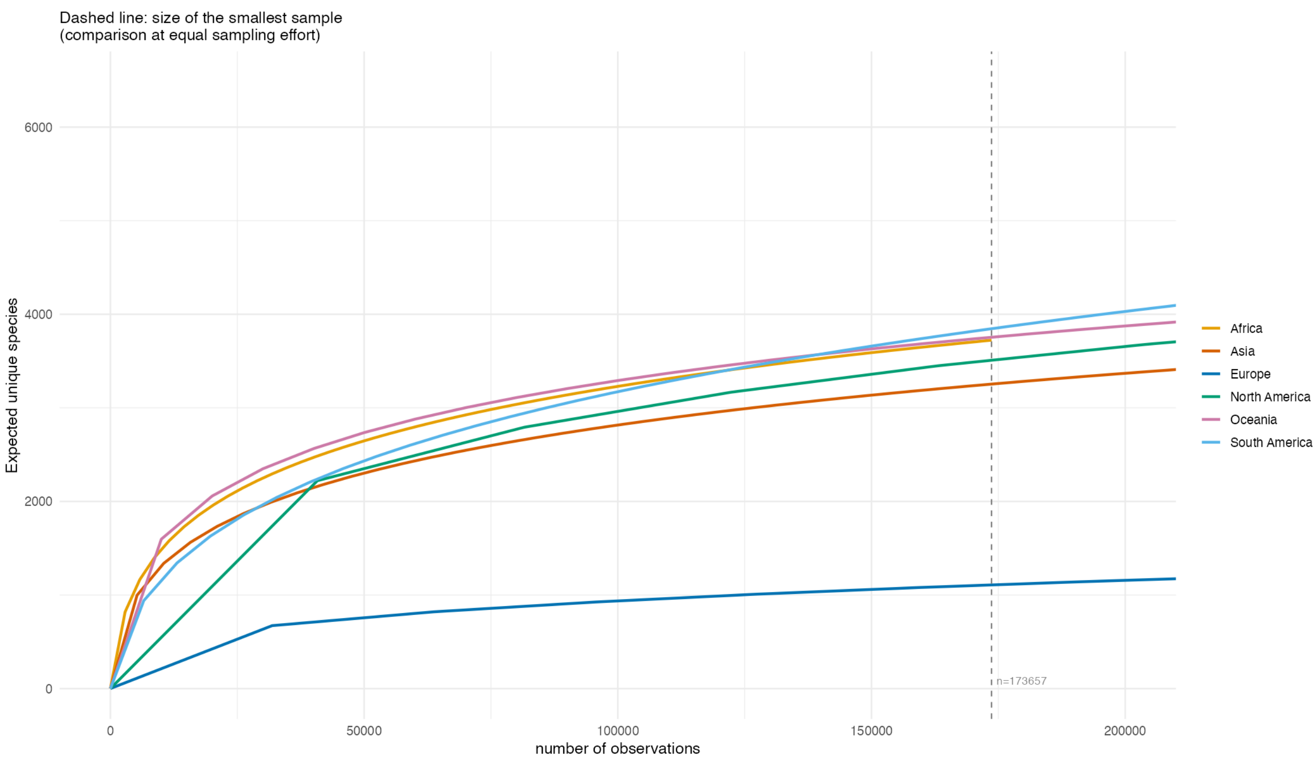


Figure 13: Rarefaction curves by continent

Figure 13 provides an equal-observation comparison that substantially revises the picture suggested by the raw counts in Figure 12. At the shared sampling effort marked by the dashed line (n=173,657, the size of Africa's full sample), the ranking of recorded richness changes considerably. Oceania and South America reach close to 3,800 expected recorded species, followed by Africa (around 3,700). North America and Asia form an intermediate group (roughly 3,500 and 3,300 respectively), while Europe has around 1,100 expected recorded species at this sample size.

The most striking change concerns North America. While it has the highest raw species count in Figure 12 (6,484 species), this advantage largely disappears once observation count is equalised. Its rarefaction curve is overtaken by Oceania, South America, and Africa before reaching the comparison point. This suggests that North America's raw lead is strongly influenced by its larger sample size (Figure 7), without establishing how the continents compare in underlying biological diversity.

Europe's rarefaction curve remains below the other continents across the range shown. This indicates lower recorded richness even at equal observation counts. The difference may be consistent with ecological patterns, but it may also reflect source composition, geographic concentration, detectability, or other sampling differences that rarefaction does not remove.

Finally, Africa, Oceania, and South America remain fairly closely clustered across the observed range, suggesting comparable recorded richness after standardising observation count. This adjustment addresses sample size rather than all forms of sampling bias.

7.5 Changes in the distribution over time

The previous subsections treated the data as a static snapshot, without considering the recorded year. Given the strong reliance on citizen-science platforms such as iNaturalist, the number and coverage of dated observations are unlikely to be constant across the period. The figures below do not show when records were ingested into GlobalGeoTree.

Figure 14 tracks observations dated to each year alongside the number of distinct countries represented, and Figure 15 breaks that trend down by continent. Figure 16 then maps sampled observations over successive two-year periods, showing whether the spatial imbalances seen above are a stable feature of the dataset or are narrowing as it grows.

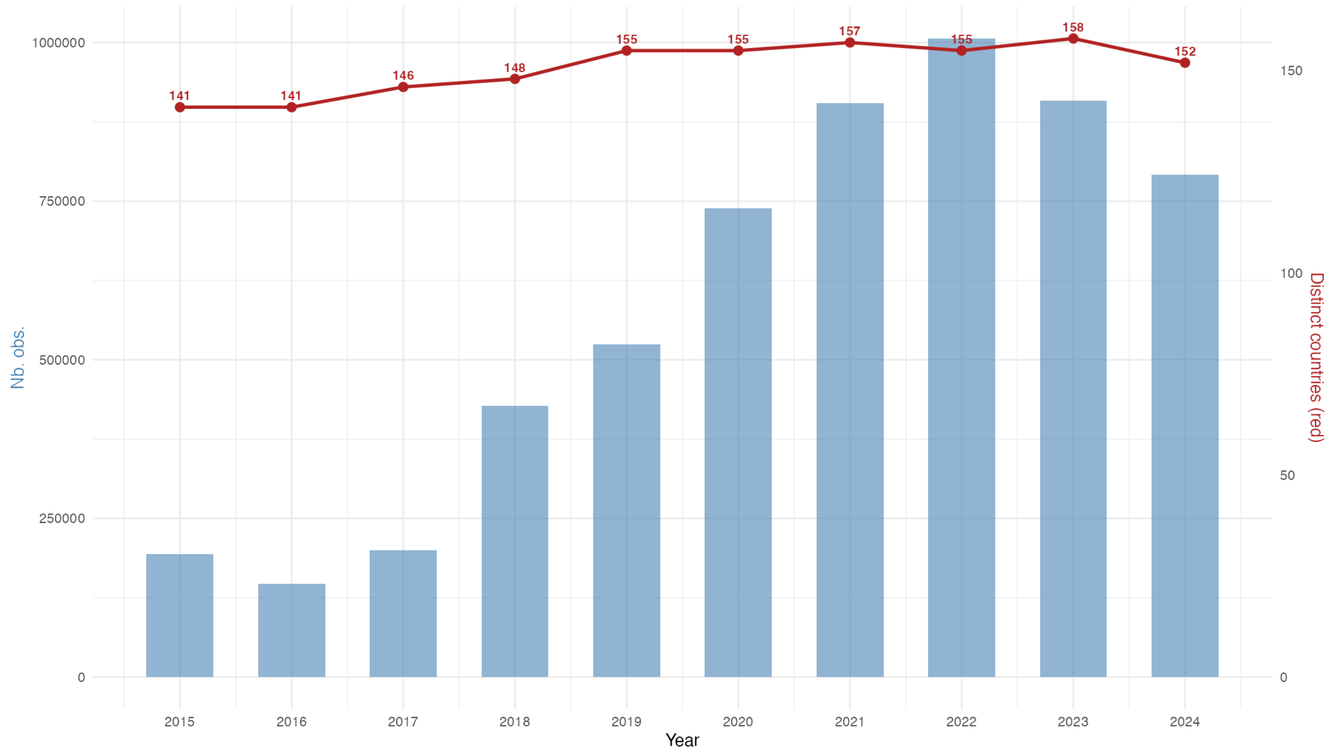


Figure 14: Observation volume and represented countries by observation year

Figure 14 reveals a marked increase in observations dated from 2018 onward. After remaining between roughly 150,000 and 430,000 observations per year from 2015 to 2018, the count rises sharply and peaks in 2022 at just over 1,000,000. This pattern is consistent with changes in citizen-science activity and source coverage, but observation year alone cannot establish when or why records were added to the combined dataset.

The number of distinct countries represented by dated observations (red line) also increases, from 141 in 2015 to over 157 by 2021, before stabilising around 152–158 through 2024. This broad representation should be interpreted carefully alongside the sharp difference in observation volume: one observation and hundreds of thousands of observations both count as country presence. The chart therefore shows breadth of representation by observation year, not equal sampling within or between countries.

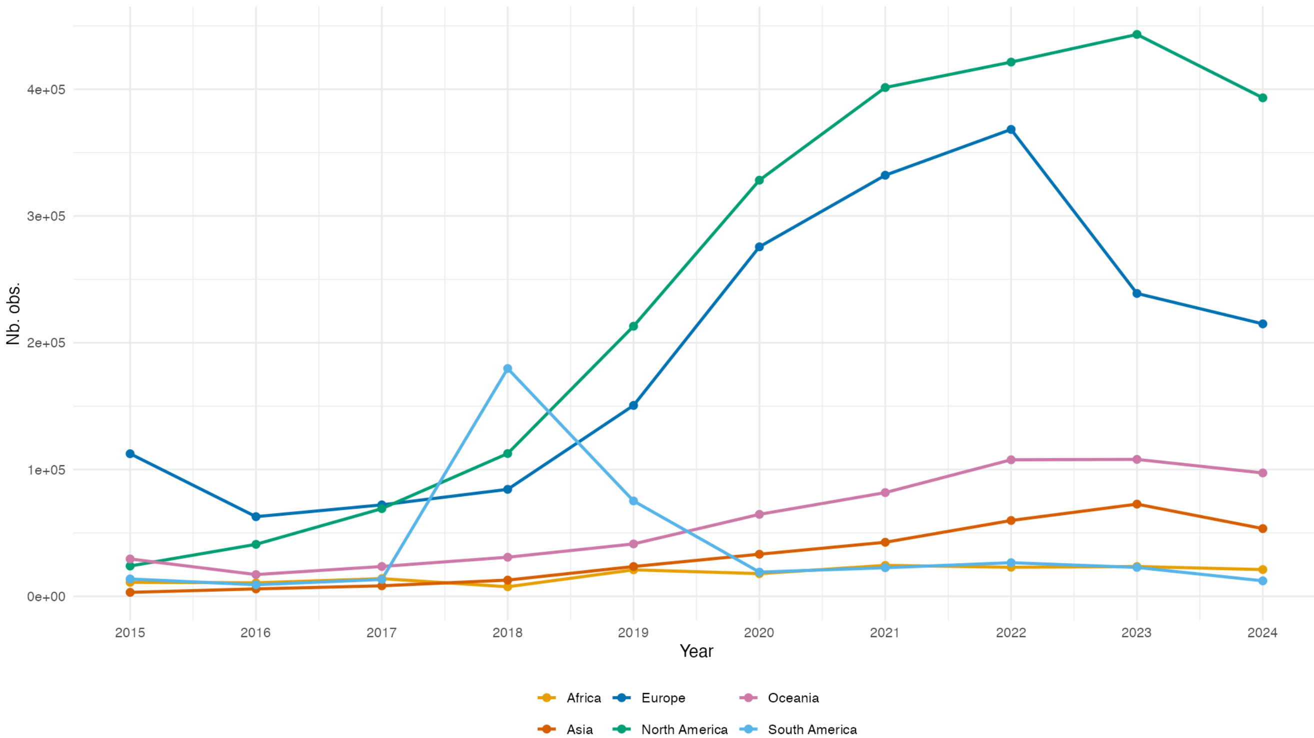


Figure 15: Observation volume by continent and observation year

Figure 15 refines the pattern in the previous figure. The increase in observations dated since 2018 is concentrated mainly in North America and Europe, while the other four continents remain comparatively flat. North America rises from roughly 24,000 observations dated 2015 to over 440,000 dated 2023, followed by a slight decline in 2024. Europe follows a similar trajectory until 2022 and then declines. Asia, Africa, and Oceania remain at lower levels, while South America has a distinct spike in 2018. These differences show that the imbalance between continents persists across observation years. They may reflect changes in platform activity, source coverage, or the temporal completeness of contributed records rather than changes in tree populations.

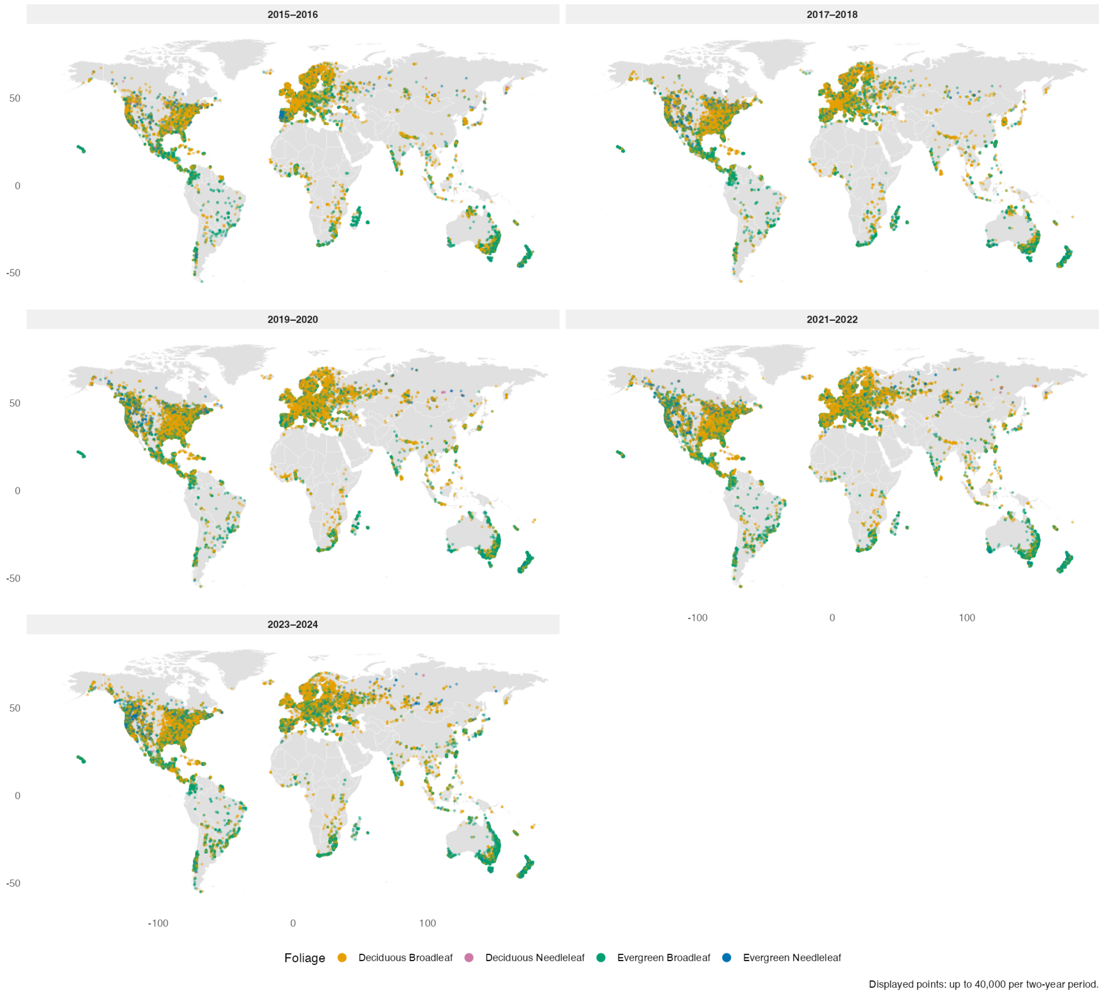


Figure 16: Geographical scope of data collection by period (5 two-year periods)

Figure 16 confirms the trends identified in the two previous figures spatially, while adding a further dimension: foliage type. Across all five period panels, the geographical footprint of the dataset remains remarkably stable — the same broad regions (North America, Europe, coastal South America, parts of Africa and Southeast Asia/Oceania) are already represented as early as 2015–2016, with no major new region appearing later. This is consistent with the near-total country coverage observed in Figure 14: geographical breadth was established early, and subsequent growth in volume (Figure 15) has taken the form of densification within already-sampled areas rather than expansion into new ones. Point coverage becomes progressively denser and more continuous over North America and Europe, particularly for Deciduous Broadleaf (orange) and Evergreen Broadleaf (green), while central Africa, the Amazon interior, and Central Asia remain visibly sparse throughout all five periods.

Overall, this final figure shows that the geographic imbalance is visible throughout the full range of observation years. This is consistent with a persistent structural bias in the dated records, while not describing when the contributing records were actually ingested into GlobalGeoTree.

Part II: Germany and Berlin Analysis

Part II examines tree observations within Germany.

Table 6: Germany data check

item	value
Total observations (all countries)	5,843,499
Germany observations used in Part II	203,728

8 Germany: Species and Common-Name Lookup

To make charts easier to read, English common names are shown next to Latin names wherever possible. The GlobalGeoTree dataset contains `species_key`, a unique GBIF identifier for each species. This key is used to query the GBIF vernacular-names API once and cache the result in `data/external/species_common_names.csv`. On every subsequent render, the report reads from that file.

Table 7: German species overview

metric	value
Unique species in German observations	464
Species with a valid GBIF species key	464

Table 8: GBIF common names lookup result

metric	value
Species queried from GBIF	464
Species with an English common name	421

Species without an English name in GBIF are shown with their Latin name only in charts.

9 Germany: National Patterns

This section examines how tree observations are distributed across Germany as a whole: spatial concentration, temporal trends, source structure, and federal state-level variation.

9.1 Pre-Processing

Polygons for the federal states (Bundesländer) are needed to draw state boundaries on Germany maps and to assign observations to states. The code first looks for a local GeoJSON file; if it is not found, the public GISCO NUTS1 file is downloaded.

Table 9: Federal state boundary check

item	value
Boundary status	Loaded federal state / NUTS1 boundaries from data/external/germany_nuts1_2021.geojson
Number of federal state polygons	16

9.2 Distribution of observations in Germany

- *Where are observations spatially concentrated across Germany?*

Two complementary views are shown below. Figure 17 is a smooth filled density surface, which highlights where observations concentrate into hotspots; Figure 18 plots the individual observation points the surface is derived from, revealing the raw recording locations behind those contours.

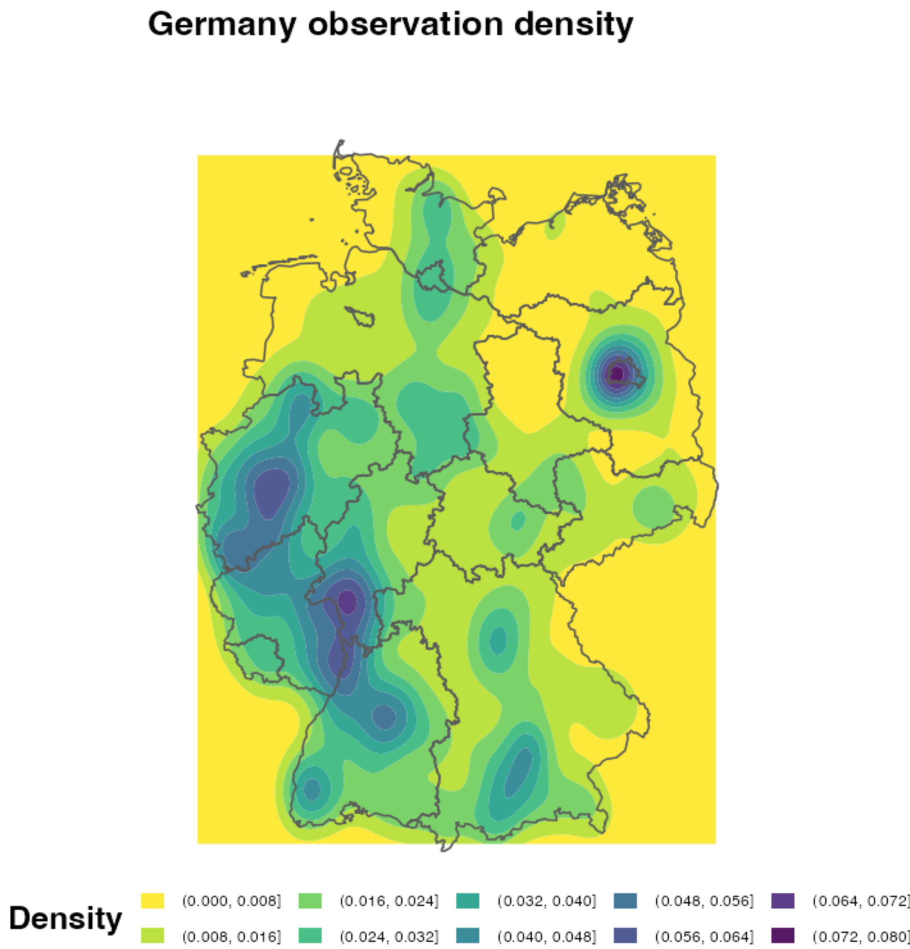


Figure 17: Smoothed density of Germany observations

Darker areas show where more observations were recorded. A location can appear dense because observers are active there, because a platform has strong coverage, or simply because access is easier.

Recording is highest in western and southwestern Germany, with a separate strong hotspot around Berlin. Large parts of eastern and northeastern Germany have much lower density.

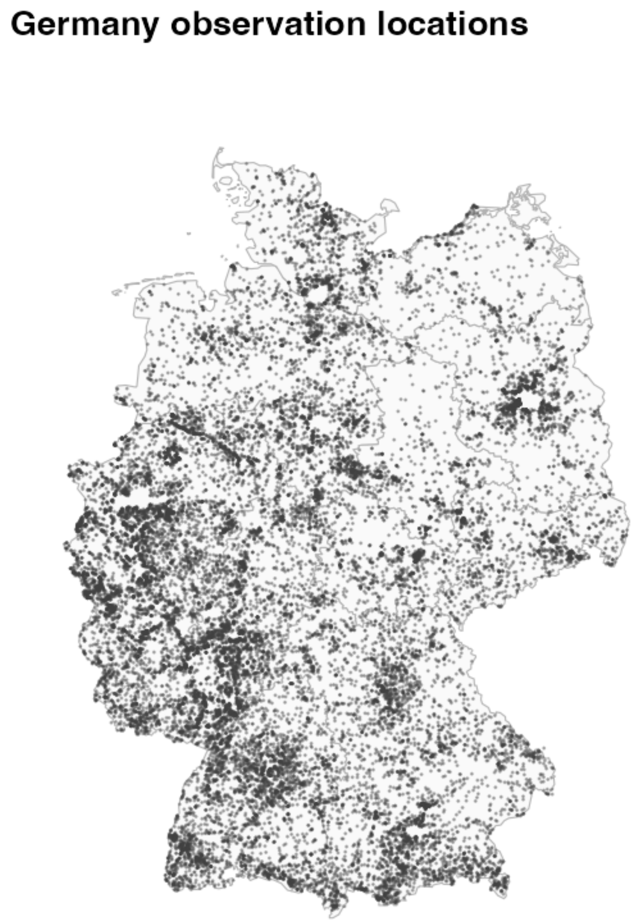


Figure 18: Locations of Germany observations

Figure 18 confirms the same pattern: observations cluster strongly in the west and south, while the east is more sparsely covered outside Berlin.

9.3 Observations by year

- *How has recording activity changed over time in Germany?*

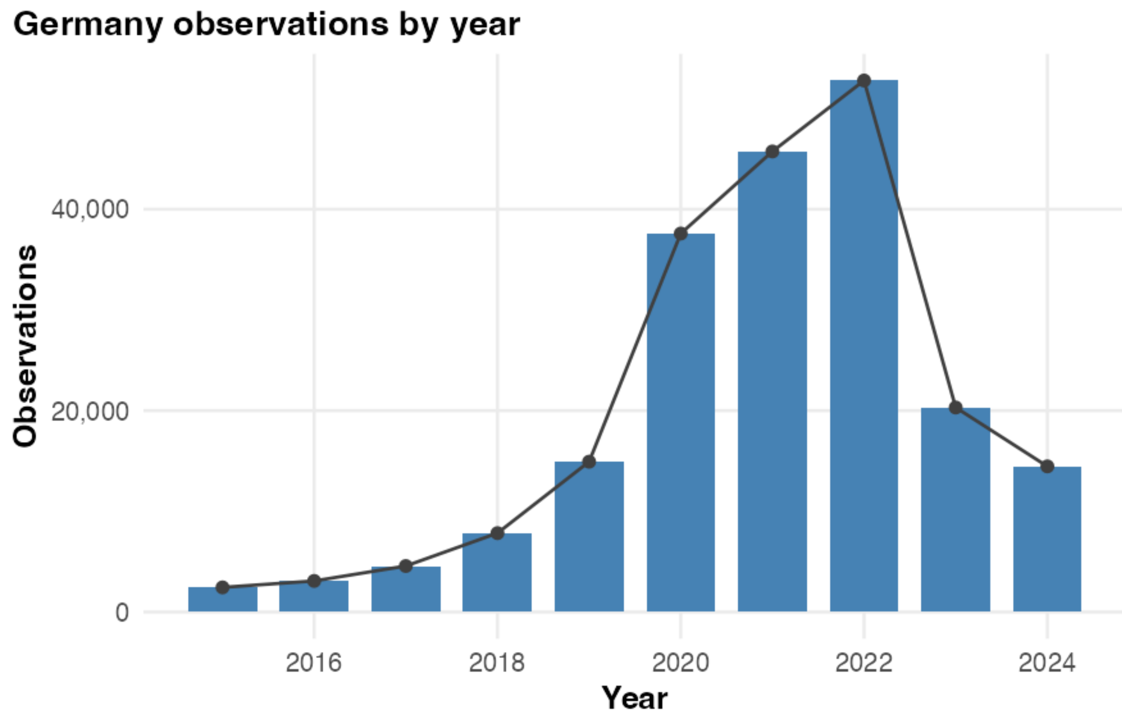


Figure 19: Germany observations by year

Recording grows steeply through the second half of the period, which is common in citizen-science datasets as more users join the platforms. Volume peaks in **2022** and then falls over the final two years.

That decline is most likely an artefact of how the dataset was assembled rather than a real collapse in German recording. Every record reaches GlobalGeoTree through GBIF, and there is a lag between an observation being made, being verified on a platform, and appearing in a GBIF snapshot — so the most recent years may still be filling up.

9.4 Source footprints across Germany

- *Which platforms contribute the most records in Germany, and do they cover the country differently?*

Figure 19 shows *when* Germany observations were recorded; this subsection examines *who* recorded them. Each panel of Figure 20 shows where one source collected its records.

Note that "source" here means an individual platform, not one of the six source groups defined in Part I. Germany is dominated by so few platforms that the individual names are more informative than the groups; the five largest are shown separately and everything else is pooled into "Other sources".

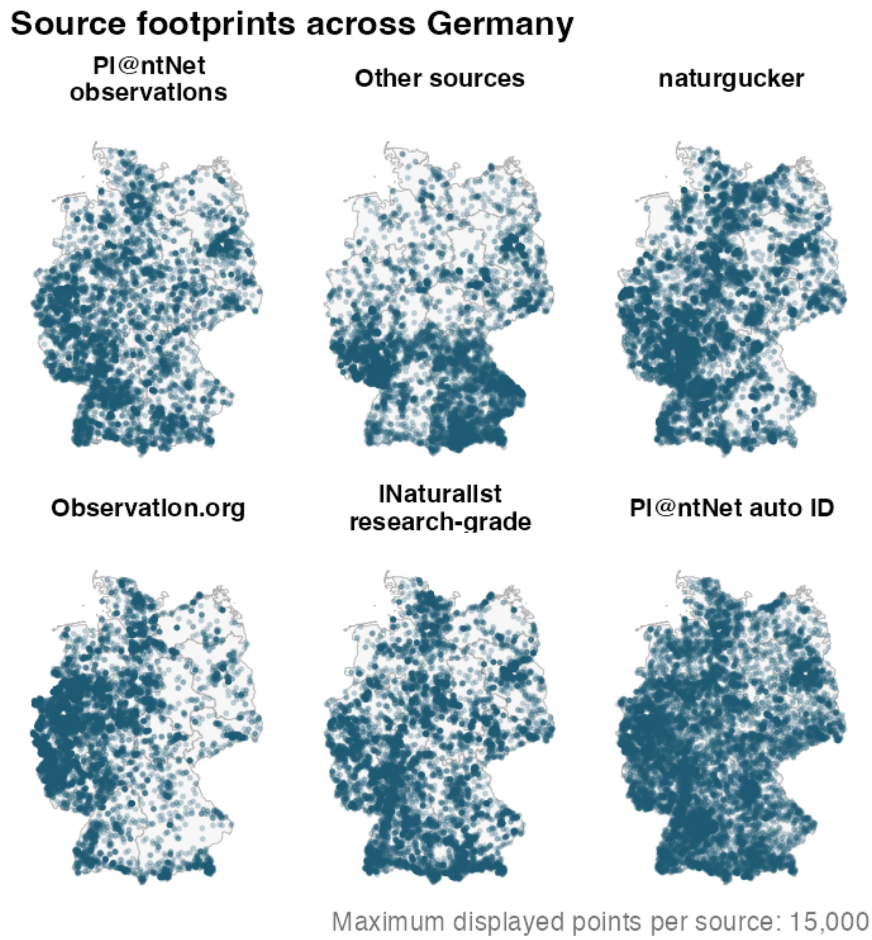


Figure 20: Source footprints in Germany

Figure 20 shows clear differences in where each source records observations. Observation.org and Pl@ntNet are more concentrated in western and southern Germany, while naturgucker and iNaturalist have broader national coverage. Pl@ntNet auto ID appears the most evenly spread, although the 15,000-point limit per panel means these show coverage patterns rather than total record numbers.

9.5 Observations by source

Figure 20 shows *where* each source records; Figure 21 shows *how many* records each source contributes.

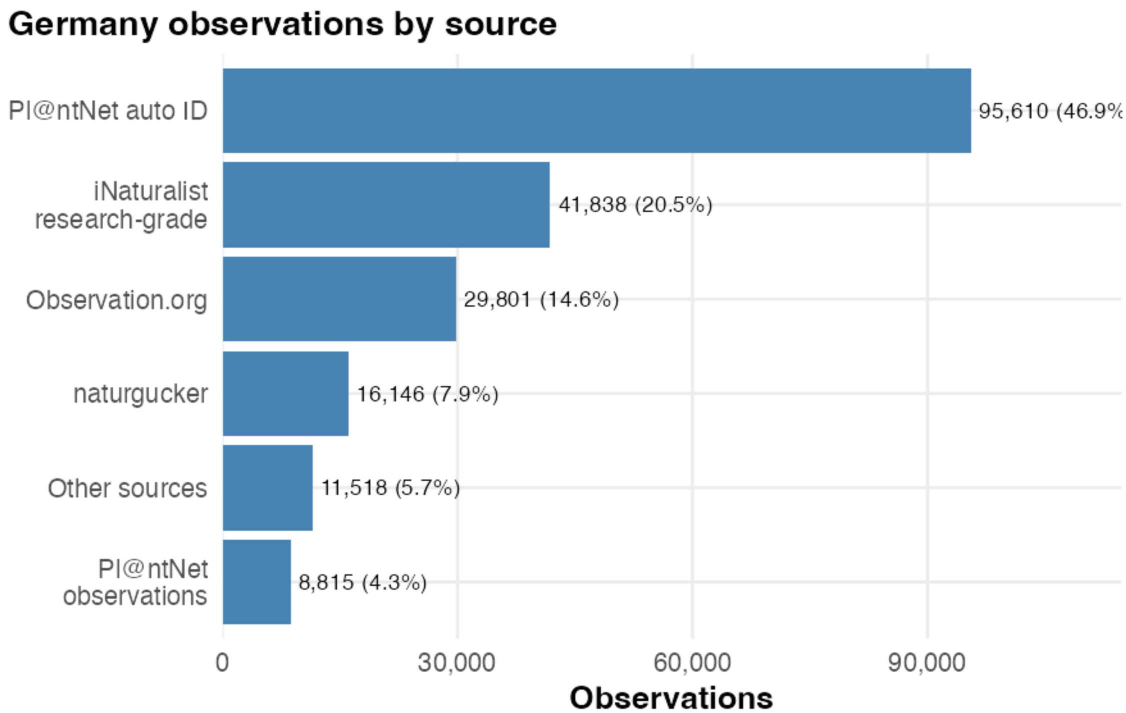


Figure 21: Germany observations by source

Pl@ntNet auto ID contributes 46.9% of all German records — almost half. iNaturalist research-grade and Observation.org add another 35.2%, so the three largest sources together account for 82.1% of the dataset. The remaining sources make much smaller contributions, meaning national patterns are shaped mainly by these three platforms.

This inverts the global picture. Part I showed that iNaturalist alone contributes 64.4% of all observations worldwide and is by some margin the largest single source. In Germany it is only the second largest, behind an automated identification pipeline.

`Pl@ntNet observations` and `Pl@ntNet auto ID` are both Pl@ntNet-derived but represent different validation routes: one is the regular shared-observation dataset, and the other contains automatically identified occurrences.

9.6 Observations mapped to federal states

- *How are observations distributed across Germany's sixteen federal states?*

Observations are assigned to federal states using a spatial join — each observation's coordinates are matched to the boundary polygon it falls within.

Table 10: Federal state spatial join check

metric	value
Germany observations before join	203,728
Matched to a federal state	203,234

metric	value
Federal states with observations	16

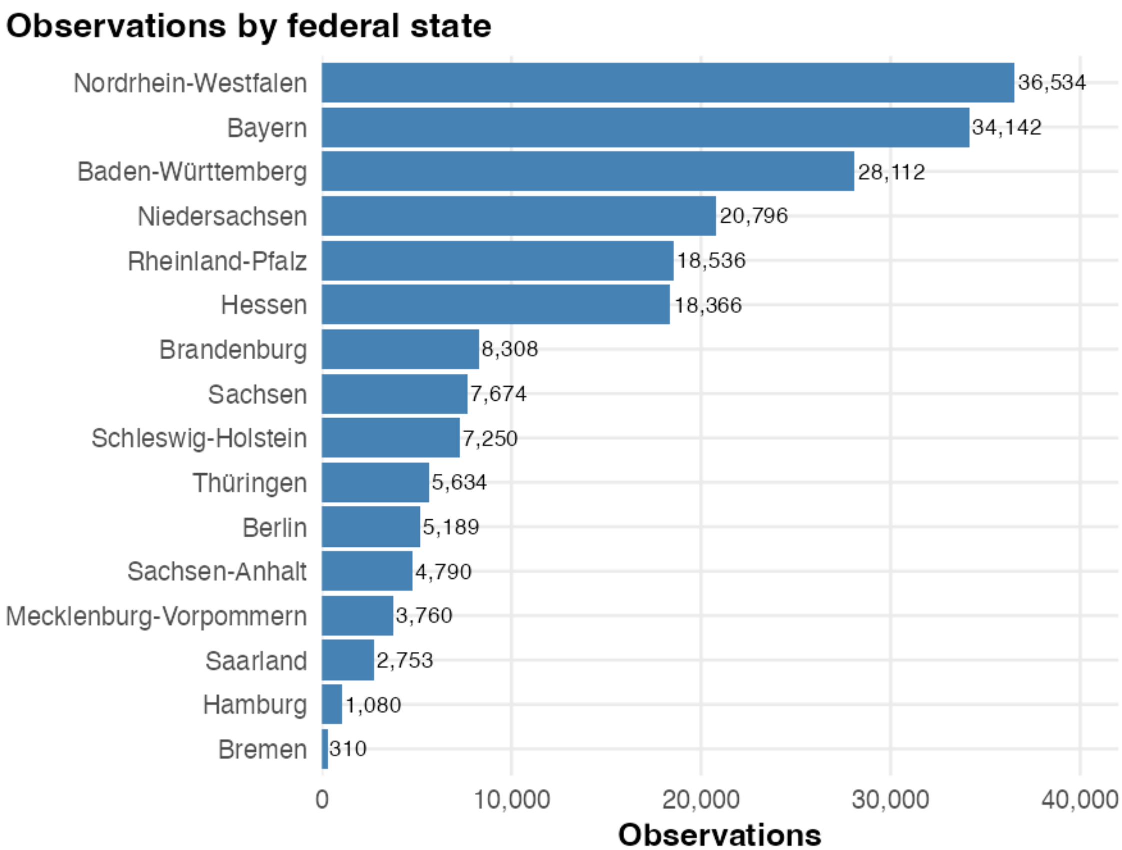


Figure 22: Germany observations by federal state

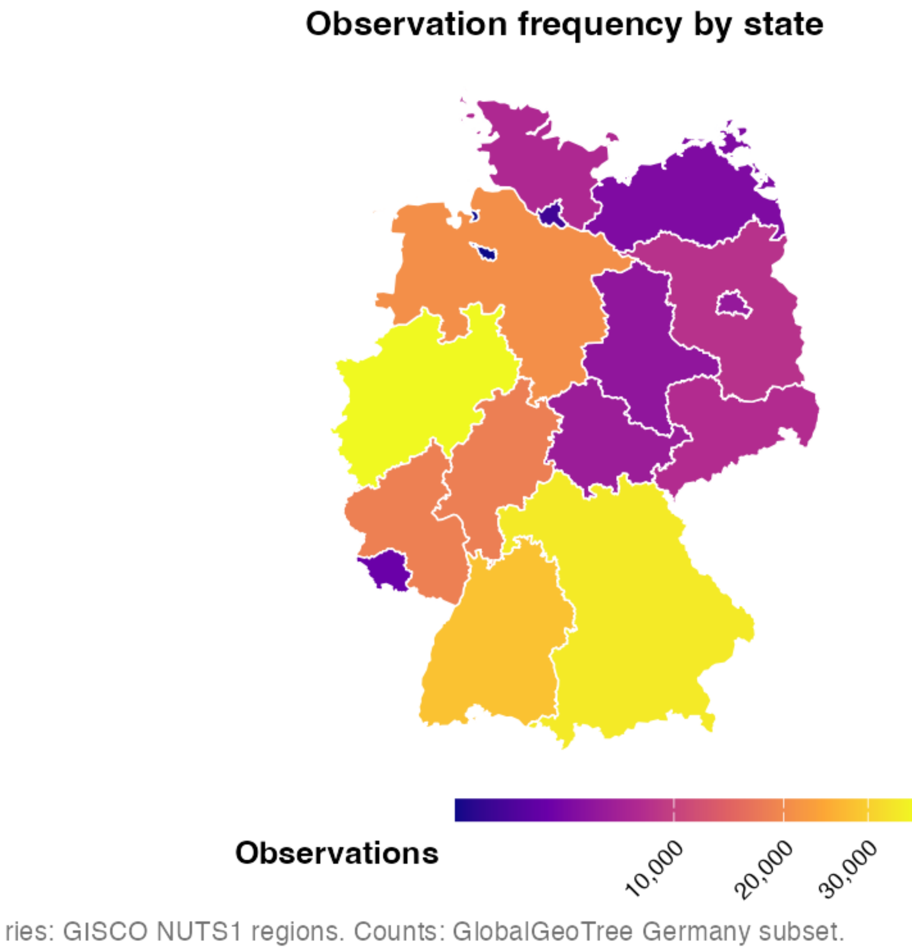


Figure 23: Observation frequency by federal state

Figure 22 provides exact rankings, while Figure 23 shows whether high-record states cluster in one part of Germany.

Observations are concentrated in western and southern Germany. North Rhine-Westphalia has the highest count, with Bavaria, Baden-Württemberg and Lower Saxony also recording many observations. Eastern states and Saarland generally have fewer records, showing that sampling effort is uneven across Germany.

That ranking is worth a second look, because it closely follows the states' populations rather than tree density: North Rhine-Westphalia, Bavaria, Baden-Württemberg and Lower Saxony are also Germany's four most populous states, in similar order. The counts here are raw totals, not normalised by area or population, so the chart cannot separate "more trees" from "more people who record trees" — and the population ordering suggests the second reading is the stronger one.

10 Berlin

Berlin is the city of our university, which is why it was selected as the main target for regional analysis in Germany. As a city-state, Berlin's entire area is urban, which may produce recording patterns quite different from larger, more rural federal states.

10.1 Pre-Processing

Berlin is handled separately because a finer boundary layer than the federal states is needed for city-level analysis. The boundaries come from an open-data project and provide polygons for Berlin's **local areas (Ortsteile)** — the small sub-units nested inside Berlin's twelve boroughs (Bezirke). "Local area" always refers to an Ortsteil in this part of the report.

Table 11: Berlin boundary check

item	value
Berlin boundary status	Loaded Berlin local-area boundaries from data/external/berlin_districts.geojson
Number of Berlin polygons	97

10.2 Berlin observations

Table 12: Berlin observation summary

metric	value
Berlin filtering method	Spatial join to Berlin district/local-area polygons
Berlin observations	5,192
Berlin recorded species	158
Berlin recorded genera	85
Berlin recorded families	43
Available source values	13
Available years	2015-2024

10.3 Berlin observation sources

- *Which data sources contribute to Berlin's observations, and does one platform dominate?*

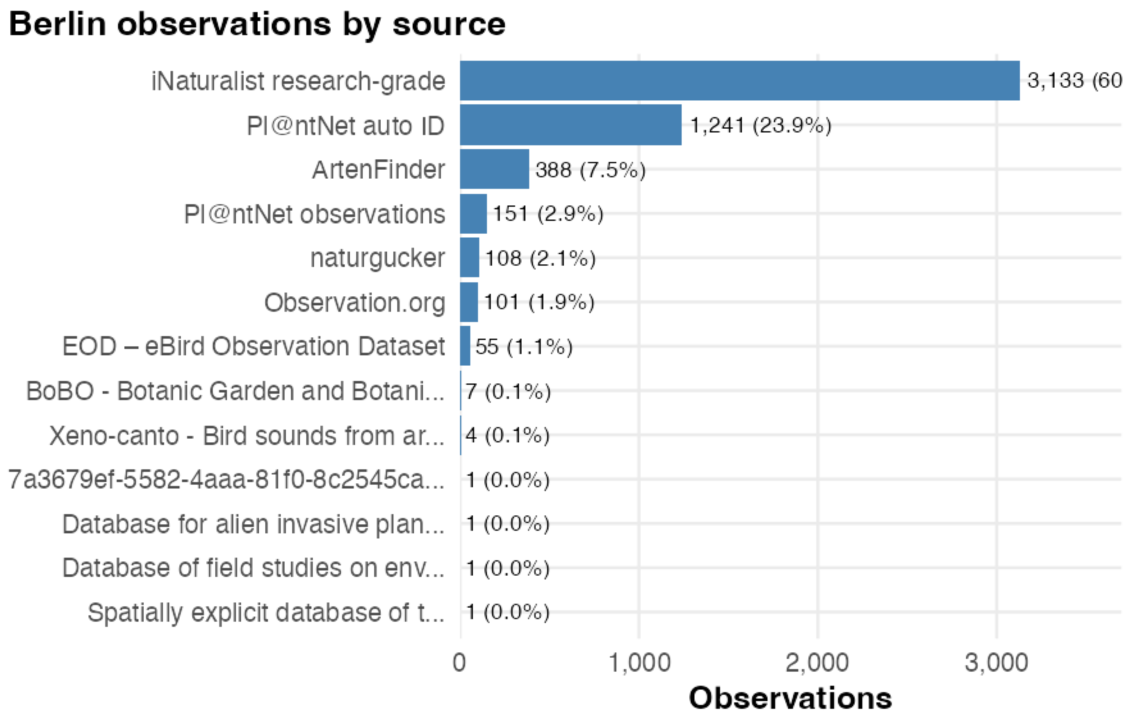


Figure 24: Berlin observations by source

As Figure 24 shows, Berlin's records are strongly dominated by iNaturalist research-grade, which contributes 60.3% of observations. Pl@ntNet auto ID adds another 23.9%, while ArtenFinder contributes 7.5%. Together, these three sources provide 91.7% of the Berlin dataset.

The majority of Berlin observations come from citizen-science platforms — both global ones, such as iNaturalist and Pl@ntNet, and local ones, such as ArtenFinder and naturgucker.

The source mix has now flipped twice. iNaturalist leads globally (Part I), Pl@ntNet auto ID leads nationally in Germany (Figure 21), and Berlin reverts to iNaturalist research-grade at 60.3%. Berlin is not representative of Germany in the way that Germany is not representative of the world. Each nested scale is dominated by a different platform, so the composition of any GlobalGeoTree subset has to be checked at the scale being analysed rather than inherited from the level above.

10.4 Spatial distribution of Berlin observations

- *Which parts of Berlin show the highest recording density?*

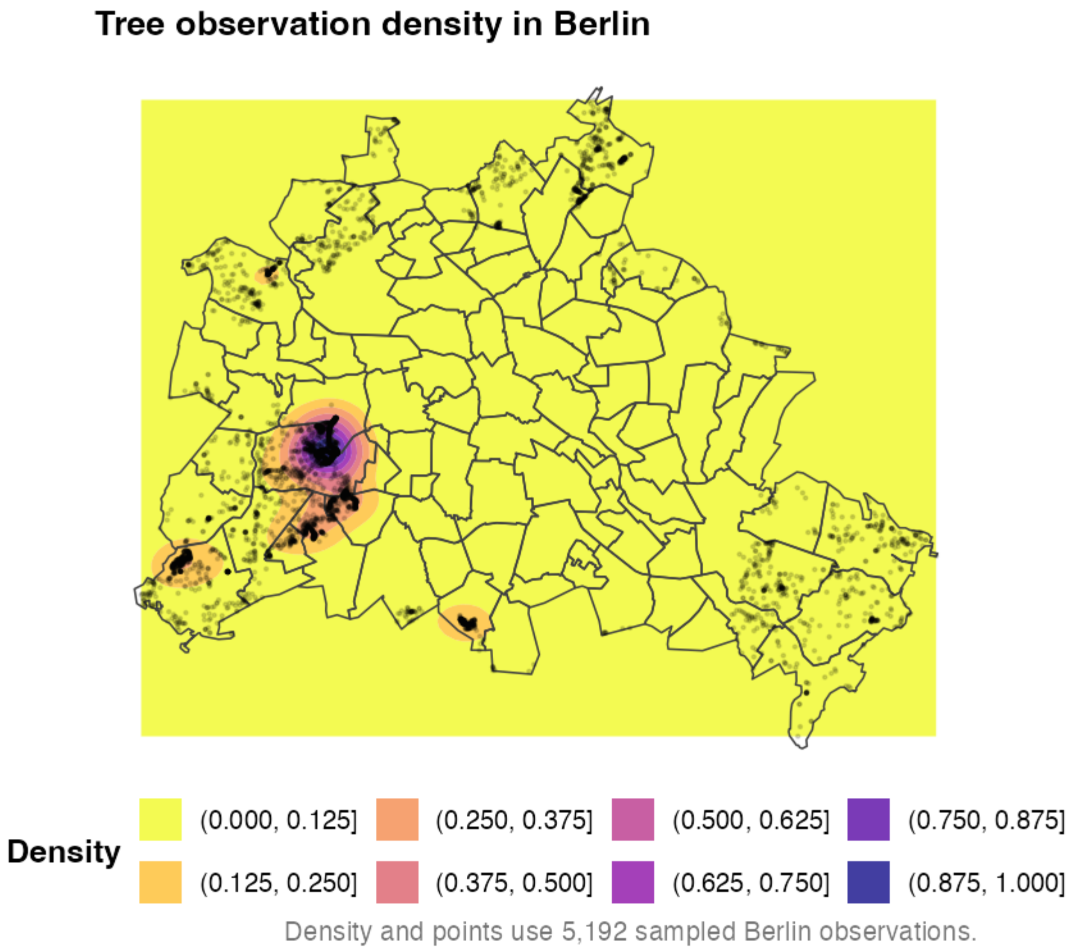


Figure 25: Tree observation density in Berlin

Figure 25 highlights which parts of Berlin have the most recorded tree observations. As always in this part, high-density areas correspond to parks, green corridors, and neighbourhoods with active observers rather than to places that necessarily hold more trees.

Observations are strongly concentrated in western Berlin, with the clearest hotspot just west of the city centre, in Grunewald. Smaller clusters appear in the southwest and along the southern edge, while most central and eastern local areas have much lower recording density.

10.5 Berlin local areas: observations and species

- *Do the local areas with the most observations also have the most recorded species?*

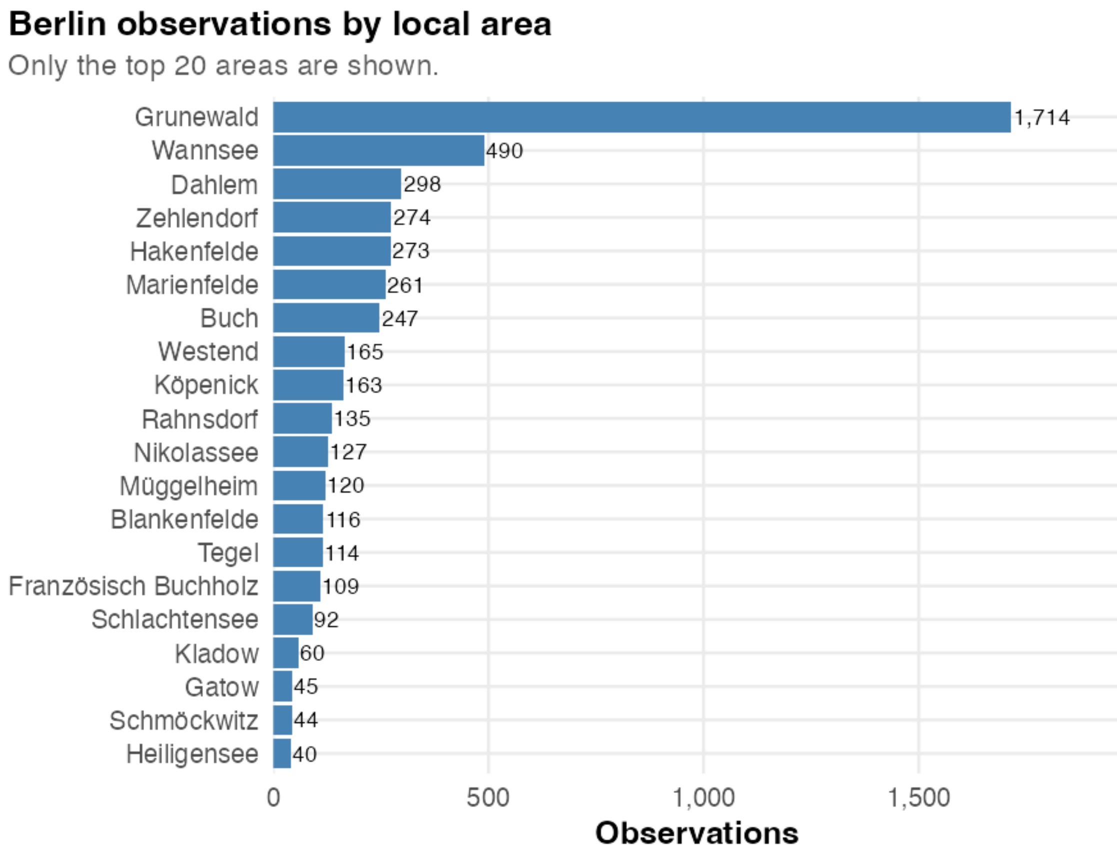


Figure 26: Berlin observations by local area

Grunewald dominates the dataset with 1,714 observations — 3.5 times the count for Wannsee, the next local area. Third-placed Dahlem has only 298, and every area below it fewer still, so recording is heavily concentrated in the southwest. Moreover, the four most represented local areas — Grunewald, Wannsee, Dahlem, and Zehlendorf — are all located in the south-western part of the city.

This is where the idea introduced at the start of the section becomes less convincing. Berlin was chosen partly because the whole city is included within one administrative boundary. However, this area also contains large forests, lakes, and quiet outer districts.

Most observations in the dataset come from Grunewald, Wannsee, and Dahlem. This means the data do not give a good picture of Berlin's street trees, nor do they represent all of Berlin's forests. The Berlin Environmental Atlas identifies large wooded areas in the northwest around Spandau and Tegel and in the southeast around Köpenick, in addition to the Grunewald and Düppel forests in the southwest (Senate Department for Urban Development, 2011). These additional forest areas are only sparsely covered in our dataset.

The data therefore appear to be biased toward a small number of popular green areas in western Berlin. Although these observations fall within the city boundary, they represent neither Berlin's urban tree stock nor the full extent of its forests.

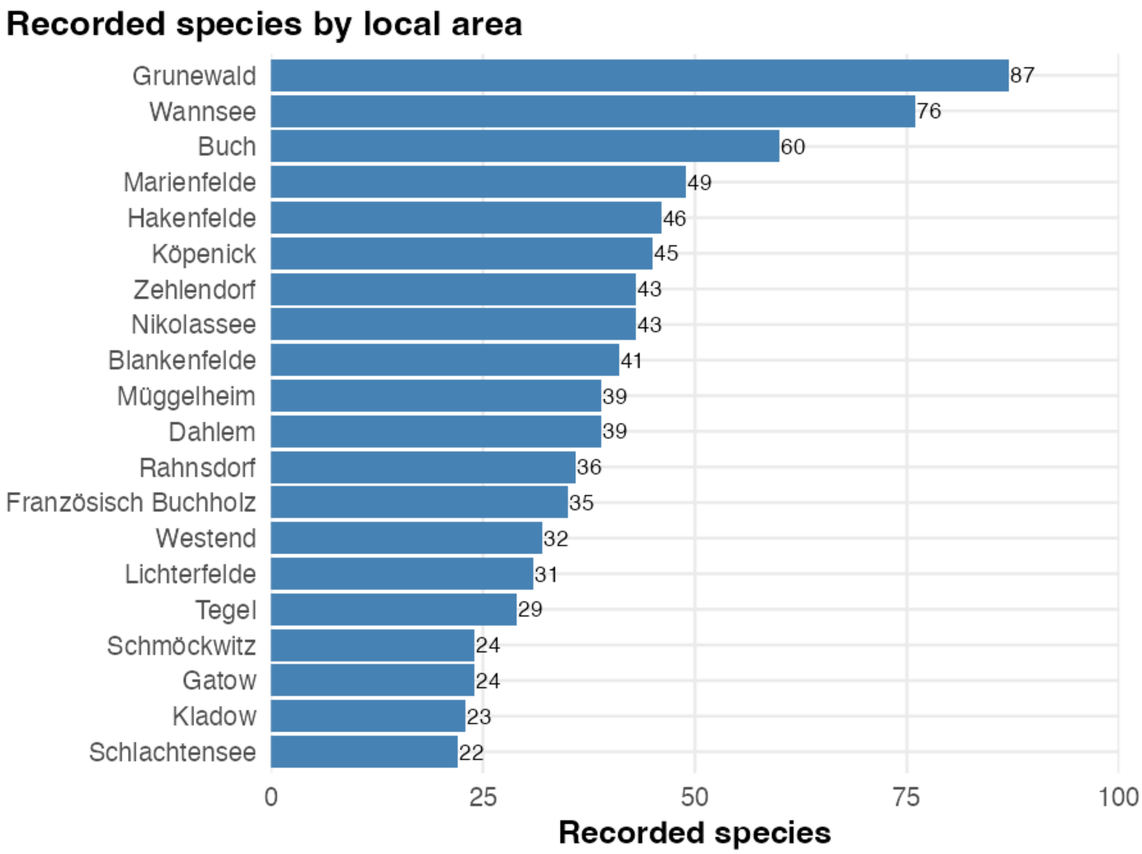


Figure 27: Recorded species by Berlin local area

Grunewald and Wannsee also have the highest recorded species counts. Buch stands out with 60 species despite having fewer observations than Dahlem, suggesting that observation volume and recorded richness do not always rise together.

The gap between Figure 26 and Figure 27 is itself informative. Grunewald's lead in species is nowhere near as extreme as its lead in observations, which is what one would expect from saturation: once an area has been visited often enough, additional visits mostly re-record species already on the list. Recorded richness is bounded by the local flora, whereas observation counts are bounded only by observer enthusiasm.

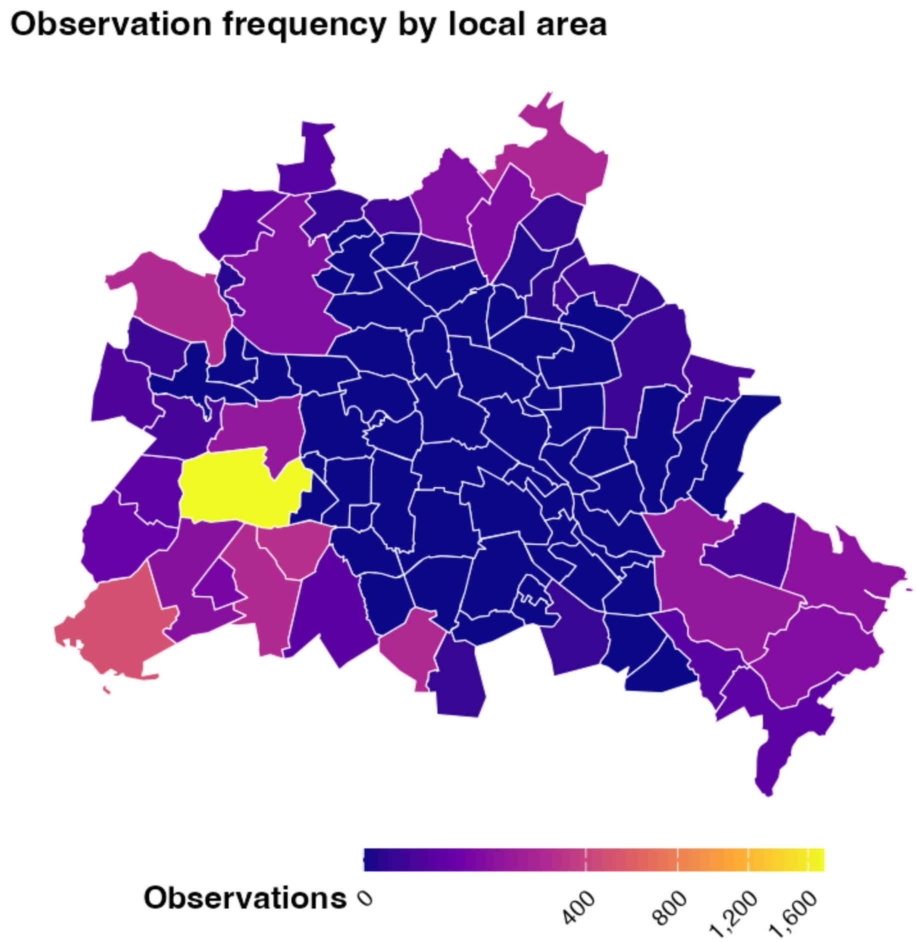


Figure 28: Observation frequency by Berlin local area

Figure 28 shows the same counts in their spatial arrangement. Square-root scaling keeps both high- and medium-count local areas visible.

The map confirms a strong hotspot around Grunewald in western Berlin. Most central areas have low counts, while several outer areas in the southwest, north and southeast show moderate recording activity.

10.6 Top recorded species in Berlin

- *Which tree species are recorded most often in Berlin, and are the species labels trustworthy?*

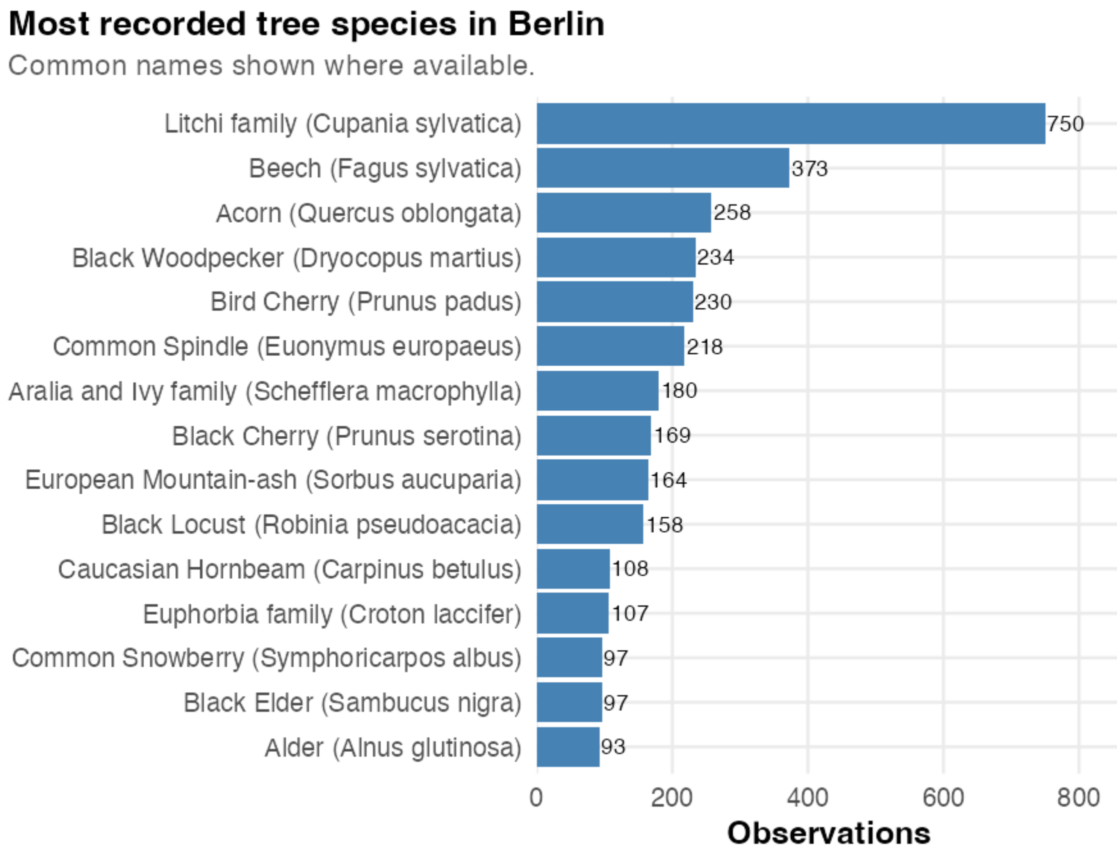


Figure 29: Most recorded tree species in Berlin

Figure 29 should be read with considerable caution, because it is the one figure in this part where the *labels* may be wrong.

Cupania sylvatica tops the ranking, with roughly 2 times the observations of the next species. That result is not credible: *Cupania* is a Neotropical genus that does not grow wild in Germany. The explanation is visible on the chart itself. The common names come from GBIF via `add_species_display_name()`, and the label rendered for this taxon is **"Litchi family"** — a *family* vernacular, not a species name. Its `species_key` (6657) is a family-rank GBIF identifier for Sapindaceae, not a species key.

The same fault recurs. `Croton laccifer` resolves to "Euphorbia family", `Schefflera macrophylla` to "Aralia and Ivy family", `Dombeya acerifolia` to "Mallow Family", `Salix arbutifolia` to "Willow Family". In every case a four-digit family-rank key sits where a seven-digit species key should be. The pattern suggests that where GlobalGeoTree could resolve a record only to family, it back-filled `level3_species` with an arbitrary — usually tropical — member of that family instead of leaving it blank.

The results can therefore be interpreted as follows: the Sapindaceae records are almost certainly maples, the Araliaceae records ivy, and the Malvaceae records limes — all thoroughly unremarkable Berlin trees wearing the wrong names. Across Germany these family-rank rows account for 11.4% of observations.

A second, unrelated problem sits alongside it: *Dryocopus martius*, the black woodpecker, is the second most recorded taxon among the German species carrying a GBIF key, with 11,353 records — behind only beech. It is a bird. Whatever filter was meant to restrict GlobalGeoTree to trees has let at least one animal through at high volume.

Both faults are inherited from the original GlobalGeoTree dataset rather than introduced here, and neither is repaired in this project. The consequence is that this ranking cannot be read as a description of Berlin's tree stock: it is a description of Berlin's *labels*, and some of them are wrong. Species-level use of the German subset would need the taxonomy validated against GBIF ranks first.

10.7 Berlin oak tree groups

- *Where are recorded oak trees concentrated in Berlin, and how could these records support a follow-up analysis?*

This section demonstrates a possible practical application of the dataset. The oak processionary moth (*Thaumetopoea processionea*) can cause allergic reactions and skin irritation near affected oak trees. Mapping recorded oak locations provides a starting point that could be compared with official tree inventories and pest-surveillance data; GlobalGeoTree observations alone do not show whether the moth is present or whether a public warning is needed.

Table 13: Berlin oak observation summary

metric	value
Berlin oak observations	258
Berlin recorded oak species	1
Share of Berlin observations that are oak	5.0%

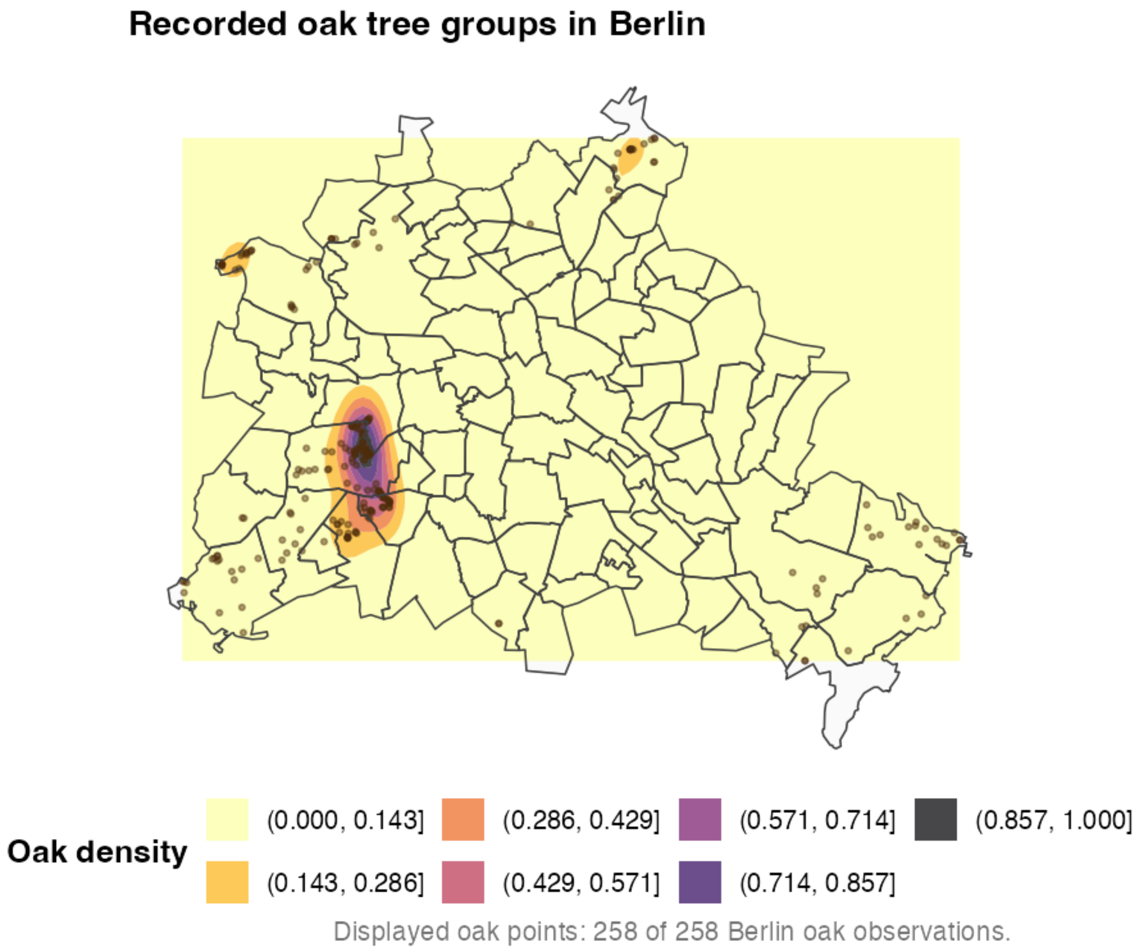


Figure 30: Recorded oak tree groups in Berlin

The oak pattern in Figure 30 closely follows the overall pattern in Figure 25, with the strongest concentration in the west. The oak map therefore mainly reflects where recording is most intensive, not where oaks are most common.

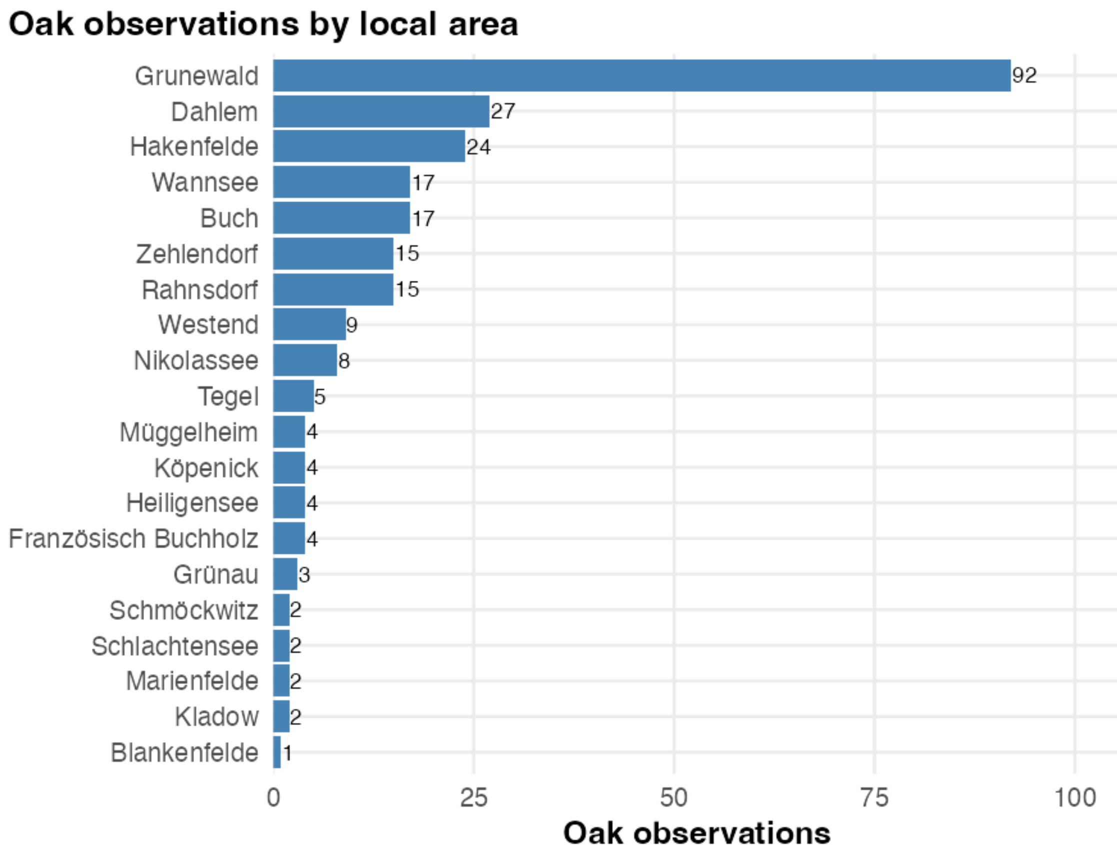


Figure 31: Recorded oak observations by Berlin local area

Grunewald, Dahlem and other highly sampled areas also contain most oak records. Because the full dataset is spatially biased, the oak subset inherits the same bias, so these counts should not be read as a ranking of true oak abundance.

A processionary-moth warning system needs to know where oaks *are*; Figure 31 shows where oaks were *looked at*. An area with many oaks and no observers is indistinguishable here from an area with no oaks, and those are exactly the areas a public-health warning would most need to reach.

GlobalGeoTree is therefore not representative for this use case — the oak records are usable as a supplementary layer over an official tree inventory, which has known coverage, but not as a substitute for one. The value of this section is in showing what the dataset would need to be joined against, rather than in the map itself.

11 Baden-Württemberg and North Rhine-Westphalia

- *Is Berlin's spatial concentration of observations a Berlin-specific pattern, or is it typical for German regions?*

The Berlin analysis above revealed that observations are heavily concentrated in a single location. To investigate, we compare Berlin with North Rhine-Westphalia (NRW) and Baden-Württemberg

— the first and third largest federal states by observation count. For each state, a density map and a NUTS3 district choropleth show where observations are concentrated.

One caveat applies to the whole comparison. Berlin covers about 890 km^2 and is divided into small local areas; Baden-Württemberg and NRW each cover roughly 35,000 km^2 and are divided into much larger NUTS3 districts. Measured concentration depends on the size of the units doing the measuring — the comparison below rests on the *shape* of the density surfaces, which are computed from coordinates rather than administrative units, and treats the choropleths as supporting detail only.

11.1 Baden-Württemberg

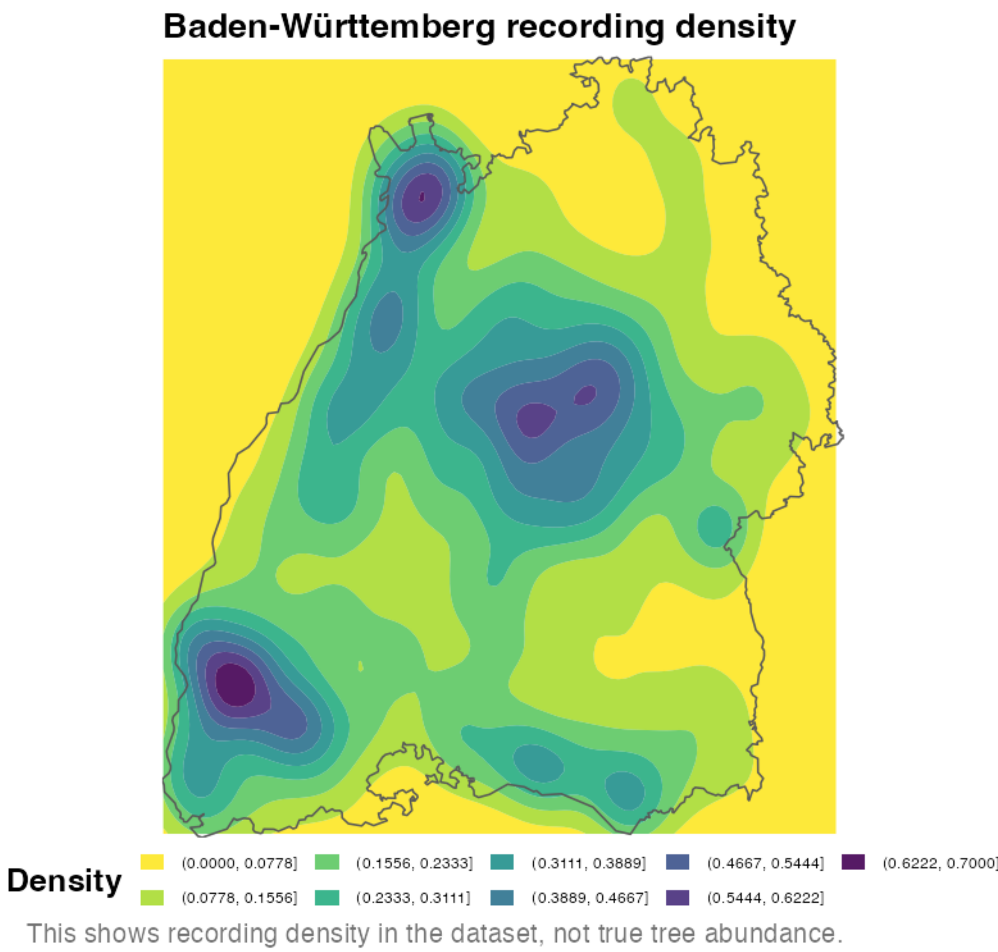


Figure 32: Baden-Württemberg observation density

In Baden-Württemberg recording is stronger in the west and south than in many eastern areas. Several hotspots are visible, especially in the southwest, northwest and central part of the state. Unlike Berlin, where observations are concentrated mainly around one western hotspot, Baden-Württemberg has several separate centres of recording activity.

11.1.1 Baden-Württemberg observations by NUTS3 district

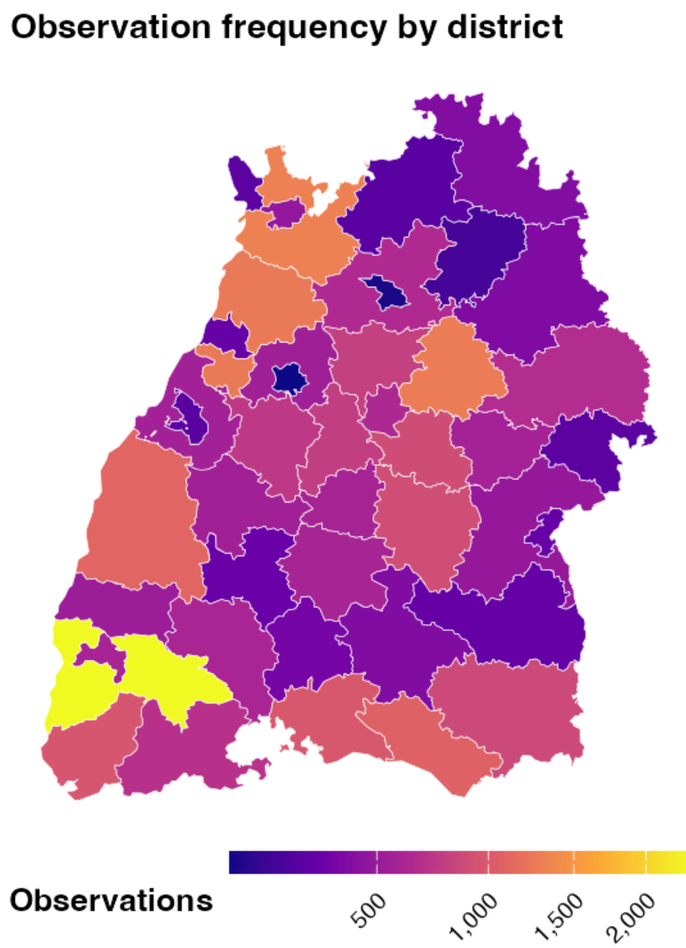


Figure 33: Baden-Württemberg observation frequency by NUTS3 district

Square-root scaling is applied so that a few very high-count districts do not wash out variation among the rest.

Figure 33 confirms that observations are unevenly distributed. The highest counts occur in the far southwest, while several western, northern and southern districts also have relatively high totals. Many central and eastern districts contribute fewer records.

11.2 North Rhine-Westphalia

NRW serves the same role as Baden-Württemberg: a state-level comparison point for Berlin.

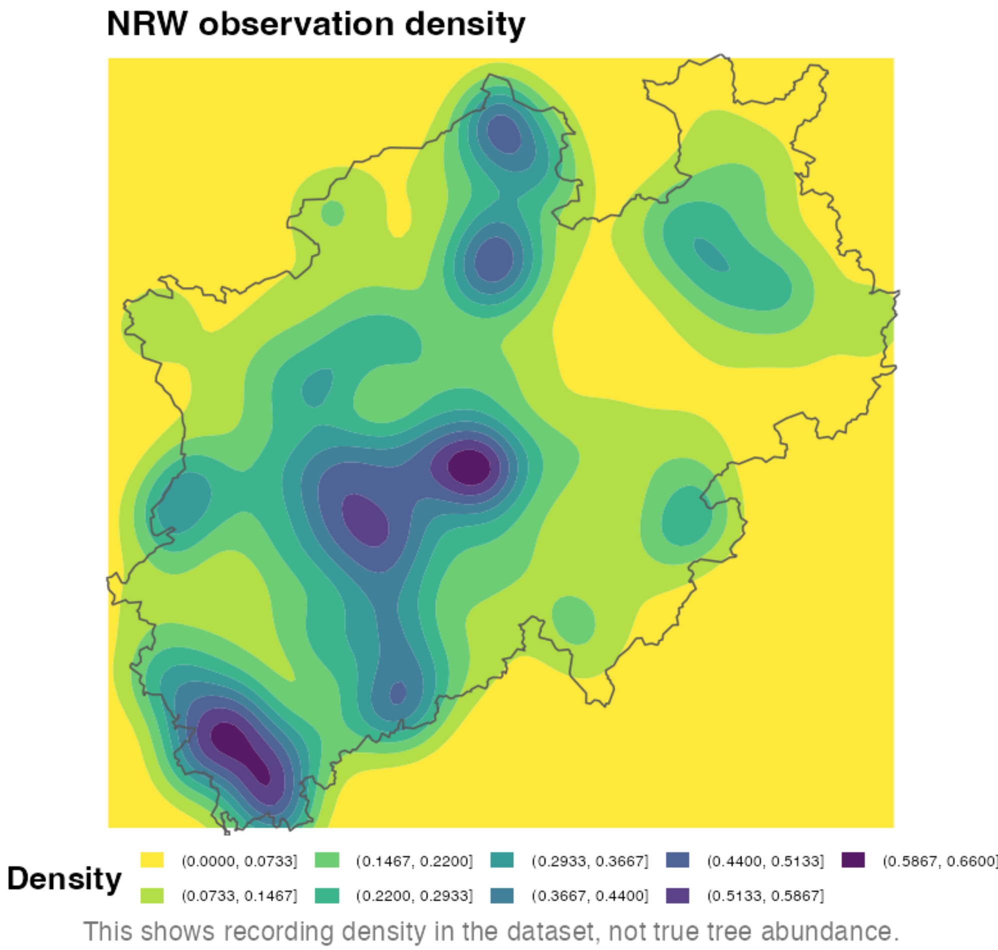


Figure 34: NRW observation density

NRW has several observation hotspots rather than one dominant centre. High recording density appears in the southwest, central-west and north, with smaller clusters in the east. This is more dispersed than Berlin's strongly concentrated western hotspot and helps explain the broad area of high density seen in western Germany in Figure 17.

11.2.1 NRW observations by NUTS3 district

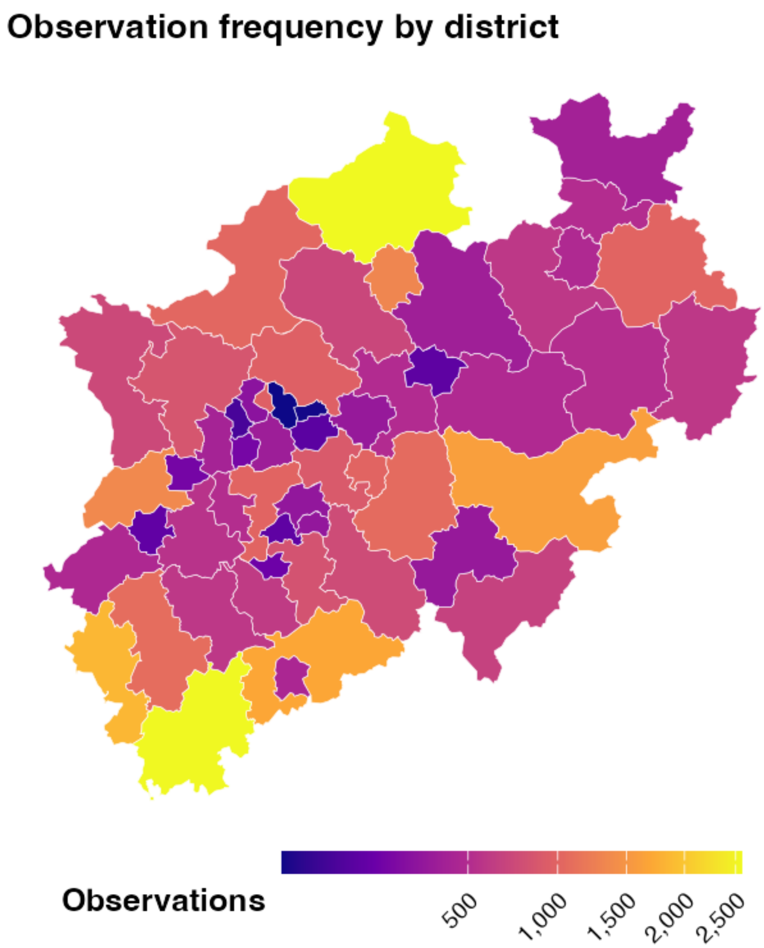


Figure 35: NRW observation frequency by NUTS3 district

As above, square-root scaling is used so that dominant districts do not obscure the rest of the distribution.

Figure 35 confirms that records are uneven but spread across much of NRW. A few northern and southern districts have the highest totals, while many central districts show medium counts. Compared with Baden-Württemberg, NRW has a similarly multi-centred pattern; compared with Berlin, its observations are less dependent on one local area.

Comparing Figure 32 and Figure 34 with Figure 25 answers the question this section opened with. Observations in Baden-Württemberg and NRW are spread much more evenly across their territories, with multiple distinct hotspots corresponding to major cities and populated areas, whereas Berlin's observations are heavily concentrated in a single location.

12 Germany: Conclusions

Part II asked where Germany's tree observations come from, who records them, and whether the national pattern also appears at city level. Five findings stand out.

Recording is concentrated in western and southwestern Germany, largely where people are. Figure 17 and Figure 18 show the same pattern: recording is strongest in the west and south, with a separate hotspot in Berlin and sparse coverage across the east and northeast.

No single platform represents Germany, and the platform mix changes by scale. Pl@ntNet auto ID provides 46.9% of German records, while the three largest sources account for 82.1%. National patterns therefore mainly reflect three platforms. This differs from the global dataset, where iNaturalist leads with 64.4%, and from Berlin, where iNaturalist again ranks first with 60.3%. Source composition must therefore be checked at each scale rather than assumed from the level above.

Berlin's records describe woodland more than the city itself. Grunewald alone contains 1,714 observations — 3.5 times as many as the next area. The Berlin data mostly capture recreational observations in forests and greener outer areas. The oak analysis carries the same bias, so it cannot support a processionary-moth warning system without an official tree inventory to fill the gaps in GlobalGeoTree.

Berlin's level of concentration is unusual. Baden-Württemberg and North Rhine-Westphalia have several hotspots around major population centres rather than one dominant site (Figure 32, Figure 34). Berlin differs in kind, not only in degree.

Species labels in the German subset are unreliable. Around 11.4% of German observations use family-level GBIF keys despite carrying plausible species names. This likely explains why apparently tropical trees top Berlin's ranking and why a black woodpecker appears second nationally. These problems come from the source dataset and are not corrected here.

Overall, Part II measures **recording behaviour in Germany, not German tree cover**. The western skew, state rankings, Berlin hotspot and oak distribution are better explained by where observers go and which apps they use than by where trees grow. The data are still useful, but mainly as evidence of a sampling process.

Part III: Distribution of the data in the United States

This part focuses on the US records and asks whether their distribution reflects ecological patterns, the way the data was collected, or a combination of both. Knowing which places and years contribute most of the observations is necessary before drawing conclusions about tree diversity from the dataset.

The analysis first describes national coverage and its regional imbalance. It then compares foliage categories, latitude profiles, and taxonomic richness. The final figures add the time dimension to show whether the geographical pattern changed as the number of records increased.

sample_id	level0	level1_family	level2_genus	level3_species	year	longitude	latitude	state	region
2638129	Deciduous Broadleaf	Actinidiaceae	Actinidia	Actinidia arguta	2024	-68.19505	44.37374	maine	Northeast
2638136	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2021	-73.93528	44.60028	new york	Northeast
2638137	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2020	-69.90992	44.22425	maine	Northeast
2638204	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-73.08378	44.52931	vermont	Northeast
2638205	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-73.01671	44.49624	vermont	Northeast
2638206	Deciduous Broadleaf	Amaranthaceae	Chenopodium	Chenopodium album	2024	-72.91761	44.99943	vermont	Northeast

Table 14: First observations after preparing the US subset

12.1 Pre-Processing

Rows labelled with the `US` country code were retained. Their coordinates were matched to state polygons from the `maps` package; points missed by the simplified coastal boundaries were linked to the closest state centre. The state names were then converted into the four Census regions — Northeast, South, Midwest, and West — so that comparisons would not be fragmented across fifty individual states.

Table 14 displays the working variables after this preparation.

The regional colours in Figure 36 remain unchanged from one graph to another. They come from the colour-blind-friendly Okabe-Ito palette.

Figure 36: Regional colours used throughout the US analysis

12.2 United States overview

The initial overview separates national reach from sampling intensity. A dataset may include all major parts of the country and still be dominated by a few places.

For that reason, Figure 37 shows the individual coordinates, Figure 38 compares the regional totals, and Figure 39 identifies the cells where records accumulate most strongly.

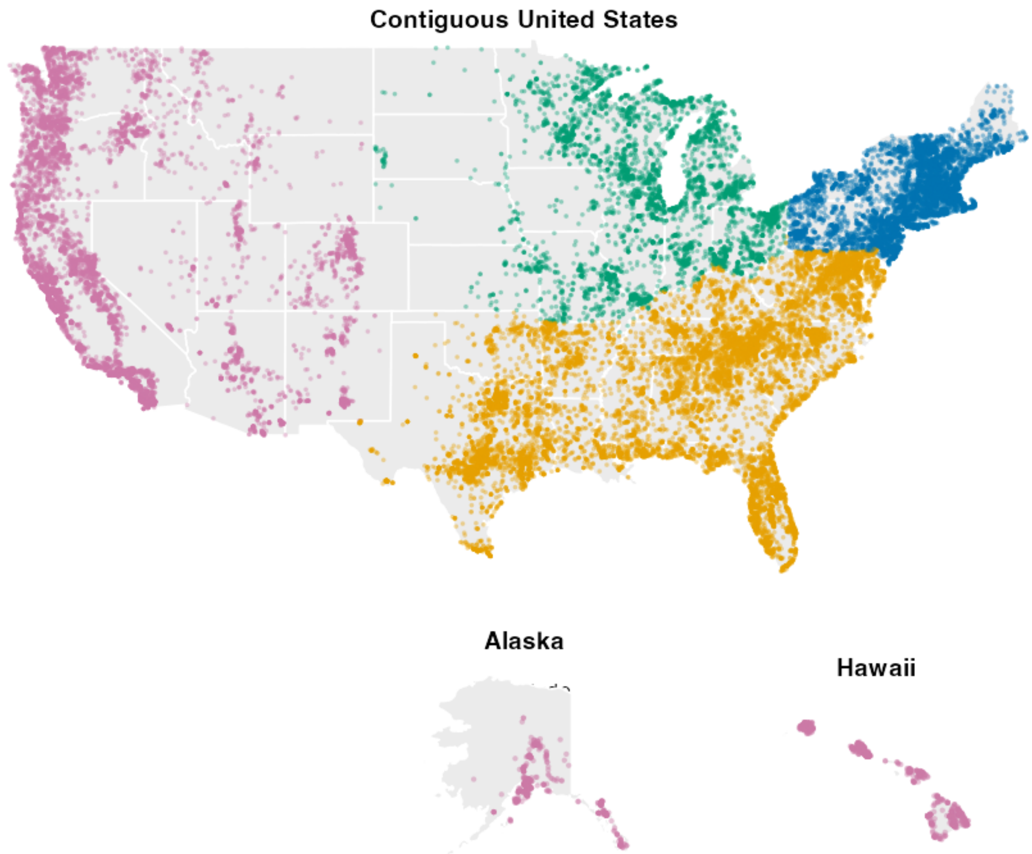


Figure 37: Location of US observations, coloured by Census region

The map displays a proportional sample of 50,000 from 1,871,976 mapped US observations. The sample is allocated across the contiguous states, Alaska, and Hawaii before drawing, with a small minimum that keeps the sparsely recorded Alaska and Hawaii areas visible above their true proportional share.

The points in Figure 37 cover most of the country, but not with the same intensity. Texas, California, Florida, and the Northeast form visible clusters, whereas the Great Plains and much of the interior West are sparsely represented. Accessibility and the location of contributors probably explain part of this contrast, so the map cannot be read as a map of tree abundance.

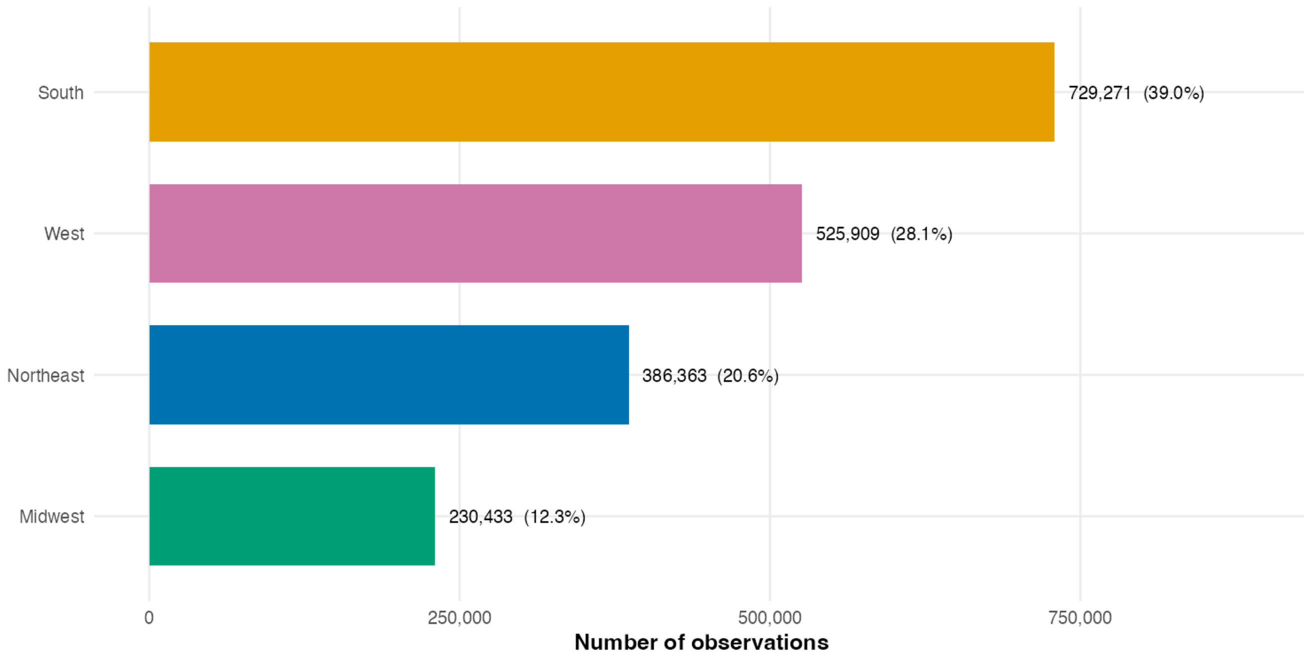


Figure 38: Contribution of each US region to the complete US subset

The regional counts in Figure 38 make the imbalance more explicit. The South supplies the largest share of records, followed by the West. The Northeast and Midwest have considerably fewer rows. These percentages refer to dataset participation and should not be interpreted as regional forest shares.

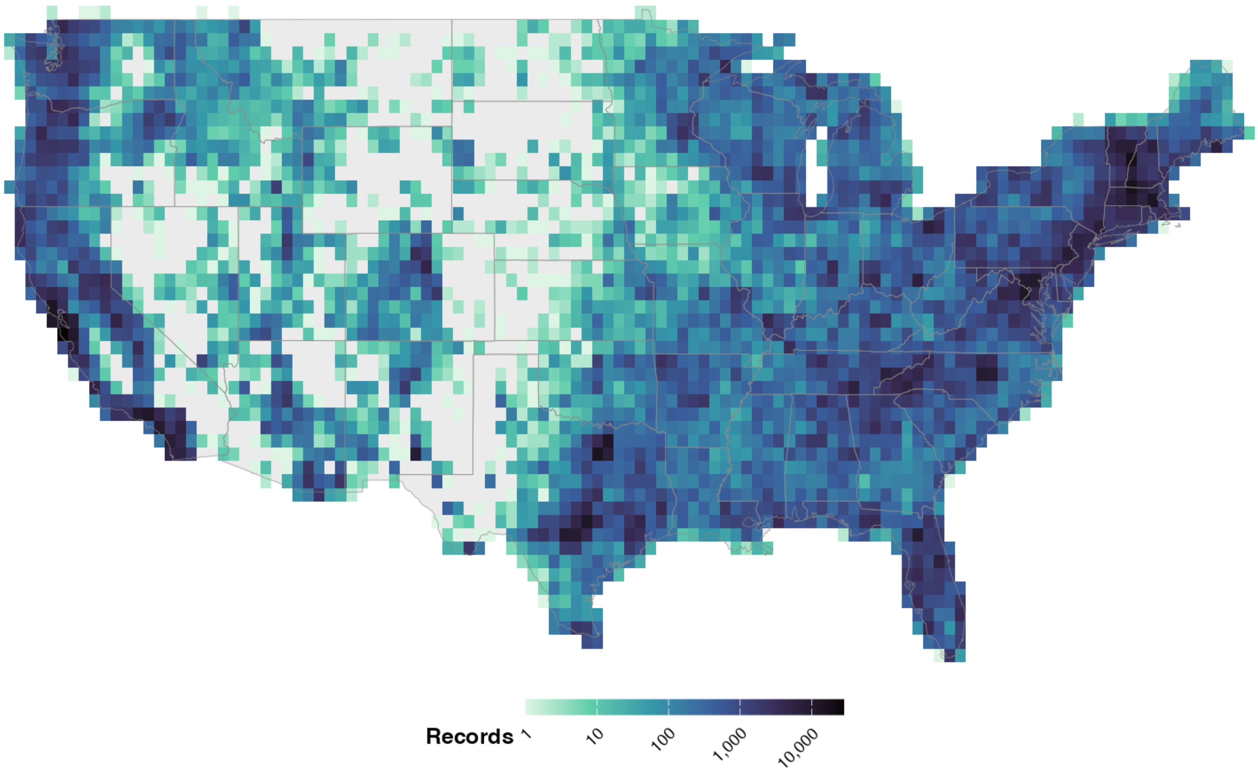


Figure 39: Local concentration of records across the contiguous United States

The grid representation in Figure 39 isolates the main hotspots around central Texas, southern California, Florida, and the northeastern corridor. Pale inland cells indicate limited sampling rather than an absence of trees.

Overall, the US subset has a wide footprint but an uneven internal balance. The following comparisons are therefore interpreted as patterns within the available records, not as a national census.

12.3 Spatial distribution of foliage types

The next step is to compare the kinds of trees recorded in different places. Foliage categories summarise important ecological strategies: broad or needle-shaped leaves and evergreen or deciduous behaviour respond differently to cold, drought, and seasonality.

The small maps in Figure 40 preserve the spatial detail. Figure 41 converts those records into regional percentages, while Figure 42 removes the administrative boundaries and follows the north–south gradient directly.

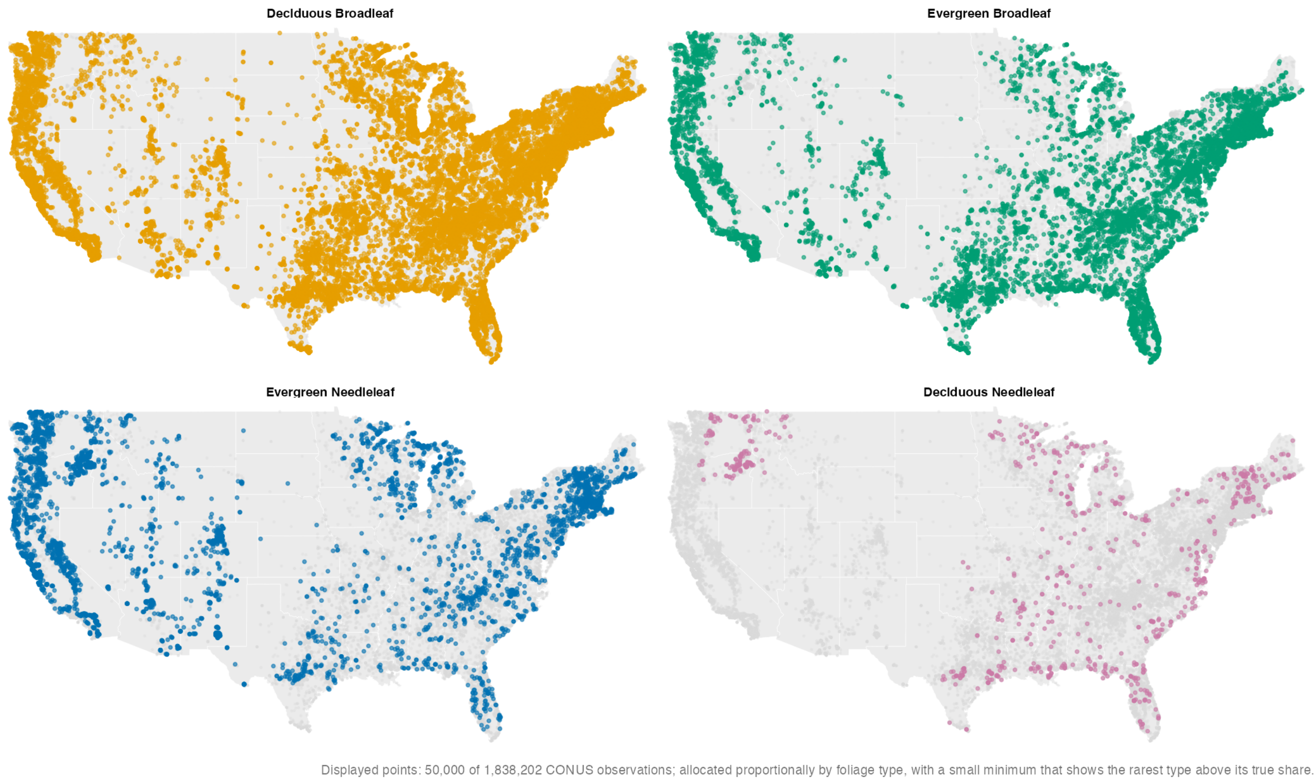


Figure 40: US locations recorded for each foliage category

The categories in Figure 40 do not occupy identical spaces. Deciduous Broadleaf is found throughout the contiguous states and is particularly visible in the East and South. Evergreen Broadleaf is concentrated farther south and on the Pacific coast, while Evergreen Needleleaf forms strong western and northern clusters. Deciduous Needleleaf is present in pink, but its 10,951 records make it much rarer than the other three categories.

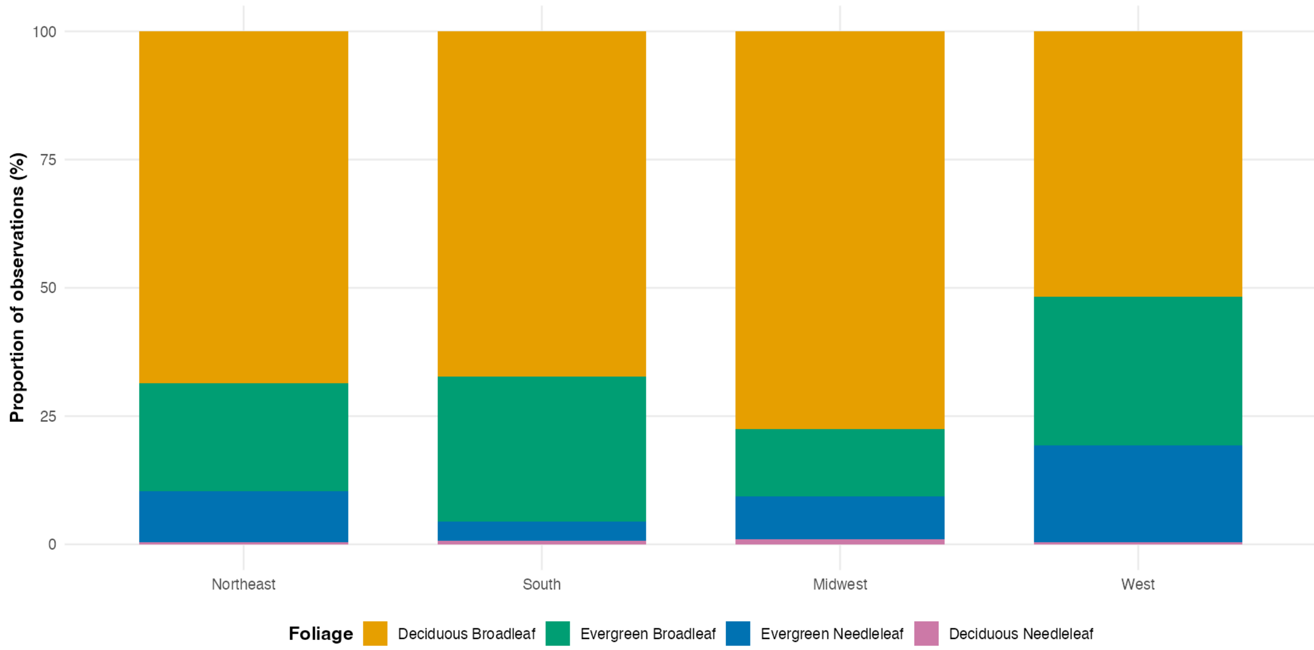


Figure 41: Foliage composition within each US region

Using percentages in Figure 41 prevents the South's larger sample from dominating the comparison. Deciduous Broadleaf remains the largest group everywhere, although its share varies greatly. Evergreen Broadleaf reaches its largest regional shares in the South and West, while Evergreen Needleleaf is especially prominent in the West. Deciduous Needleleaf occurs in every region but never exceeds a little over 1%. Regional climate is one plausible contributor to these differences.

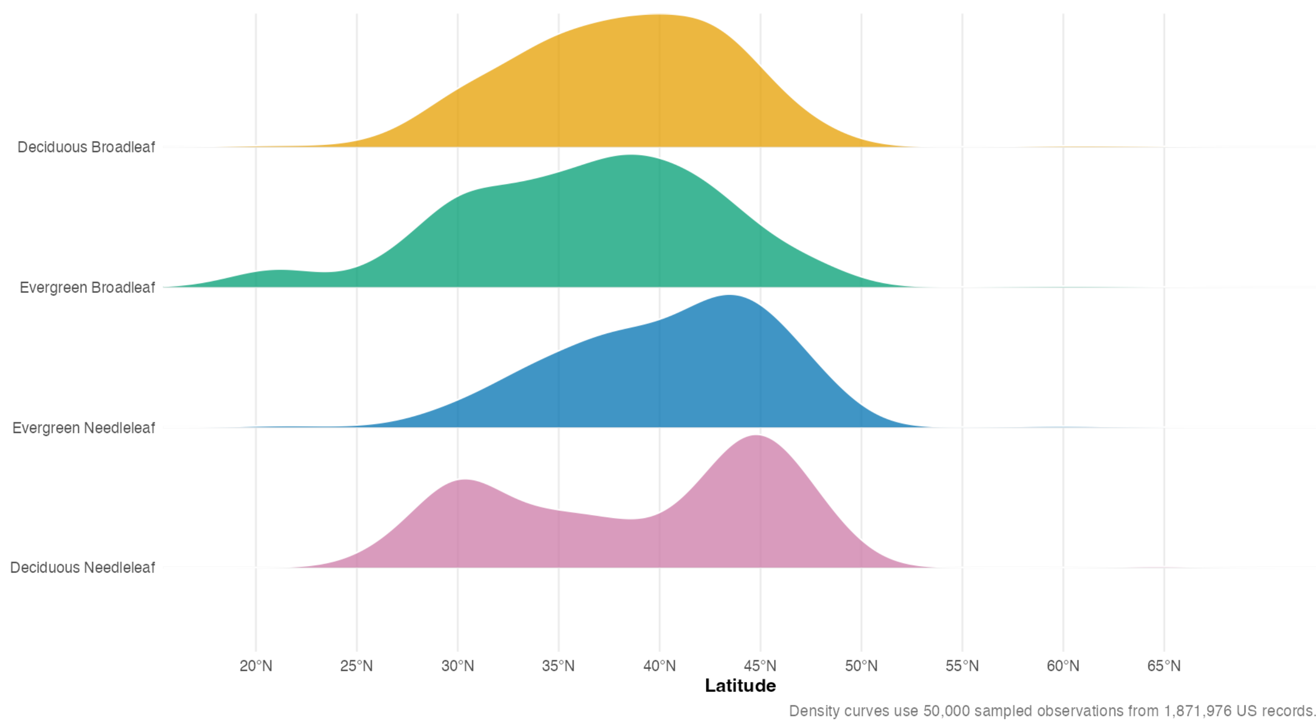


Figure 42: Latitude profiles of the foliage categories

The curves in Figure 42 overlap, but their peaks occur in different places. Evergreen Broadleaf has a clear southern component, including Hawaii, whereas Evergreen Needleleaf is weighted toward the northern half of the country. Deciduous Broadleaf covers the broadest span of the contiguous states. Deciduous Needleleaf is much less frequent but is not absent: its pink curve has concentrations around 30°N and 45°N.

These shifts support the idea that climate helps structure foliage type. Moving north usually changes temperature, season length, and frost exposure. Still, latitude does not measure climate directly, and other factors — precipitation, elevation, soil, land use, historical range, and sampling density — change at the same time. The figure therefore provides evidence of association, not a causal estimate.

12.4 Taxonomic richness by region

Taxonomic richness is measured here at three levels: family, genus, and species. Simple totals are useful, but they favour regions with more observations because additional records create more opportunities to encounter a new taxon.

Table 15 and Figure 43 present the unadjusted counts. Figure 44 then places the regions on a common sampling scale. The last visualisation, Figure 45, changes the question from "how many species?" to "how geographically concentrated is each common species?"

	Region	Foliage	Families	Genera	Species	Observations
4	South	4	138	490	1,626	729,271
3	West	4	131	497	1,811	525,909
1	Northeast	4	86	203	653	386,363
2	Midwest	4	84	199	660	230,433

Table 15: Recorded taxonomic diversity and sample size by region

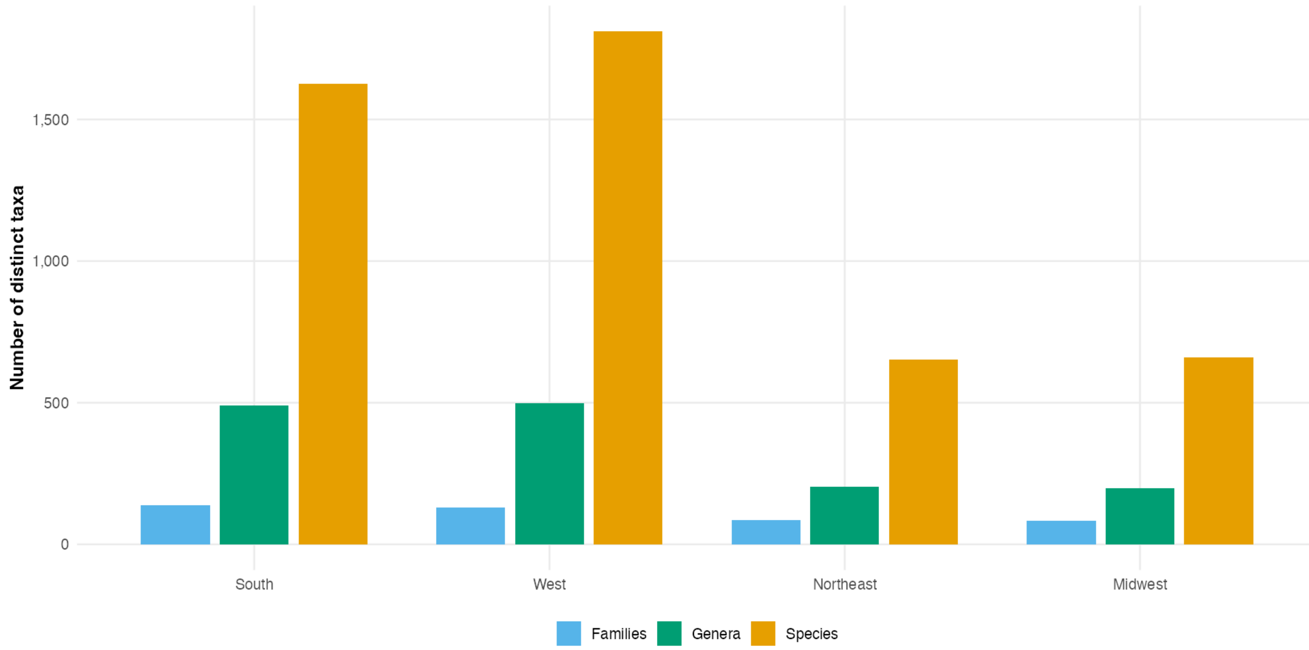


Figure 43: Unadjusted family, genus, and species counts by region

The South leads the raw counts in Table 15 and Figure 43, followed by the West. Their advantage is partly expected because these are also the two largest regional subsets. Despite having fewer observations, the Northeast and Midwest still include a notable share of the 3,001 species recorded for the United States.

Methodological note: `vegan::rarefy()` computes the expected number of recorded species in a random sample of n observations, drawn without replacement from each region's species-abundance counts. It evaluates this expectation analytically instead of averaging repeated random draws, producing one rarefaction curve per region. This standardises the number of observations being compared, but it does not correct for uneven geographic coverage, source composition, or detectability within each region.

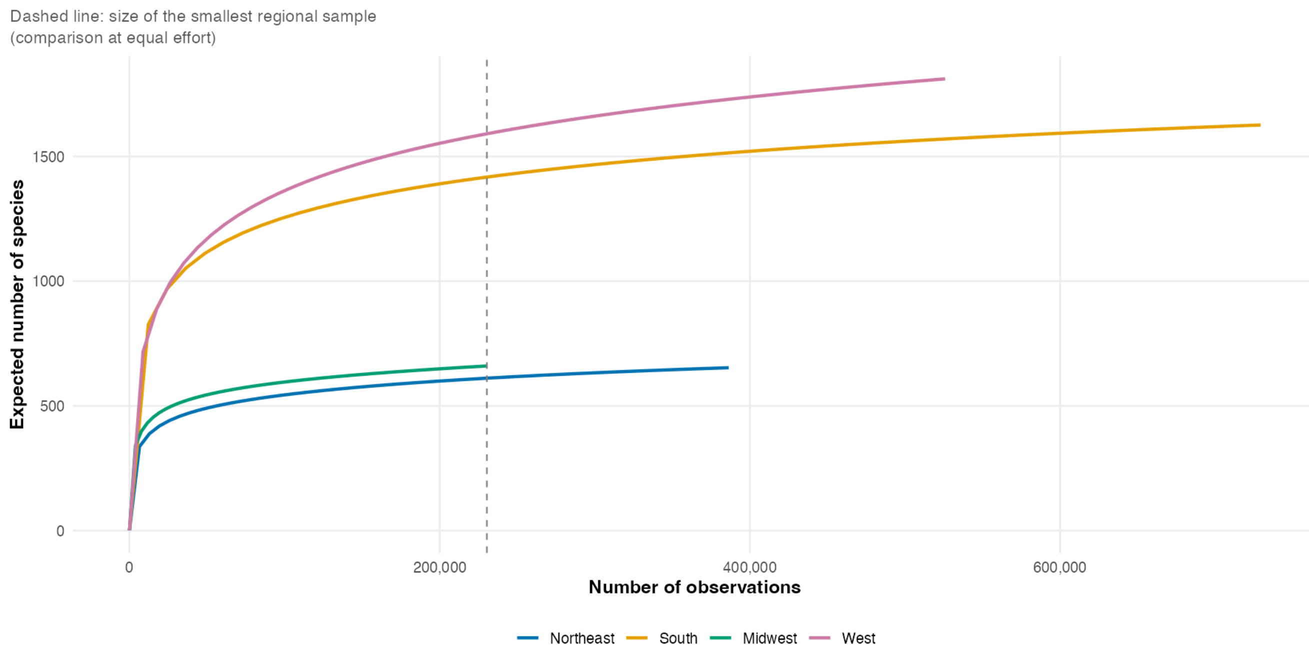


Figure 44: Growth of recorded species richness as regional sample size increases

In Figure 44, regions can be compared at the same horizontal position and therefore at the same number of sampled records. The curves reveal how quickly additional observations continue to add species in each region. Comparing them at a shared sample size separates part of the diversity pattern from the very different regional record totals.

Figure 45: Observed ranges of the 12 most frequent species in the contiguous states

There is no single answer to whether members of one species grow together. In Figure 45, several species are restricted to compact clusters on the Pacific coast, in Texas, Florida, or the Northeast, while others extend across several states or regions. Each species has its own observed range, but that range may be broad or discontinuous. Environmental requirements and dispersal history matter, and the gaps may also reflect missing observations.

12.5 Changes in the distribution over time

All previous figures combine ten observation years into one picture. This can hide changes in participation, especially for data coming mainly from citizen-science platforms.

The bars and line in Figure 46 compare annual volume with the number of represented states. Figure 47 identifies which parts of the country produce the increase, and Figure 48 checks whether later records spread into new territory or reinforce existing hotspots.

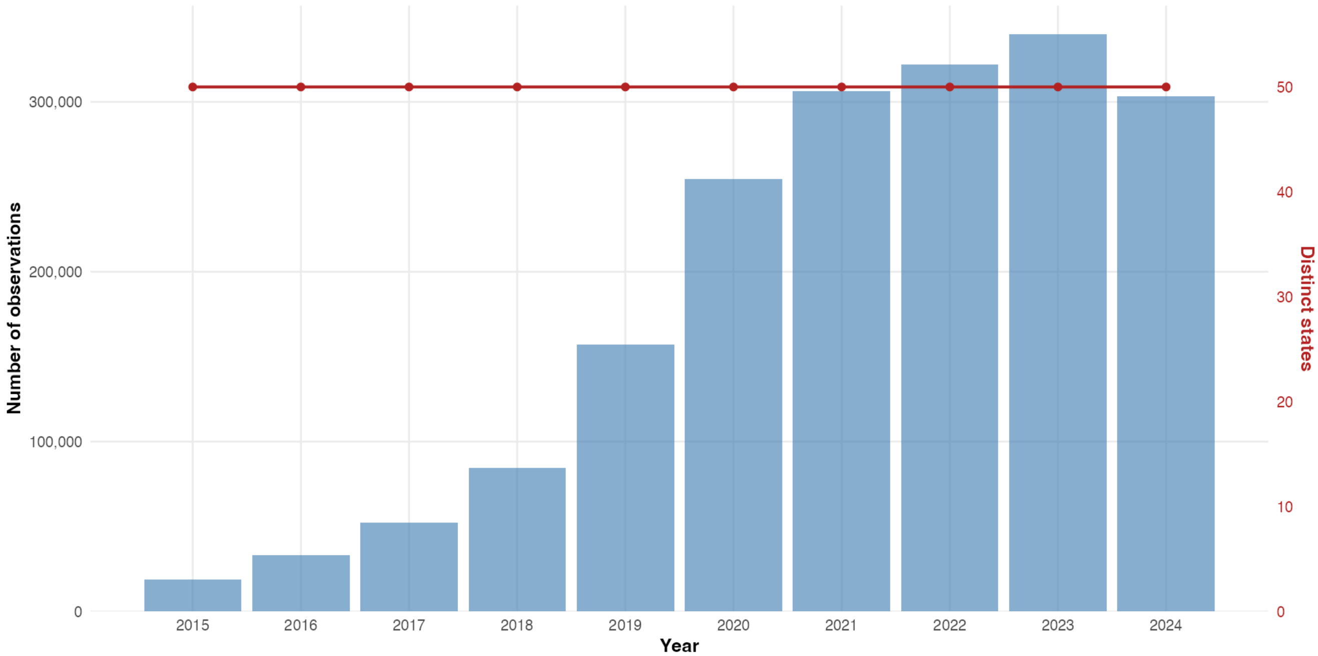


Figure 46: Annual US record volume and number of represented states

The annual series in Figure 46 rises markedly after 2018 and reaches its highest level near the end of the period. State coverage expands earlier and then changes much less. In other words, recent growth mostly adds records within an already broad national footprint. It indicates greater data-collection activity, not rapid growth in the tree population.

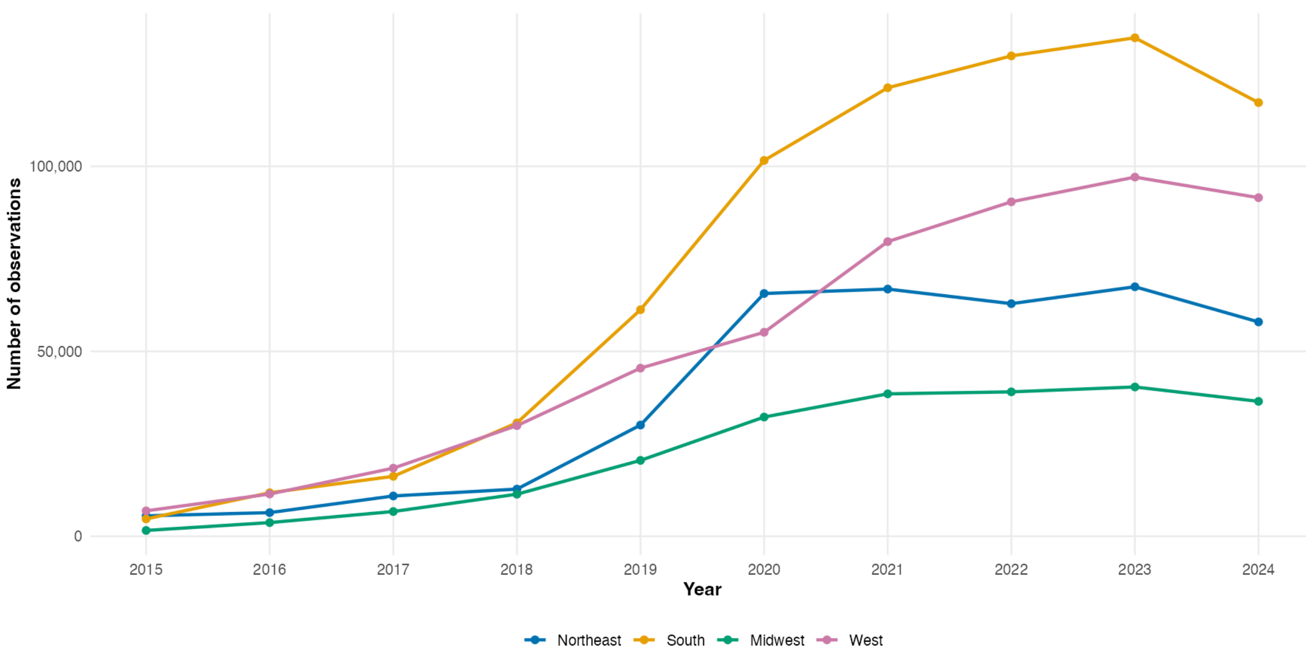


Figure 47: Yearly observation counts separated by US region

The regional lines in Figure 47 do not grow in parallel. Most of the national increase comes from the South and West, whereas the Northeast and Midwest stay at lower levels. The regional imbalance is therefore repeated across several observation years rather than being produced by one exceptional year.

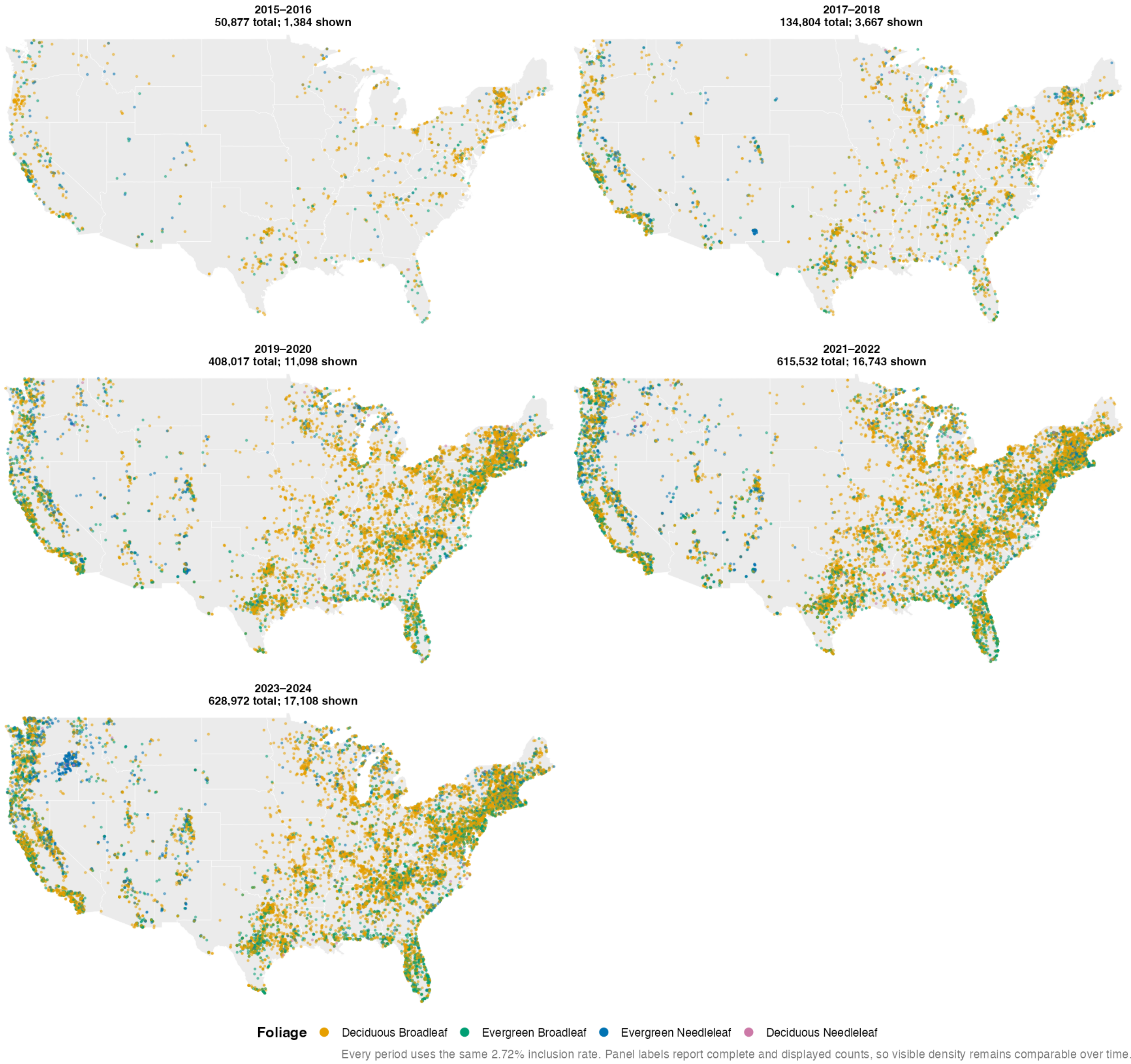


Figure 48: Spatial coverage of US observations across five observation periods

The period maps in Figure 48 repeatedly show the same geographical cores, although the number of visible points increases. Coverage becomes denser more often than it becomes geographically broader. The dataset thus preserves a similar internal bias throughout the decade.

13 Outlook and next steps

The next stage should turn the bias diagnosis in this report into a reproducible workflow for validation and quality control.

Strengthen taxonomic validation. Species names and `species_key` values should be checked against the current GBIF taxonomic backbone, with explicit flags for non-species ranks, non-plant taxa, and inconsistent family–species combinations. Corrections should be quantified before analysing richness or ranking dominant species.

Measure and adjust for sampling effort. Observation density should be assessed relative to population, land area, accessible green space, and source activity. Rarefaction, stratified subsampling, and models that include source, year, and spatial effort would help separate recording intensity from biological patterns.

Run source-sensitivity analyses. Key global, German, Berlin, and US results should be repeated after excluding the largest source, restricting the data to institutional and survey records, or balancing source groups. Results that remain stable across these variants would provide stronger evidence than patterns visible only in the full, source-dominated sample.

Validate regional use cases with independent data. Berlin observations should be compared with official street-tree and forest inventories, while national patterns should be benchmarked against forest inventories or land-cover datasets. For applications such as oak processionary-moth monitoring, GlobalGeoTree observations could supplement an authoritative oak inventory, but should not be used to infer absence in unobserved locations.

Improve source metadata and reproducibility. The keyword-based source classification could be replaced or supplemented with a documented source lookup table or a validated classifier. Raw source names, classification uncertainty, boundary-file versions, API dates, and all filtering decisions should remain available in the analytical output.

Investigate underrepresented regions. Follow-up work should focus on central Africa, the Amazon interior, Central Asia, and other sparsely represented areas, ideally with regional expertise and additional local data sources. This would help distinguish genuine observation gaps from failures of source classification or data integration.

Together, these steps would make GlobalGeoTree more useful as a bias-aware analytical resource while keeping its limitations transparent.

14 Conclusion

GlobalGeoTree is a valuable record of where and how tree observations have been collected, but it should not be treated as a systematic inventory of global tree abundance or diversity. Across the analyses, the most consistent finding is that the recorded patterns reflect an interaction between ecological reality, uneven observer activity, platform coverage, and source-specific practices.

Source composition changes with scale: iNaturalist dominates globally and in Berlin, whereas Pl@ntNet auto ID leads in Germany. Similarly, much of the apparent temporal growth is driven by citizen-science platforms and by denser recording in regions that were already well represented, rather than by broad geographic expansion. The German and US case studies reinforce the same point: areas with more observations or recorded species are often areas with more people, more active platforms, or repeated visits.

These biases do not make the dataset unusable; they define the questions it can answer reliably. GlobalGeoTree is well suited to analysing recording activity, comparing source footprints, identifying coverage gaps, and generating hypotheses. Claims about biological abundance, species richness, regional diversity, or pest risk require independent validation, correction for observation effort, and careful taxonomic quality control. The concentrated Berlin sample and the mislabelled German records demonstrate why these safeguards are essential.

AI Usage Disclaimer

We acknowledge that AI chatbots were used during this project in accordance with the course guidelines described in *Tools and Software for Our Course* and *Rules for Tools in Data Visualization* (Grömping, 2023, 2026).

The search for data and the selection of datasets were carried out entirely manually using Hugging Face, Kaggle, and Google Dataset Search. The initial brainstorming, project planning, development of visualisation ideas, and distribution of tasks were completed by the group in person during class sessions.

The report was written, revised, and reviewed manually by the authors. AI tools were used to support language editing, including the identification of grammatical errors, typographical mistakes, and inconsistencies in terminology.

AI tools were also used to assist with code development. However, code reviews, analytical and engineering decisions, as well as final implementations were completed and validated by the authors.

AI assistance helped us to:

- identify and correct coding errors more efficiently;
- resolve rendering issues, such as oversized text, overlapping legends, and cropped chart titles;
- expand the scope of the project and develop more engaging visualisations; and
- redirect effort from routine coding difficulties towards the visual design and substantive interpretation of the project.

The AI tools used in this project were:

- ChatGPT, including the GPT-5.5 and GPT-5.6 models;
- Claude, including the Sonnet and Opus models.

Examples of prompts used during the project include:

- "The legend in the ... visualisation is formatted incorrectly and the numbers overlap. How can this be fixed?"
- "The ... table is not rendered correctly in the PDF. How can this issue be resolved?"
- "Review this R code in the ... section and replace hard-coded values with calculations derived directly from the data."
- "Review and suggest improvements to the subsampling method in the ... section so that it preserves meaningful regional patterns."

- “The scales, legend, or rendering in this `ggplot2` visualisation are incorrect. How can they be fixed?”
- “Check this text for typographical errors and inconsistencies in terminology, and suggest corrections.”

The use of AI enabled faster iteration, clearer writing, more consistent visualisations, improved code quality, and earlier identification of potential errors. AI tools did not replace the authors’ analytical involvement, subject-matter judgement, or responsibility for the final results. AI-assisted text and code were reviewed and validated by the authors.

In addition to AI tools, the following software and collaboration platforms were used:

- R as the programming language of the project;
- Quarto for creating and rendering the report;
- RStudio and Visual Studio Code as IDEs;
- Git and GitHub for version control and collaboration;
- Google Docs and Google Slides for collaborative work and the exchange of project materials;
- a group chat for communication and coordination among team members.

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